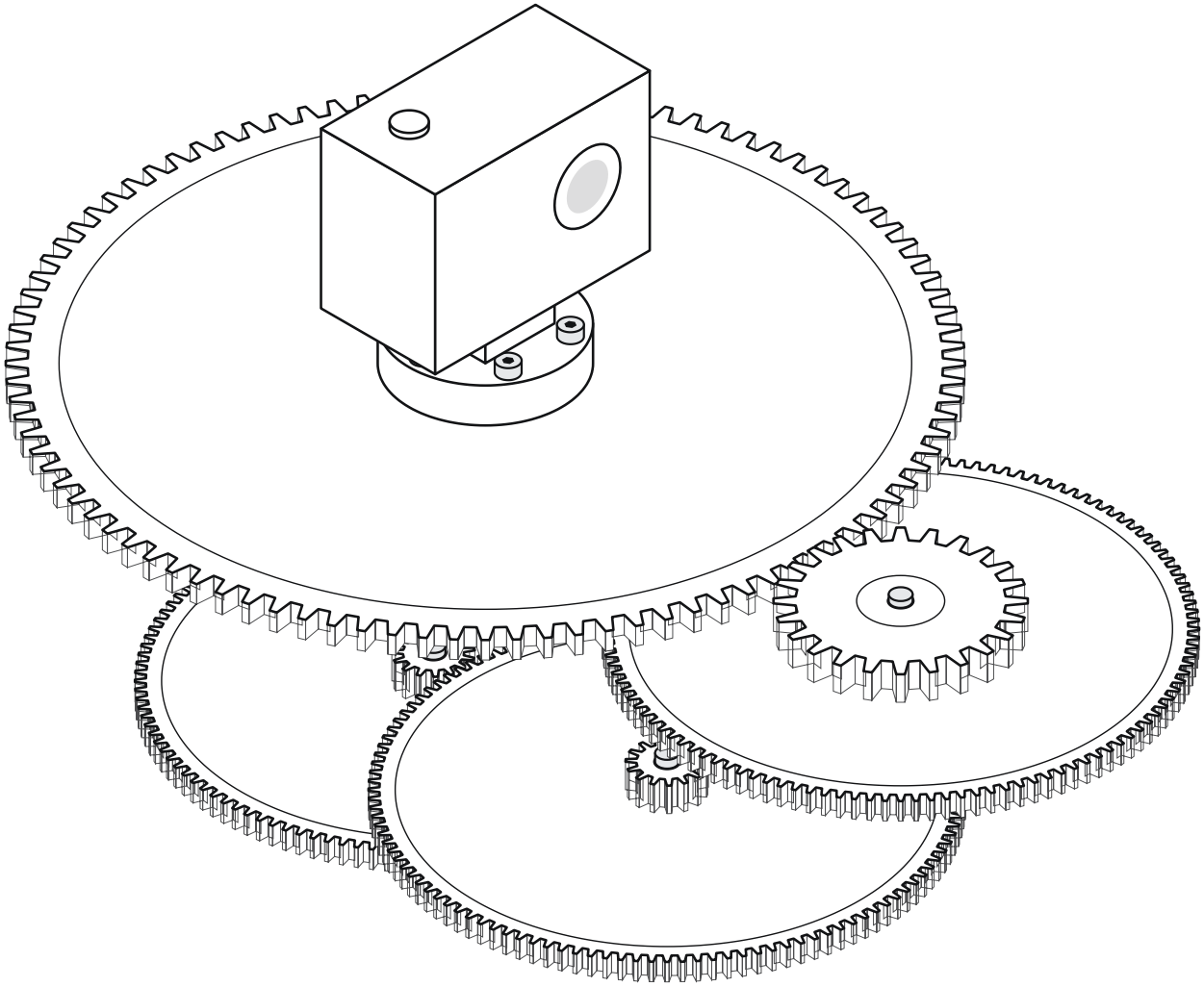
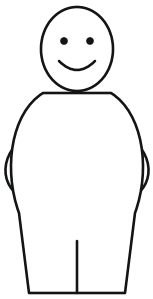
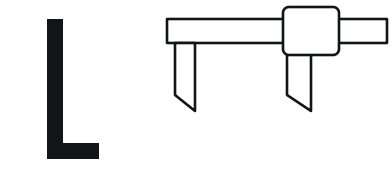
STJÄRNSPÅR

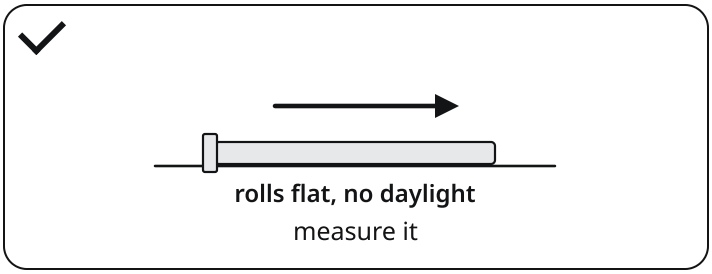
Sidereal camera tracker, 2048 : 1. Assembly, wiring, smoke test, first light.

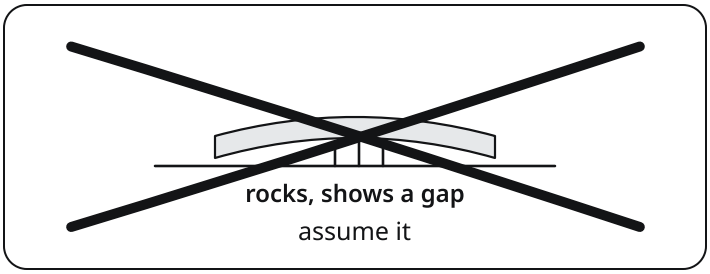


Another Amazing DIY Project
by Mike Montana









This build has three measured numbers and several guessed ones. Where a step says **measure**, the design depends on it: the printer offsets, the bearing fit and the magnet gap were all found this way, and none of them can be predicted.

Contents

A **Gearbox**
steps 1 to 20

B **Controller board**
steps 21 to 40

C **Smoke test**
steps 41 to 49

D **Wiring**
steps 50 to 54

E **First power-up, calibration**
steps 55 to 64

F **How to use**
steps 65 to 70

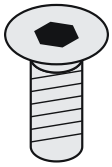
G **Appendix: schematics**
system, logical, full board

Print first, assemble second. The calibration gauges set the numbers every later step depends on: `BOREGAUGE` picks the motor pinion, `BEARINGGAUGE` sets `$BrgFit` before the tower is printed, and the fit gauge already gave the 0.50 mm post/hole offset. PRINTING.md has the order and settings.

Build the polar wedge **before** the gearbox. One degree of polar misalignment drifts about 16 arcsec a minute, which is more than every gear error in this machine put together.

1

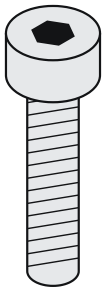
Fasteners



ISO 10642 M3 × 8

4x

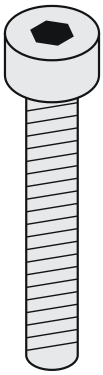
countersunk, motor



ISO 4762 M3 × 14

3x

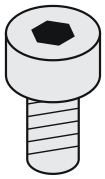
tower, into the disc



ISO 4762 M3 × 20

6x

wheel clamp



M3 × 6

2x

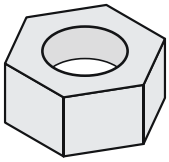
AS5600 board



ISO 4029 M3 × 4

2x

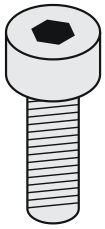
pinion, lower hub



1/4"-20 UNC hex

1x

7/16" AF, full height



M5, to suit

4x

wedge adapter

2

Metal parts



20d bright common

3x

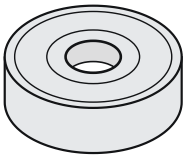
4.88 mm, buy a box



5 mm ground, h6

1x

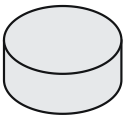
65 mm, cut to 55



625ZZ 5 × 16 × 5

2x

output shaft



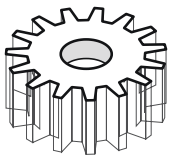
6 × 2.5 diametric

1x

must be diametric

3

Parts you will need to 3D print



pinion.stl

1x

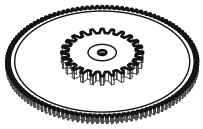
15T, D-bore variant



stage.stl

2x

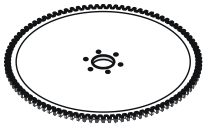
120T + 15T



final_stage_gear.stl

1x

120T + 24T m2



final_wheel_96T_m2

1x

196 mm OD



output_tower_*.stl

1+1

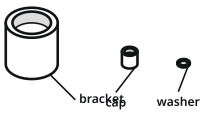
3 dowels, 11.9 long



output_hub_*.stl

1+1

lower, upper



cap, washer, bracket

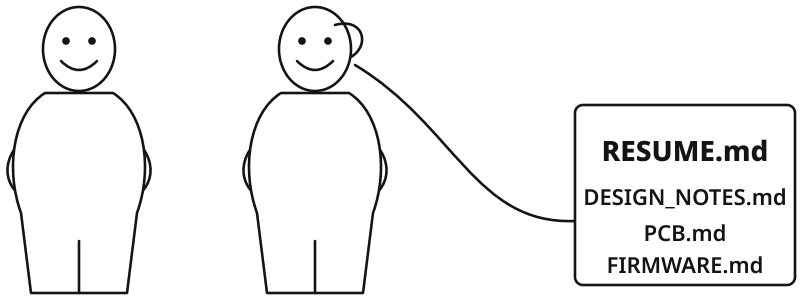
1+1+1



SPACERS_shaft_0-4

3x

use S1, S2, S3 only



What changed in Rev B

fixed in the CAD

Drawing Rev A for this guide turned up eight conflicts. All are fixed in `star_tracker_gears.scad`, rebuilt, and checked for interference in OpenSCAD. Two need your hands: **countersink the plate**, which is already cut, and **cut the shaft** to 55 mm. If you printed Rev A parts, reprint the tower halves, both hubs and the bracket.

KI 1 **Tower cone sat inside the plate.** Rev A's 45° cone was 32 to 44 mm across inside a 22.4 mm hole, so the flange could never seat. **Rev B:** the halves clamp the plate. The flange is on top, a 41 mm disc is underneath, dowels cross the gap, and 3 × M3×14 thread-form into the disc. Step 13.

KI 2 **Shaft ran through the camera nut. Rev B:** cut it to 55 mm, z −33 to 22. Step 14.

KI 3 **M3×10 motor screws bottomed out. Rev B:** M3×8 flat head, 3.7 mm of thread. Step 5.

KI 4 **Screw heads under the S1 wheel.** The wheel is 2.0 mm above the plate, over two motor screws and one M5 wedge hole. **Rev B:** countersink those holes by hand. Step 5, step 19.

KI 5 **Lower hub rubbed the tower. Rev B:** a printed 7.2 × 1.0 washer sits on the bearing's inner race, and the hub is 1 mm shorter, so the wheel stays at 21.5 mm. Step 15.

KI 6 **Bracket had no axial stop. Rev B:** it slides up until it stops on the clamp disc, so the 2.0 mm magnet gap comes from the parts. Step 18.

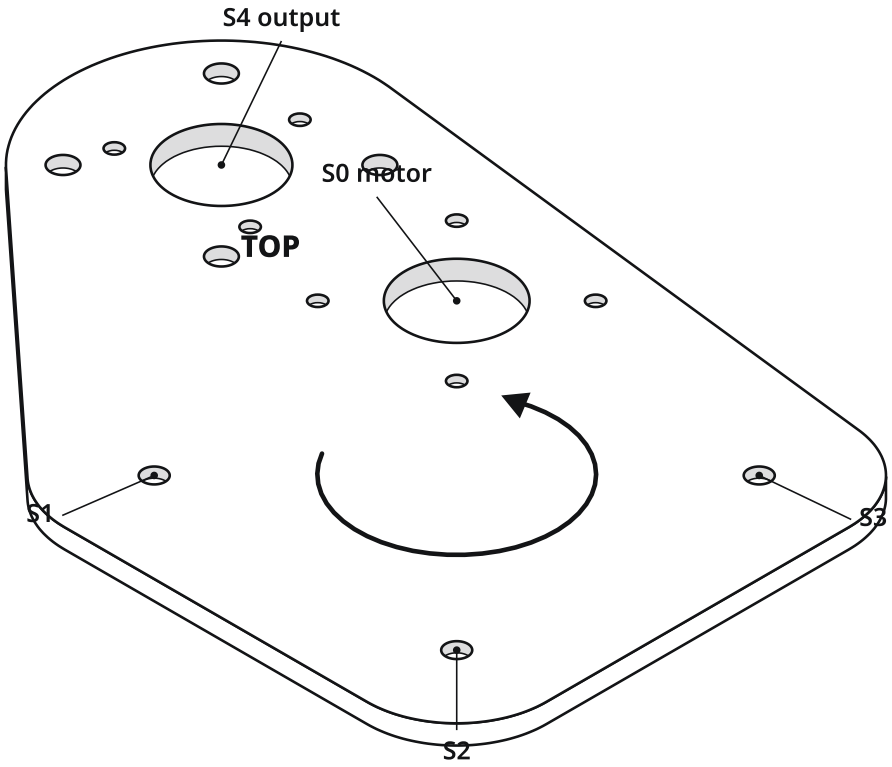
KI 7 **Grub-screw holes in gears that turn on fixed nails. Rev B:** removed from both stage gear files. On Rev A prints, leave them empty. Step 6.

KI 8 **Camera nut pocket 2.2 mm deep. Rev B:** the upper hub is 10 mm thick with a 5.9 mm pocket for a full-height nut. The wheel screws become M3×20. Step 17.

A Gearbox

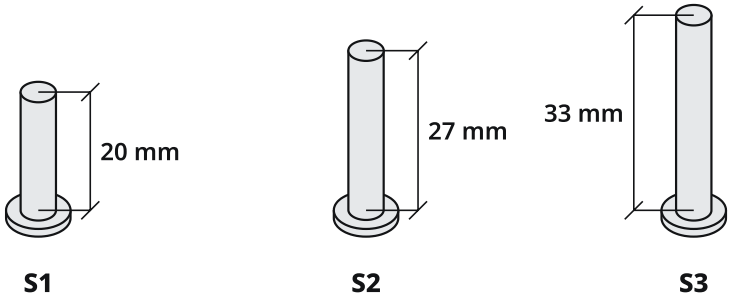
Plate, nails, motor, then each stage from the motor outwards, then the output tower. Heights all measure from the top face of the plate.

1 Check the plate, find its top

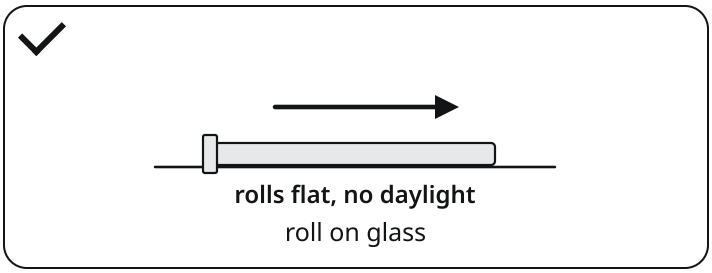


- Deburr every hole on both faces. Do not open up the three 4.9 mm nail holes; they are a light press.
- Seen from the top, S0 → S1 → S2 → S3 runs **counter-clockwise**. Mark **TOP** with a marker.
- Lay it on `baseplate_template.pdf`, printed at 100 %, and check every hole lines up.

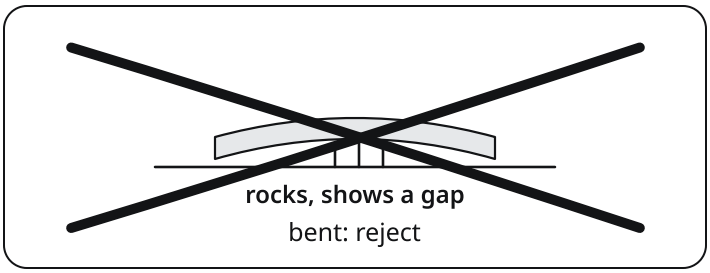
2 Pick and cut the nails



S1 **S2** **S3**



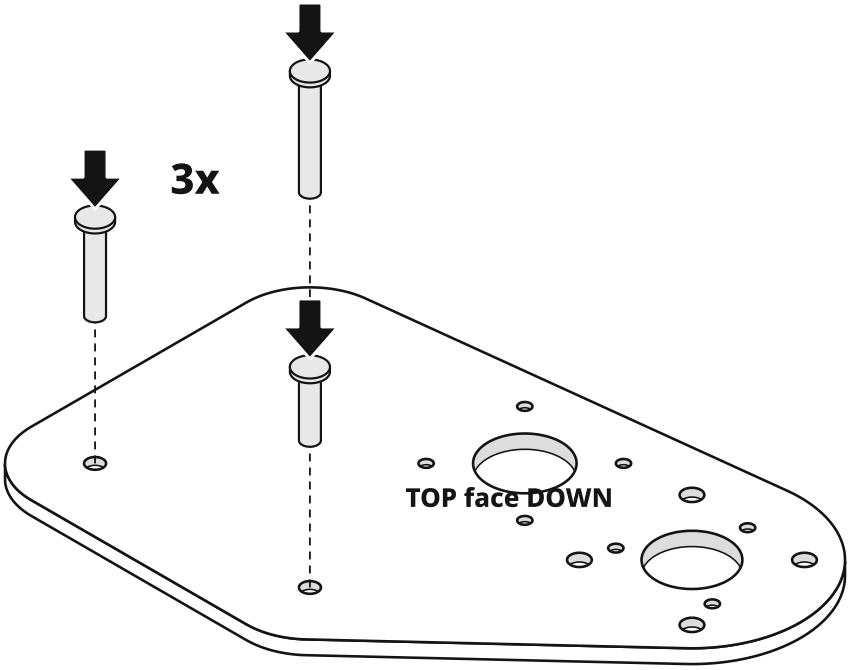
rolls flat, no daylight
roll on glass

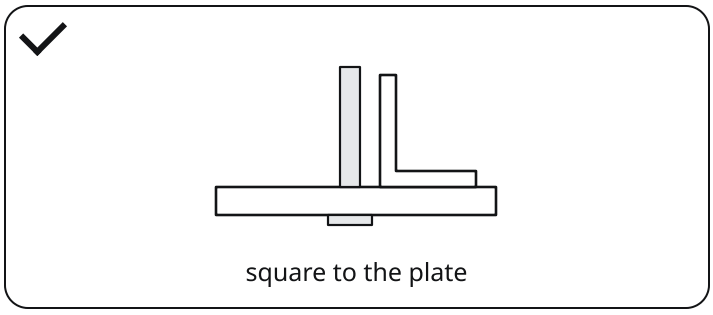


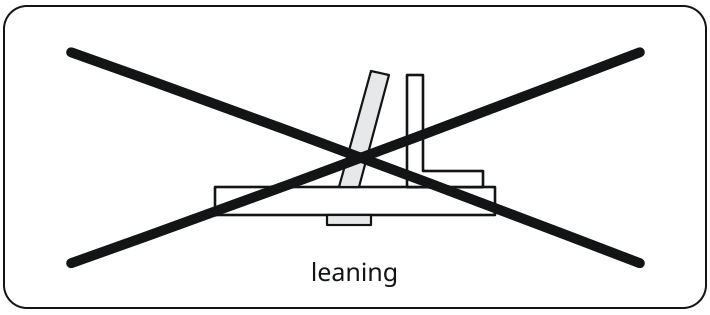
rocks, shows a gap
bent: reject

- 20d **bright common** only. Box nails and sinkers run a gauge thinner; galvanised coating is lumpy.
- Roll-test the box, keep the three straightest. The very straightest goes to **S3**: it carries the final pinion.
- Cut the point off, square and deburr the end. Lengths under the head: **S1 20**, **S2 27**, **S3 33 mm**.

3 Press the nails in from underneath



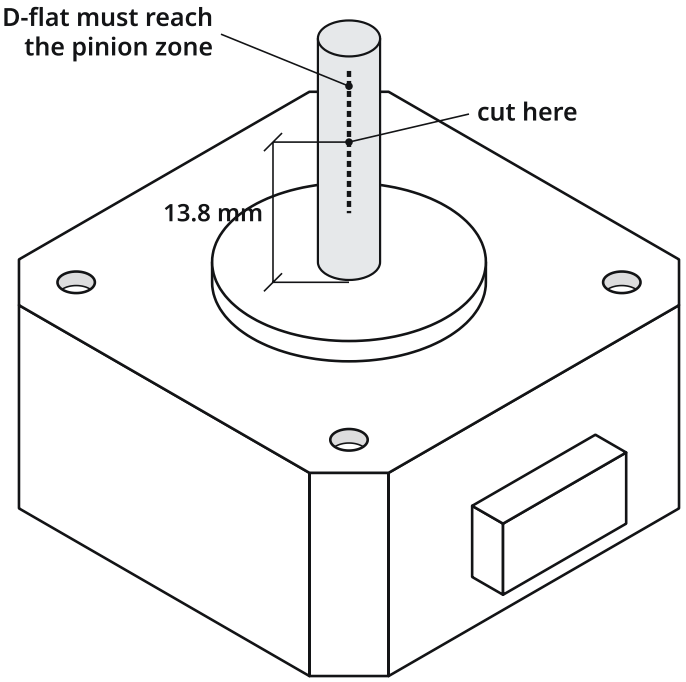




- Turn the plate TOP face down. Support each hole on an open socket so the tip has somewhere to go, then press or tap until the head seats.
- Height above the top face: S1 15.7, S2 22.7, S3 28.7 mm.

S1 must stay under 19 mm. The 96T output wheel passes over S0 and S1 with its underside at 21.5 mm.

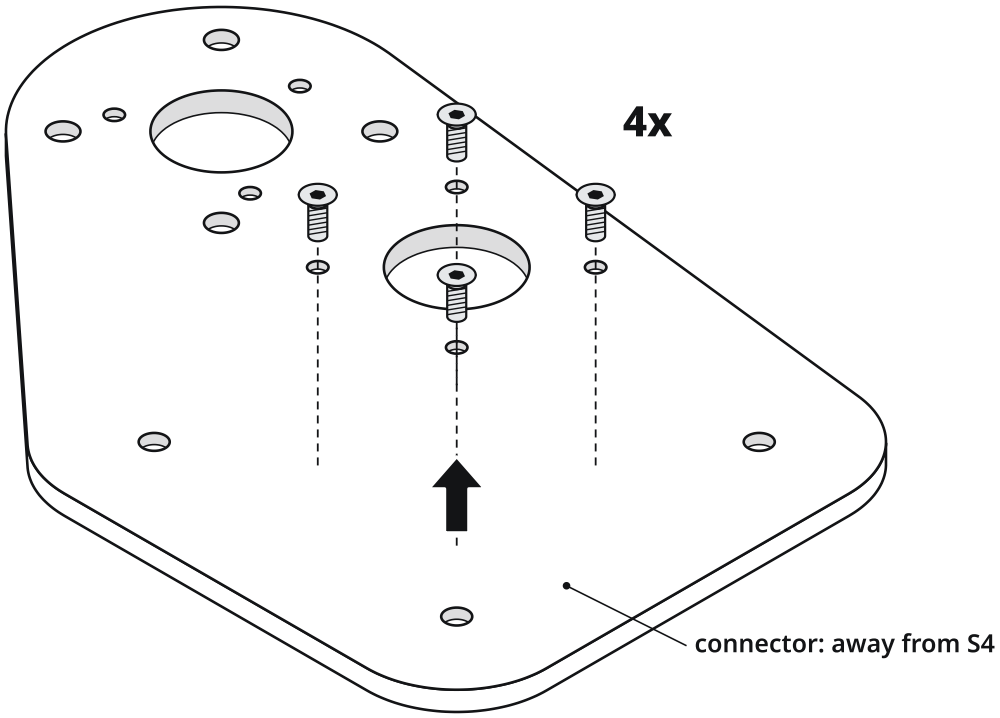
4 Trim the motor shaft

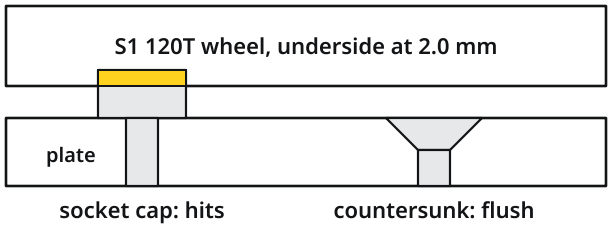


- The pinion sits 2 to 9 mm above the plate, so the shaft ends about 9.5 mm above the plate: **13.8 mm above the motor's mounting face**.
- Only cut if it is longer. Check the D-flat still covers the 6.3 to 13.3 mm band where the pinion's grub screw lands.
- Tape over the gap around the shaft first. Steel filings and a magnet rotor do not part.

5 Mount the motor

KI 3 KI 4



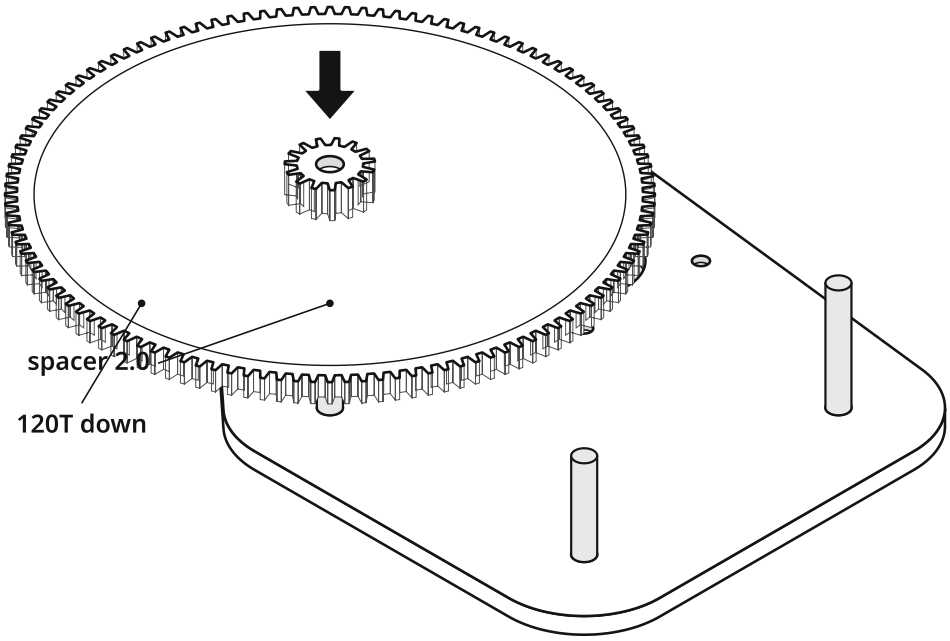


- Pilot boss into the 23 mm hole. Connector on the side **away from S4**; the encoder bracket hangs there.
- Countersink the four holes from the top (90°, 6 mm across), then `4 × M3×8 flat head`. That leaves 3.7 mm of thread in the motor.
- While the countersink is out: do the M5 wedge hole nearest S1 too (step 19).

⚠ Push a drill bit into a motor hole to gauge its depth before choosing a length. **Thread engaged = length − 4.3**, and it must be shorter than the hole. M3×10 needs 5.7 mm.

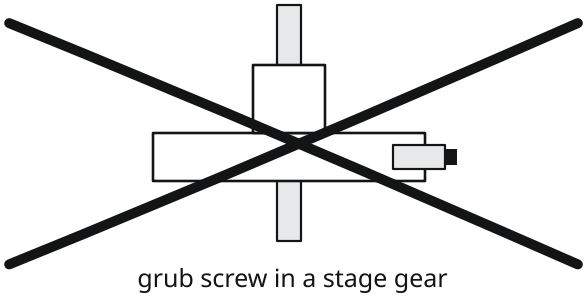
6 Shaft 1

KI 7



✓

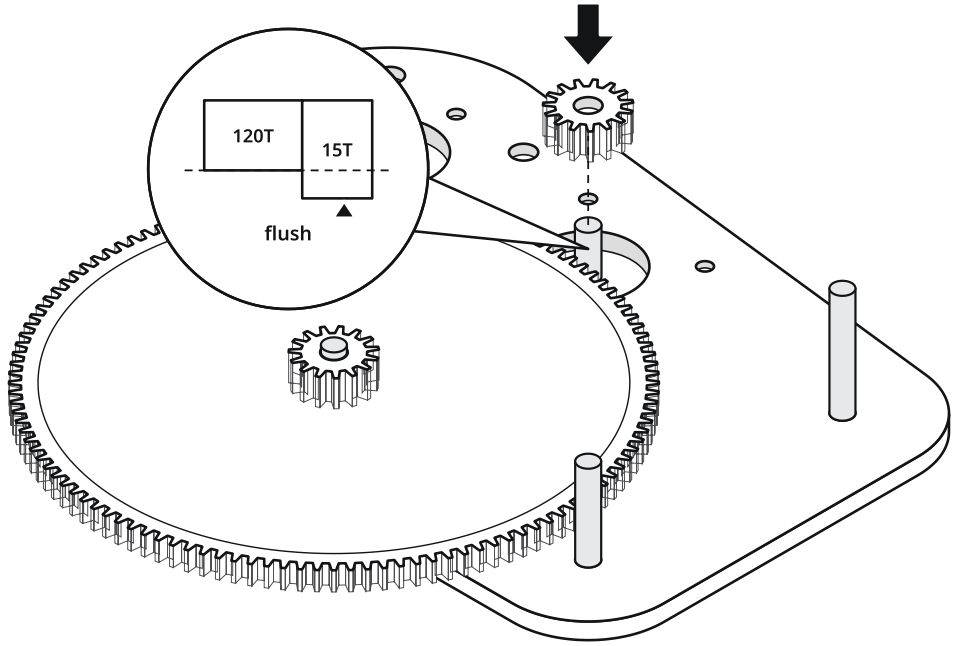
spins free on the nail



grub screw in a stage gear

- `spacer 2.0`, then `stage.stl` with the **120T wheel down**.
- It should spin a full turn from a flick and wobble only as much as the nail.

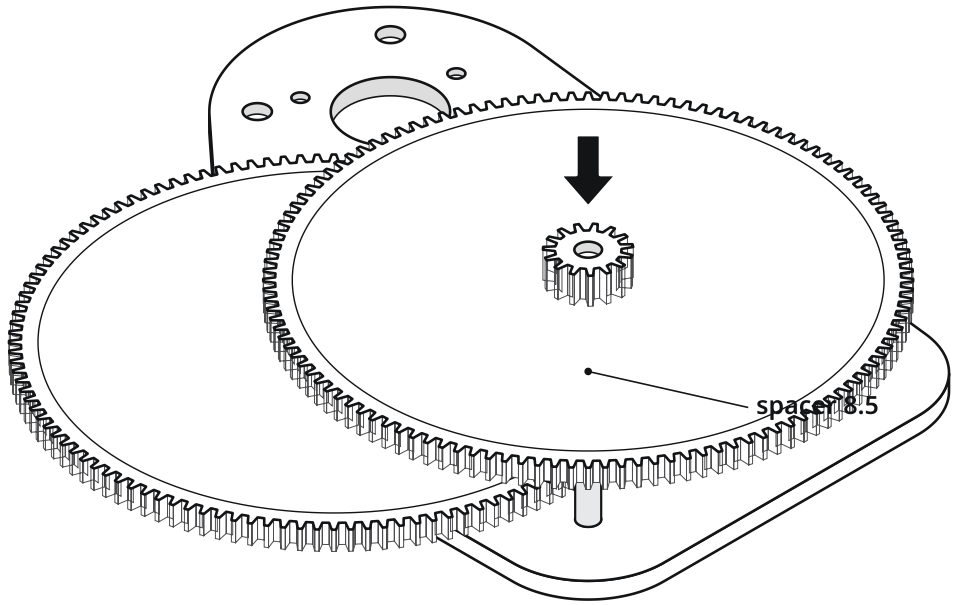
7 Motor pinion



- Use the bore variant that slid onto the motor shaft without force: `5.30`, `5.45` or `5.60`.
- Grub hole over the D-flat. Lower until the pinion's underside is **flush with the S1 wheel's underside**. It is 2 mm taller than the wheel on purpose and overhangs at the top.
- Run the `M3 grub` in and out once to form the thread, then tighten onto the flat.

Shaft 0 has no spacer. The 2 mm spacer would drop through the 23 mm motor hole, so the grub screw sets this height.

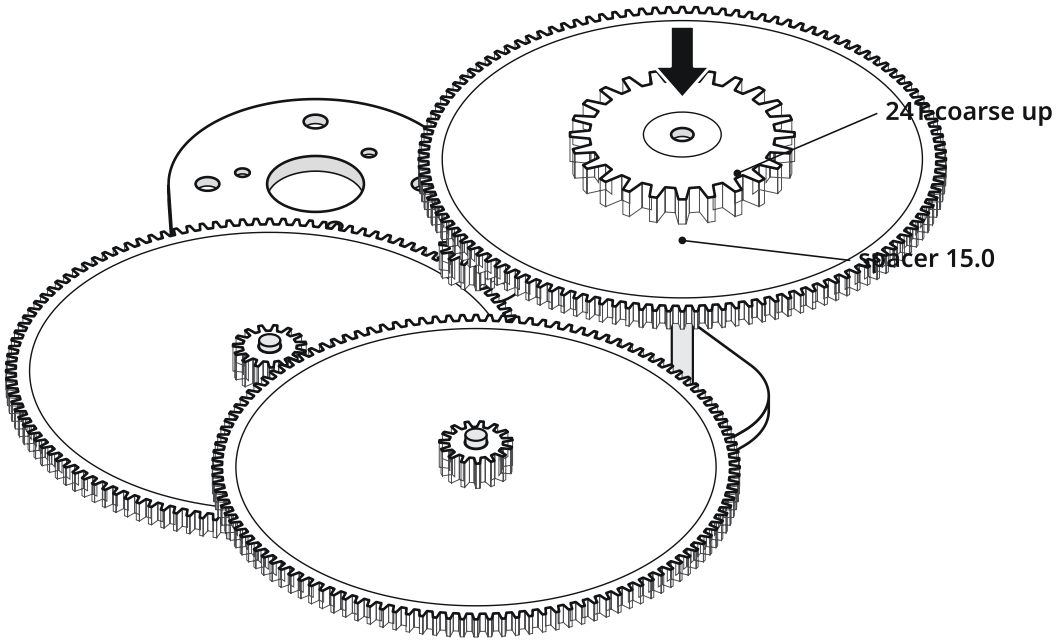
8 Shaft 2



- `spacer 8.5`, then `stage.stl`, 120T down. It meshes with S1's 15T pinion.

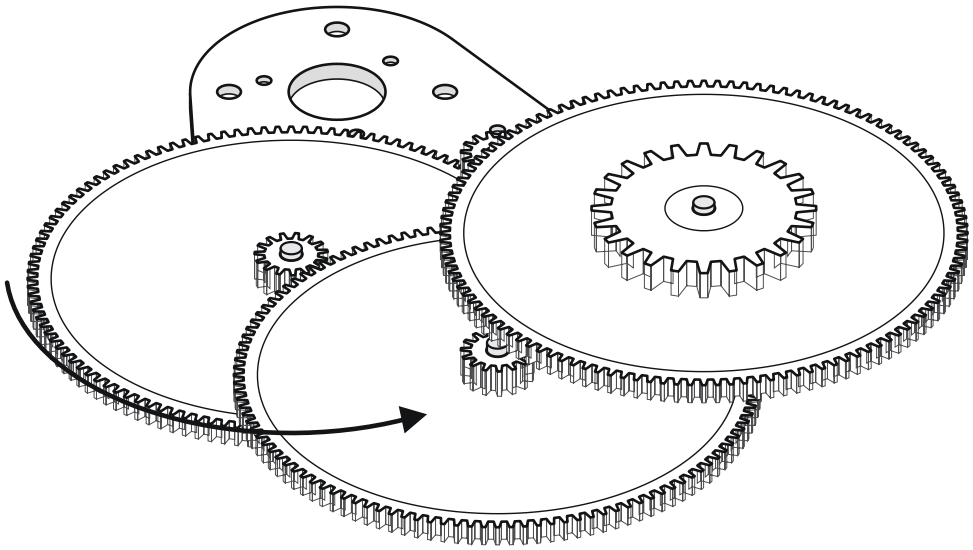
⚠ Never fit a stock shaft collar at S2 or S3. The printed spacer is 10 mm across; a spacer 13 mm across touches the neighbouring wheel's tips.

9 Shaft 3



- `spacer 15.0`, then `final_stage_gear.stl`: 120T down, **coarse 24T module-2 pinion up**.
- This pinion's radius sets short-exposure trailing. It goes on the straightest nail.

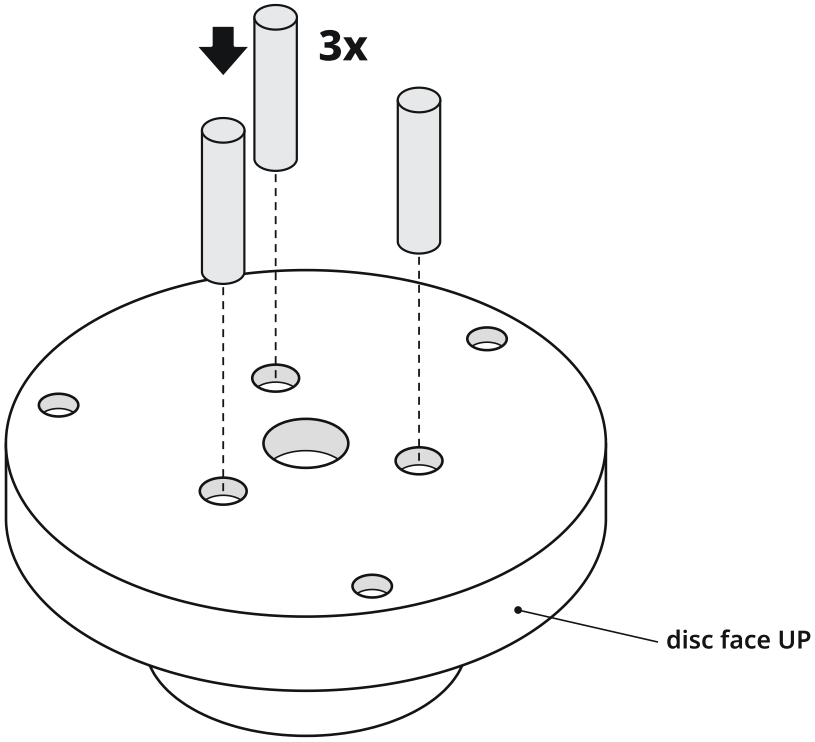
10 Turn test



- Turn the S1 wheel with a finger. Motor, S2 and S3 should follow smoothly with a small, even click of backlash at each mesh.
- A tight spot once per pinion turn is a Z seam on one tooth. Reprint that pinion with **Z Seam = Random**.
- Look along each mesh: every wheel face sits inside its pinion's face.

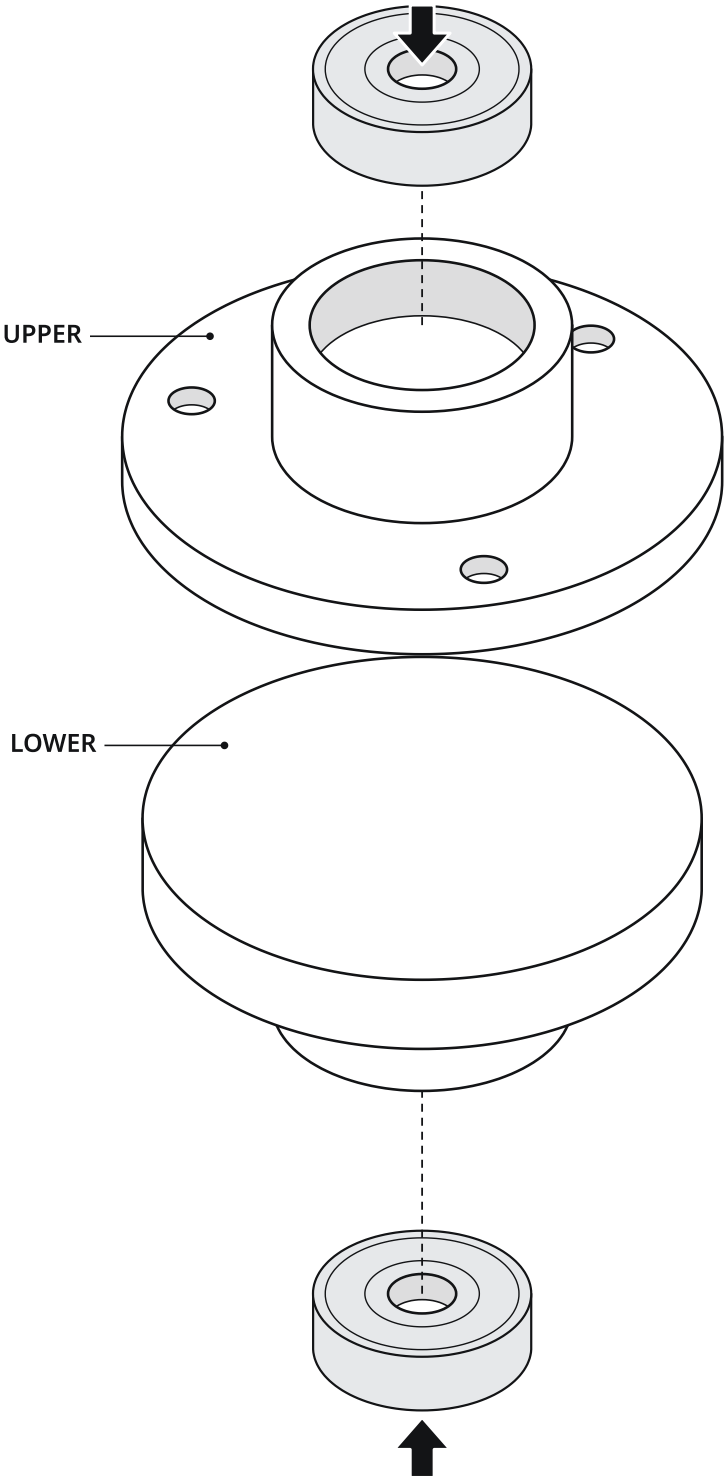
⚠ Turn the train from the S1 end only. Once the output wheel is on, never turn the output by hand: that is 2048 turns of the motor for one of the wheel, and the teeth strip first.

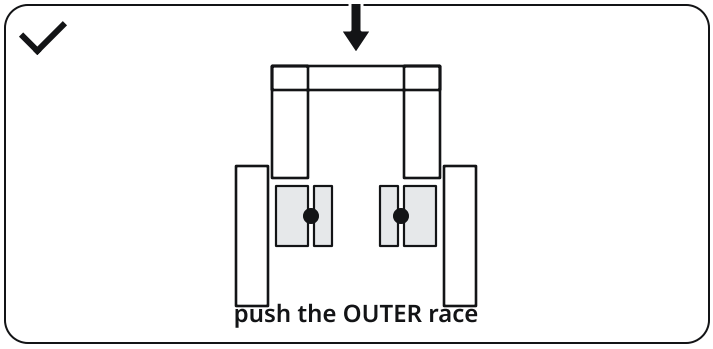
11 Dowels into the lower half

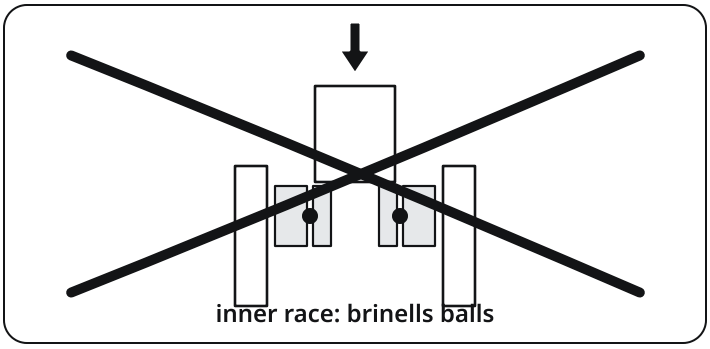


- Stand the lower half on its tube, **disc face up**. That is the face that meets the plate.
- Three dowels, **11.9 mm** long: 4 in this half, 4.3 across the plate, 4 into the upper half. A 3 mm rod cut to length works too.
- No glue anywhere in the tower. The halves never touch each other; the screws clamp them to the plate.

12 Press in the bearings



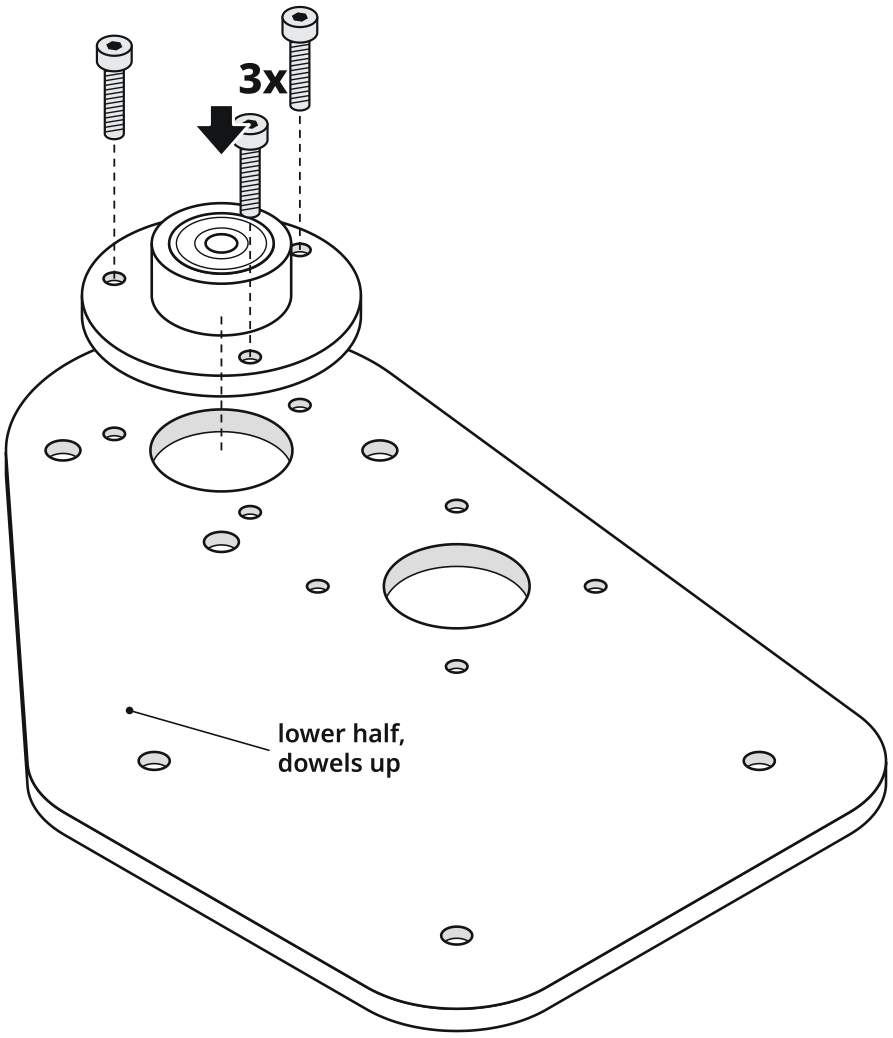


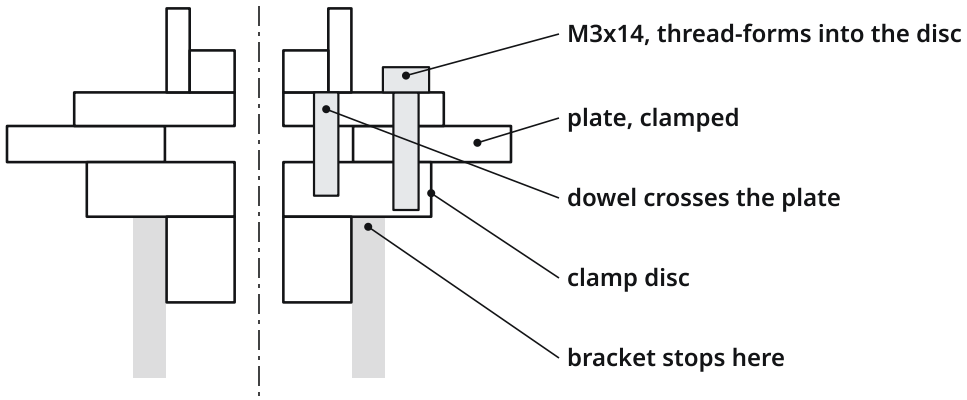


- Lower half: its pocket opens at the bottom of the tube. Upper half: at the top. Use a socket that fits the outer race, with firm thumb or vise pressure.
- Loose? Print again with a smaller `$BrgFit`. Needs a hammer? Larger.

13 Clamp the tower to the plate

KI 1

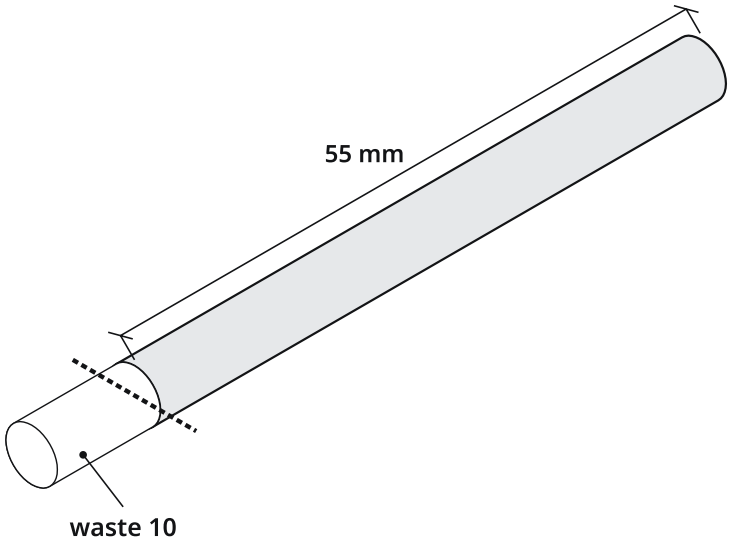


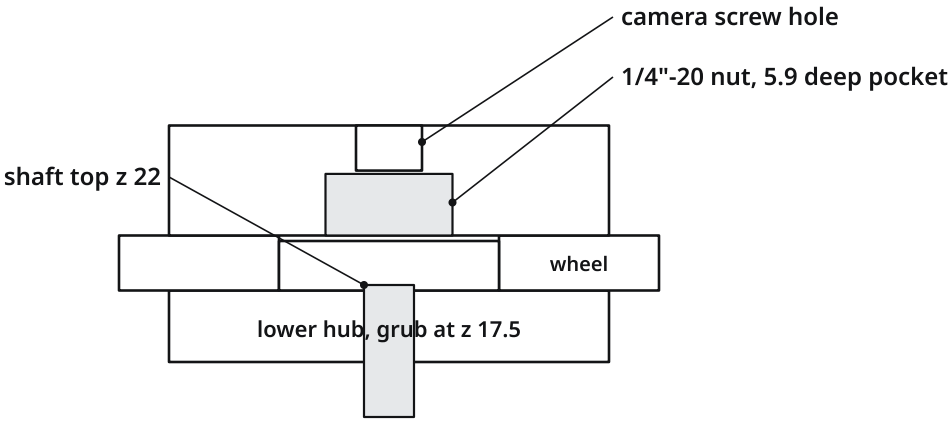


- Lower half up under the plate at S4, dowels up through the 22.4 mm hole. Upper half down onto the dowels.
- `3 × M3×14` from above: through the flange and the plate, then they thread-form 5.7 mm into the disc. No nuts.
- Snug the three in turn. Flange and disc sit flat on both faces of the plate, and the tube stands square to it.

14 Cut the output shaft

KI 2

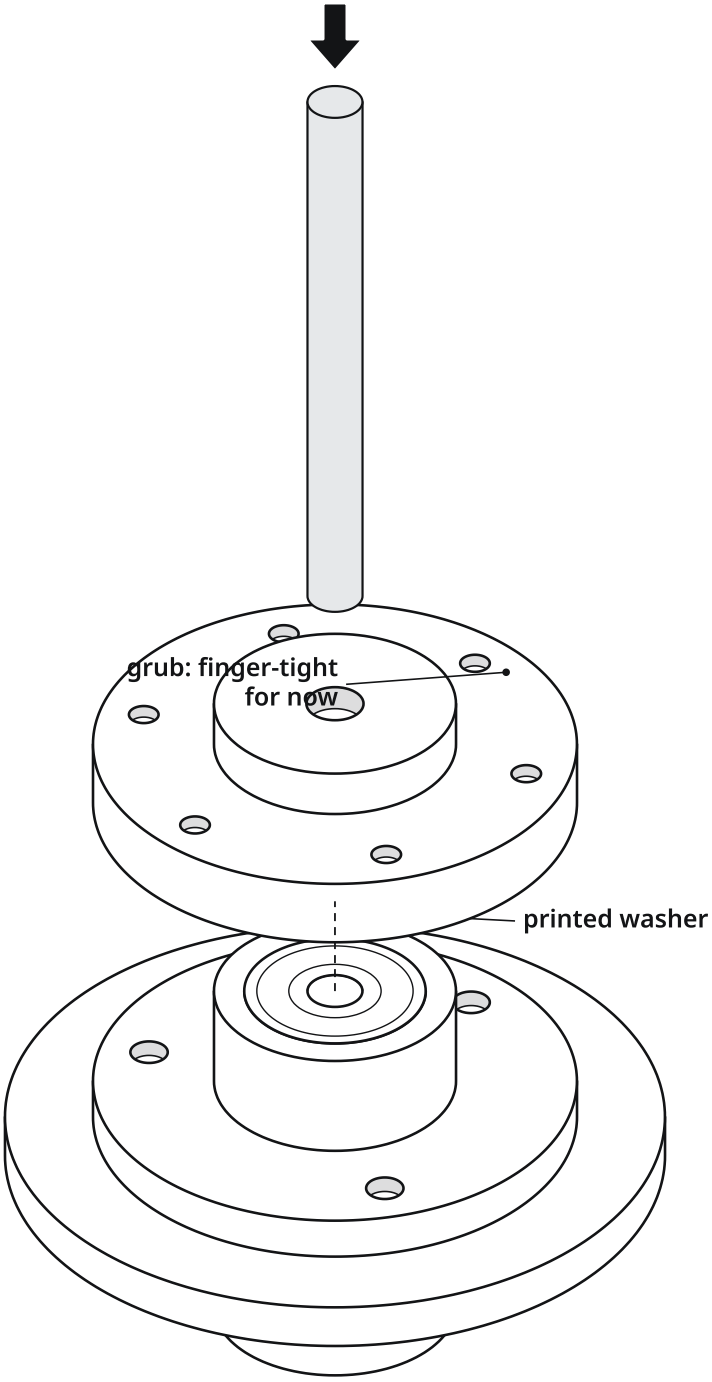


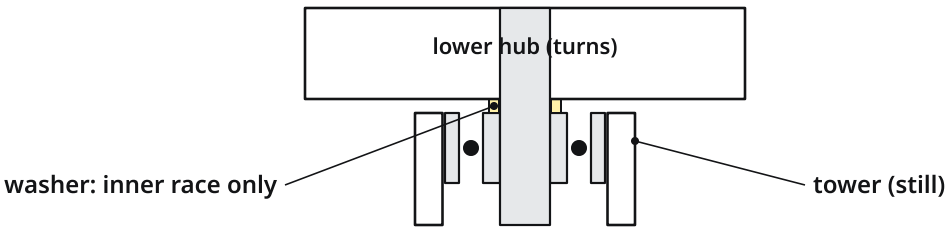


- Ground 5 mm stock, not a nail. This one runs in bearings and carries the magnet.
- **55 mm**: bottom end in the magnet cap at z −33, top end at z 22, above the hub grub (z 17.5) and below the camera nut (z 26.5). Chamfer both ends lightly.

15 Washer, lower hub and shaft

KI 5

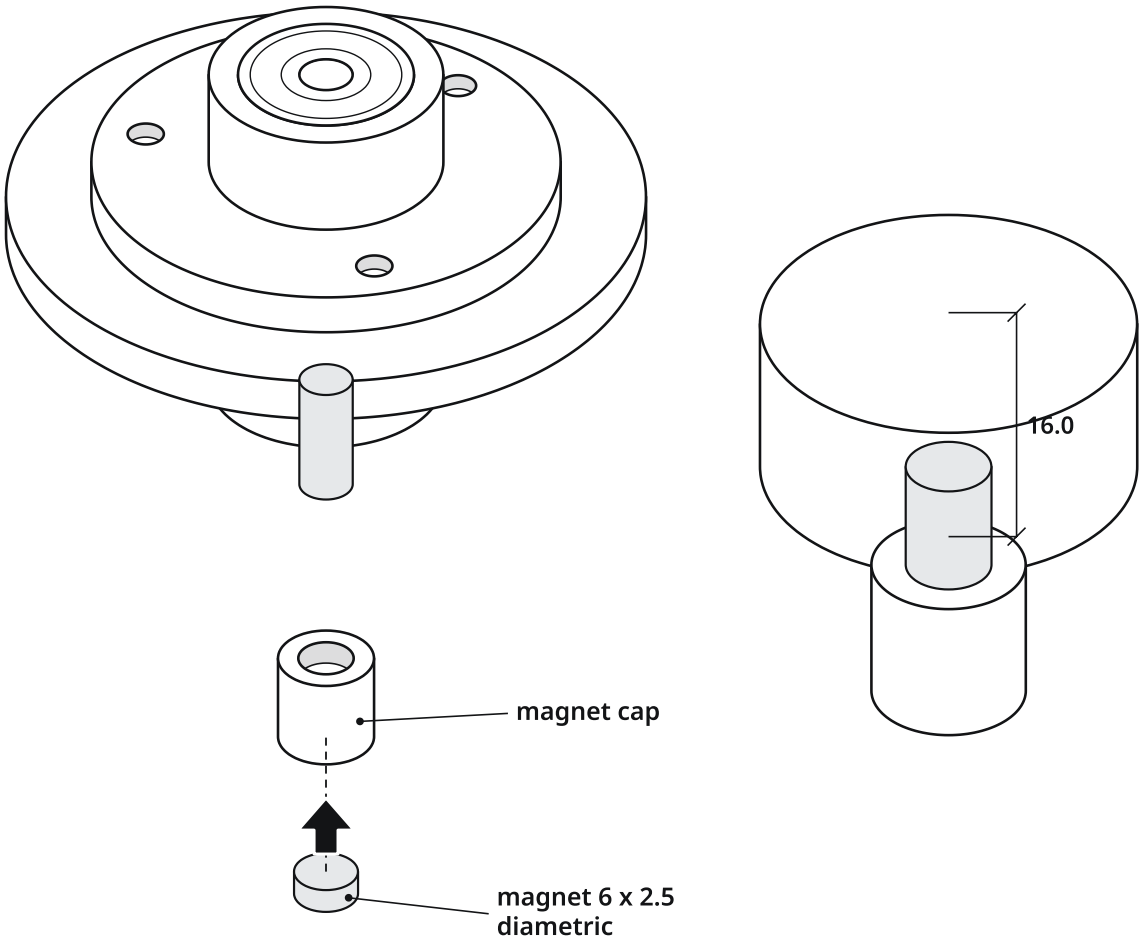




- `hub_washer.stl` (7.2 × 1.0) on the upper bearing's inner race. Hold it against the bearing first: it must not touch the shield.
- Lower hub on, register **up**. Shaft down through the hub and both bearings.
- Hub grub (`M3`, radial) finger-tight: the height is set in the next step.

Washer, inner race, hub and shaft all turn together, so the washer never wears. It only keeps the hub off the stationary tower rim.

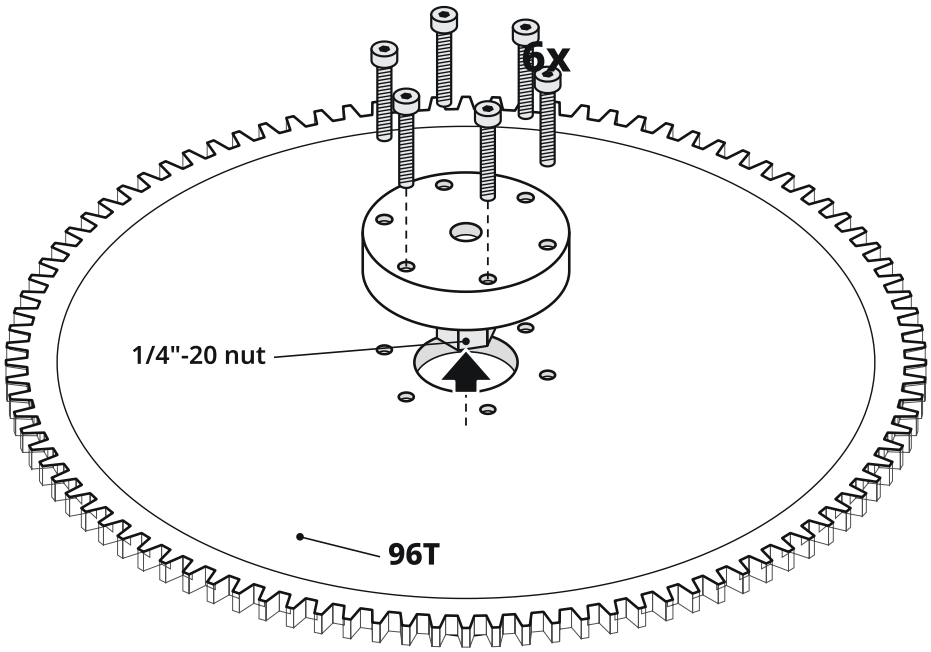
16 Magnet and cap



- Check the magnet is **diametric**: spin it flat beside a compass and the needle turns a full circle with it.
- **Either face can go toward the sensor.** A diametric magnet's north and south are on its rim, not its faces, so flipping it only shifts the reading by a fixed angle, which step 59 takes out.
- Magnet flush into the pocket in the cap's end face. Press the cap up onto the shaft until the shaft bottoms. Do not glue it yet; step 59 sets its rotation.
- Slide the shaft until the cap's face is **16.0 mm below the tower's bottom face** (caliper depth rod), then tighten the hub grub.

17 Output wheel and camera platform

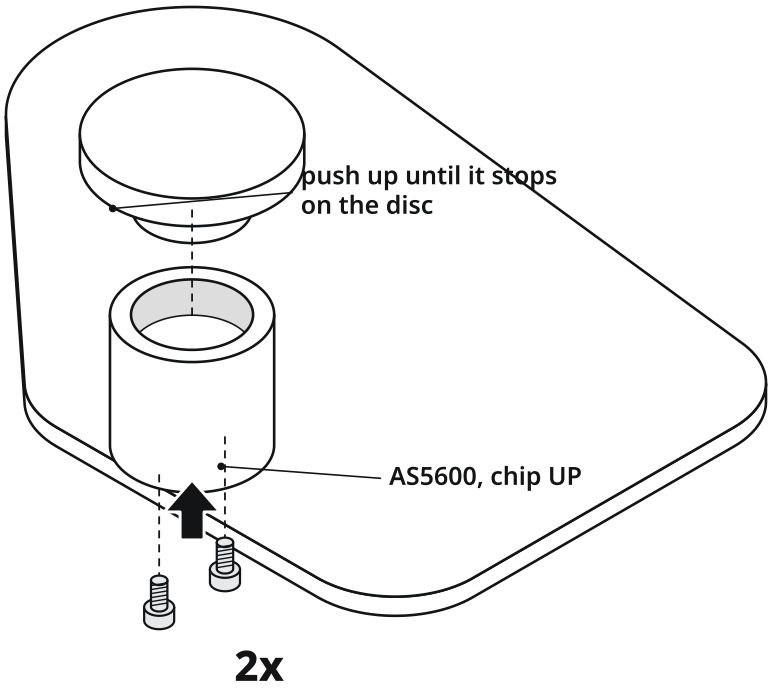
KI 8



- 96T wheel over the 22 mm register. It is located by the register, not by its own bore.
- Full-height ¼"-20 nut into the hex pocket under the upper hub, then the hub on top.
- `6 × M3×20` through hub and wheel into the lower hub (thread-forming, 5 mm engagement). Tighten in a star pattern.
- The wheel face must sit inside the 24T pinion's face: 21.5 to 26.5 mm inside 20 to 27 mm.

18 AS5600 and its bracket

KI 6



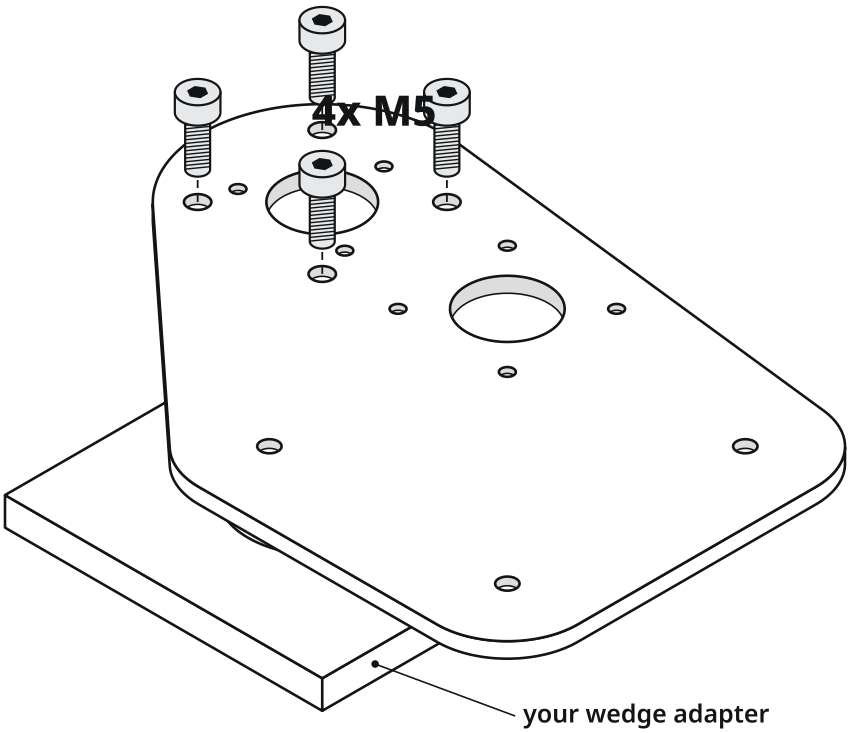
2x

- Solder a 5-wire lead to the breakout first: GND, VCC, SDA, SCL, DIR.
- Board to the bracket underside, chip **up** into the window: `2 × M3×6`, thread-forming.
- Push the bracket up the tower spigot until it **stops on the clamp disc**. That puts the **chip's top surface** 2.0 mm from the magnet's face (the datasheet's air gap is measured to the package, not the board). Step 58 confirms it with the AGC reading; then a dab of hot glue holds it there.

ⓘ At 3.3 V the AS5600 wants its VDD5V and VDD3V3 pins tied together. Many breakouts have a solder bridge for this; check yours.

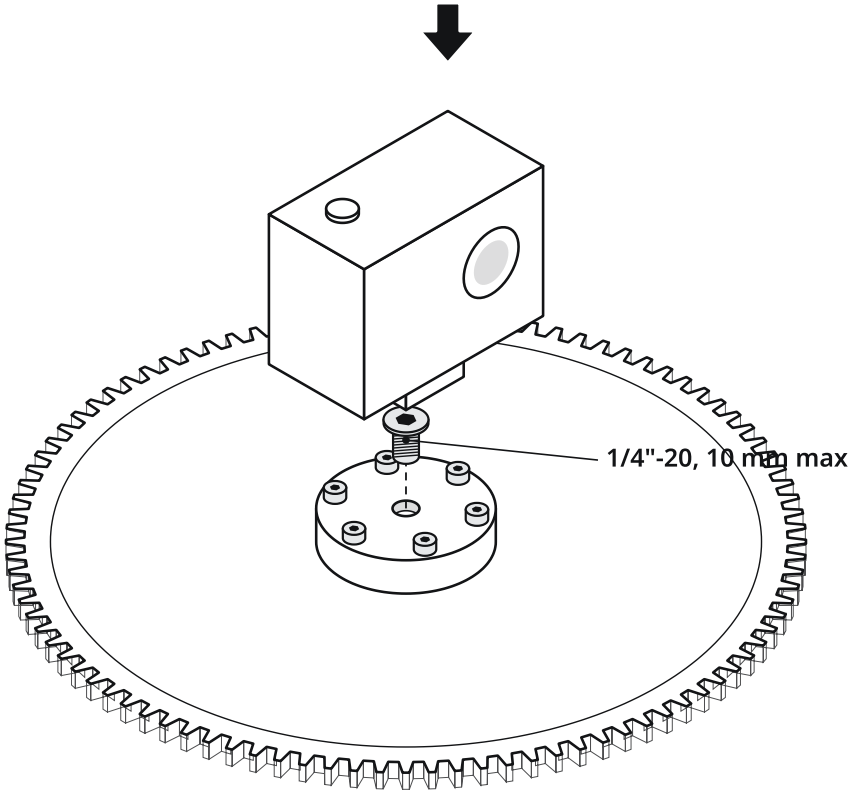
19 Wedge adapter

KI 4



- `4 × M5` on a 50 mm circle, centred on the **output** axis, not the plate's middle. That is where the camera load is.
- The adapter needs a **42 × 7 mm relief** for the tower's clamp disc, and a hole at least 32 mm across with about 45 mm of depth for the bracket and its lead.
- The M5 hole nearest S1 is under the S1 wheel. Anything above the plate there must stay under 1.5 mm: countersunk, or leave that bolt out.

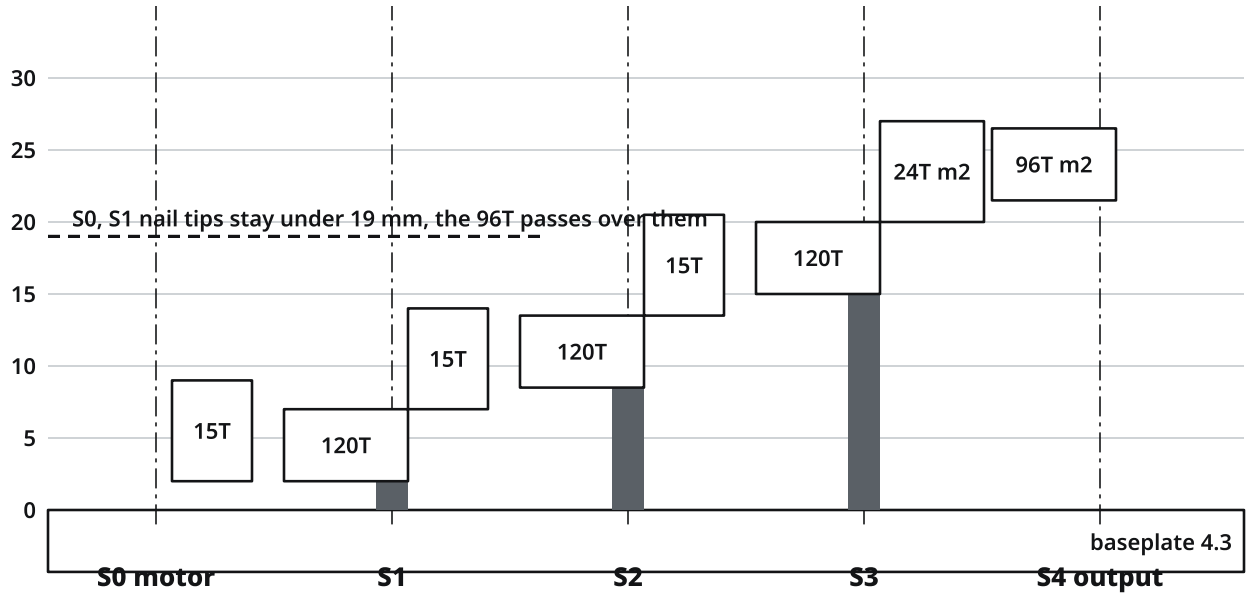
20 Camera



- GoPro frame and ¼"-20 adapter. At most **10 mm** of thread below the platform top.
- Bias the balance so the load always pulls the same way. A perfectly balanced camera lets the teeth rattle across the backlash.

From here on, the platform moves only under power (jog, slew, track). Turning it by hand back-drives 2048 : 1.

Every mesh gets its own height band

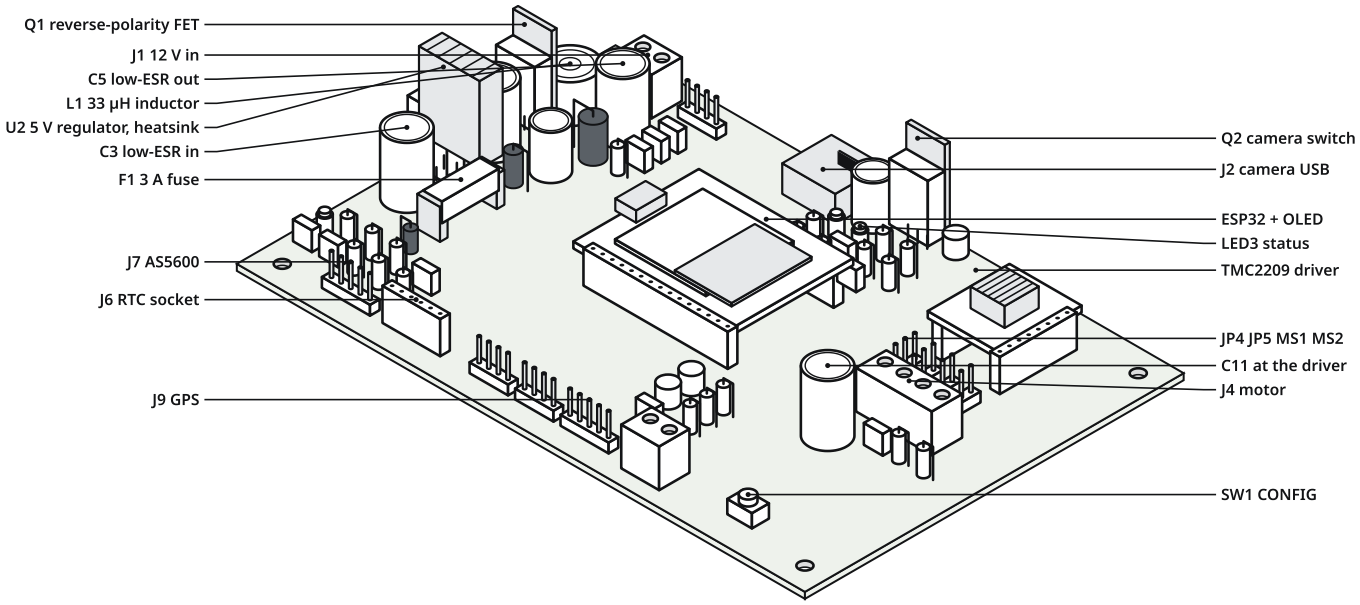


A pinion always collides with any wheel next to it, meshing or not, so four meshes need four bands. Spacers (dark) set each wheel's height. Every pinion is 7 mm tall against a 5 mm wheel so the wheel stays inside it.

B Controller board

All through-hole, every module in a socket. Written for a first board: one kind of part at a time, simplest first, each with a map of where it goes. Then test everything (section C) before any module goes in.

The finished board

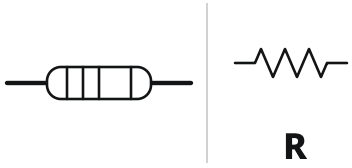


Drawn from the KiCad layout, so every part is where the board puts it. Modules shown seated in their sockets. This is board Rev C, laid out for the 30-pin ideaspark ESP32 + OLED; the drawing follows the layout on every build.

4

Electronic parts for the board

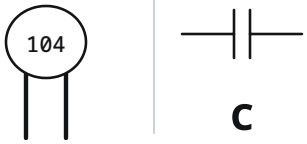
Each card shows the part, its schematic symbol, and the letter it carries on the silkscreen: **Q4** is the fourth transistor, **R12** the twelfth resistor.



R1–R19

17x

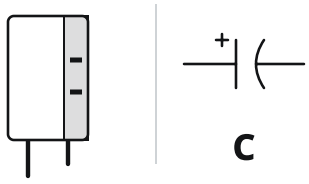
Resistors, ¼ W. 10k ×7, 1k ×4, 100k ×2, 4.7k, 330R, and the 1 % pair 100k + 18k. R10, R11 stay empty.



C2 C4 C6 C8 C10 C12 C13 C15–C17

10x

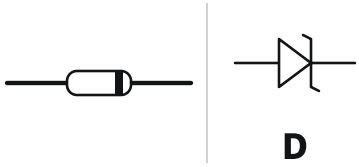
Ceramic capacitors, 100 nF, marked 104. No polarity.



C1 C3 C5 C7 C9 C11

6x

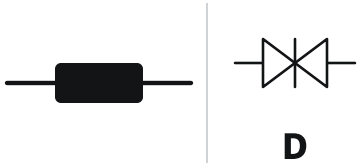
Electrolytic capacitors. 470 µF ×2 (C3 low-ESR), 220 µF low-ESR, 100 µF ×3 (C11 low-ESR). Polarised.



D1

1x

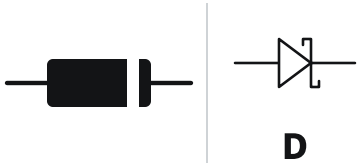
Zener diode, 1N4744A 15 V. Clamps Q1's gate.



D2

1x

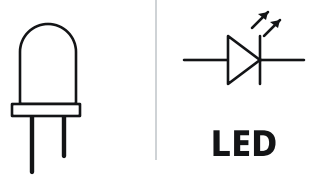
TVS diode, P6KE20CA. Bidirectional, fits either way.



D3

1x

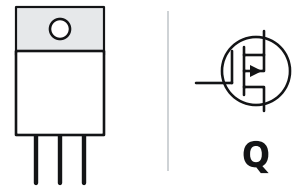
Schottky diode, 1N5822 3 A. The buck's catch diode.



LED1 LED2 LED3

3x

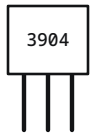
LEDs, 3 mm. Green ×2 (5 V, camera), red ×1 (status).

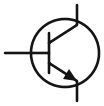


Q1 Q2

2x

P-channel MOSFET, IRF4905, TO-220. Reverse polarity and the camera switch.



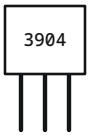


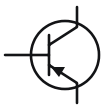
Q

2x

NPN transistor, 2N3904, TO-92. The arrow points out.

Q3 Q5





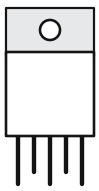
Q

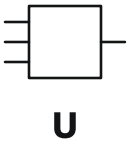
1x

PNP transistor, 2N3906, TO-92. The arrow points in. Looks just like the 2N3904; read the print.

Q4

U2



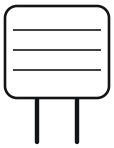


U

1x

Regulator, LM2596T-5.0, TO-220-5, with a clip-on heatsink.

L1

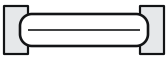


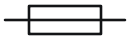
L

1x

Inductor, 33 µH, 3 A or more, 12 mm radial.

F1



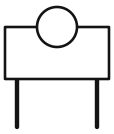


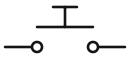
F

1x

Fuse, 3 A 5×20 mm, in two PCB clips.

SW1



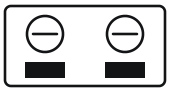


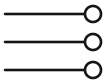
SW

1x

Tactile switch, 6 mm. CONFIG and wake.

J1 J3 J4



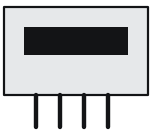


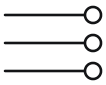
J

3x

Screw terminals, 5 mm pitch. 2-way ×2, 4-way ×1.

J2



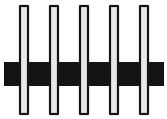


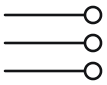
J

1x

USB-A receptacle, through-hole, horizontal. Camera power.

J5 J7–J11 JP3–JP5



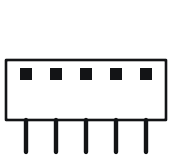


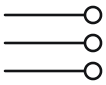
J

1x

Male header strip, 1×40, 2.54 mm. Cut to length.

J20 J21 J30 J31 J6





J

5x

Female sockets, 2.54 mm: 1×15 ×2 for the 30-pin ESP32, 1×8 ×2, 1×6.





JP4 JP5

JP

2x

Jumper shunts, 2.54 mm. MS1 and MS2 to GND.

Modules go in the sockets last, after section C: ESP32 + OLED, TMC2209, DS3231 with a CR2032. Plus four M3 standoffs for the corners.

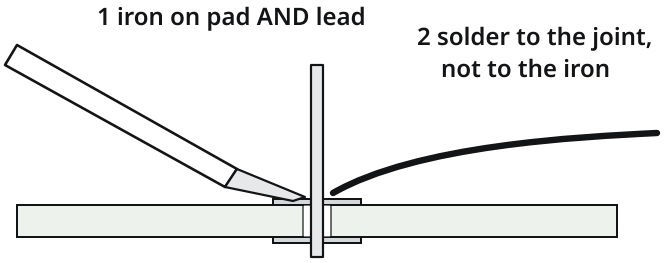
21 Which board?

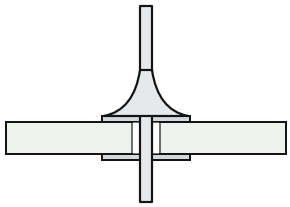
Build board Rev C. It is laid out for the module in hand: the 30-pin ideaspark ESP32 + 0.96" OLED, rows 27.94 mm apart. Rev B was drawn for a 38-pin DevKitC and does not fit it.

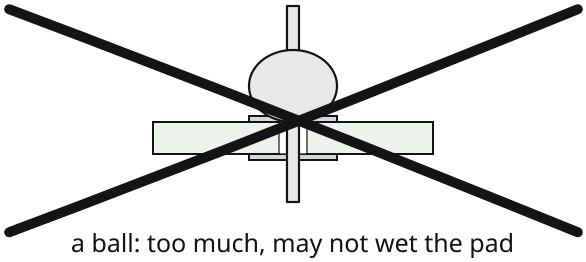
The pin map was settled by the pin check (step 55): the OLED shares the sensor I^2C bus on GPIO 21/22, so WAKE stays on GPIO 4, and the module has no PSRAM. Using a different ESP32 board? Run the pin check on it first.

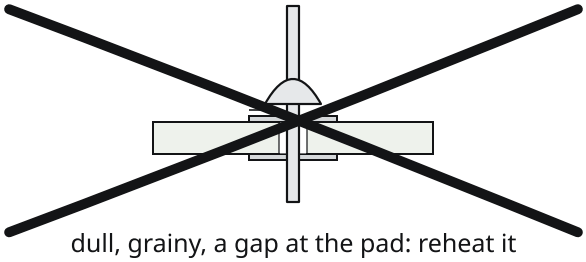
This section is written for a first board. Parts go in **one kind at a time**, simplest first. Each kind gets two pages: **how** the part goes in, then a **map** of the board with every place it goes, its value, and a box to tick.

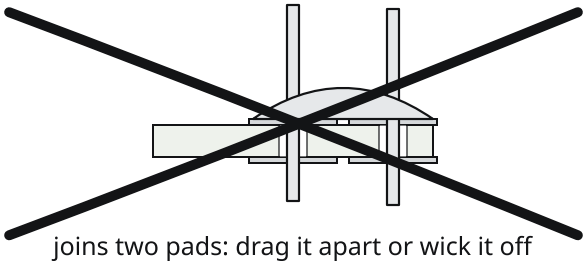
22 Soldering, the basics




a shiny cone that hugs the lead


a ball: too much, may not wet the pad

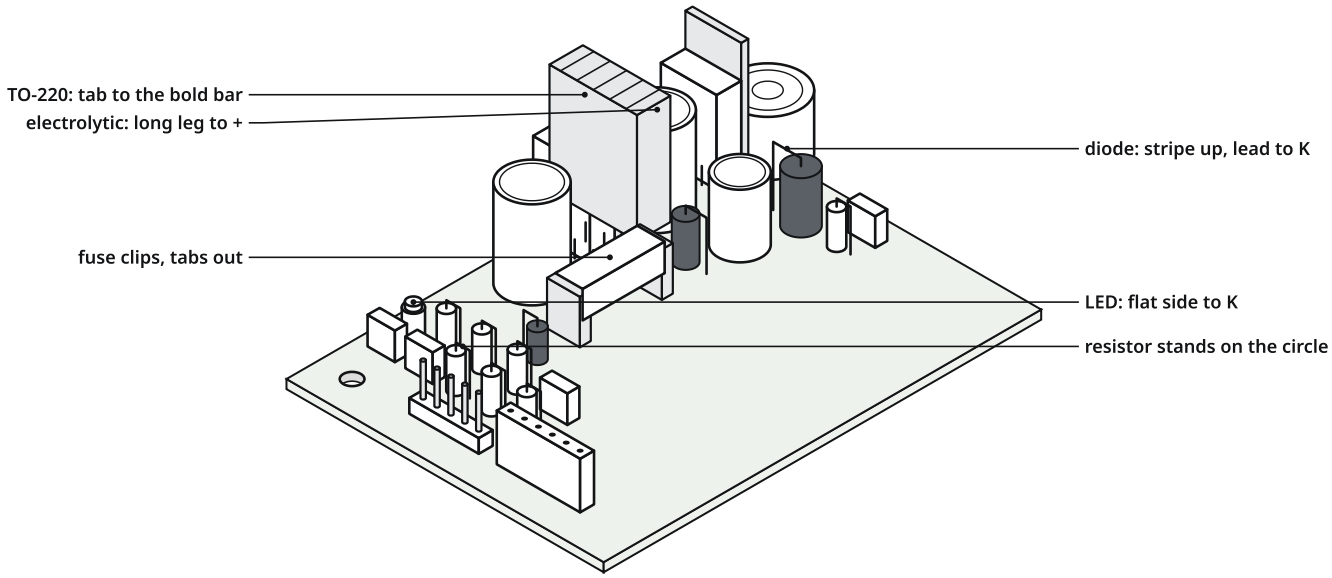

dull, grainy, a gap at the pad: reheat it

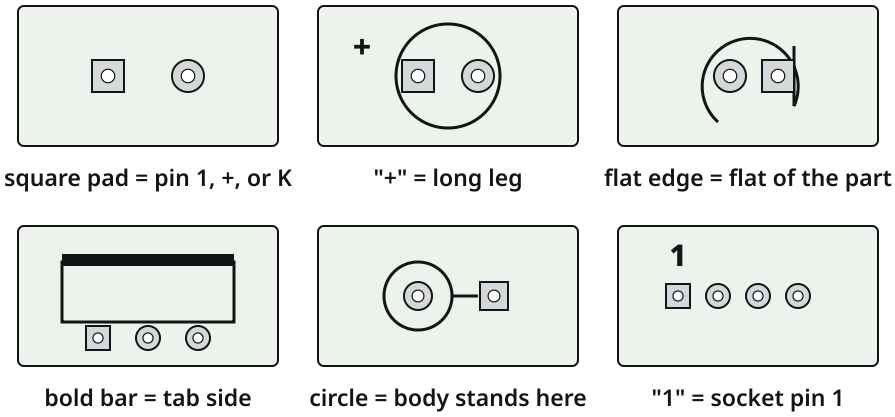

joins two pads: drag it apart or wick it off

- Iron at **330 to 350 °C**. Wipe the tip on damp sponge or brass wool, then melt a little solder onto it (**tinning**): a shiny tip moves heat.
- Use **0.8 mm rosin-core solder**. 63/37 leaded flows most easily; wash your hands after. Work with a fan or open window.
- Parts go in from the **printed side**. Solder from the back. Trim each lead just above the cone with flush cutters, after it is soldered.

Never soldered before? Practise ten joints on a scrap of perfboard and some resistor offcuts first. It takes about fifteen minutes and it is the best quarter hour of the build.

23 Read the silkscreen

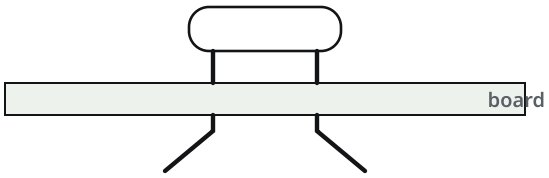




- The **square pad** is the one to find: pin 1 of a socket, **+** of an electrolytic, **K** (cathode) of a diode or LED.
- Hold the board as the maps show it: **J1 (12 V IN) at the top left**, the row of headers along the bottom.
- Nothing on this board is fitted by guessing. If a part does not match its outline, stop and check.

24 The order of work

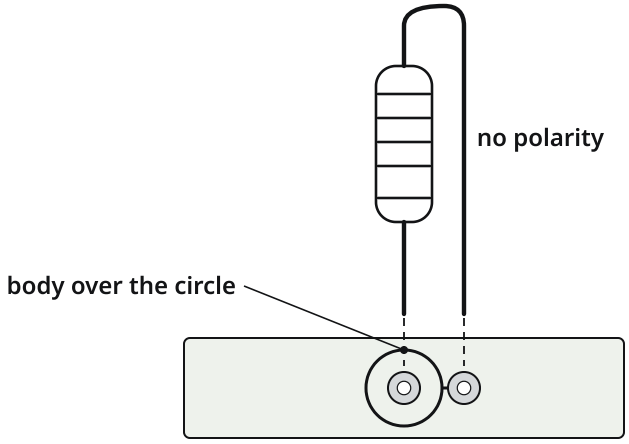
	Kind	How many	New idea
1	Resistors	17	bending a lead, reading a value
2	Ceramic capacitors	10	reading a code
3	Electrolytic capacitors	6	**polarity**: one way round only
4	Diodes and LEDs	6	polarity marked by a stripe or a flat
5	Male headers and jumpers	9	cutting a strip, keeping it upright
6	Female sockets	5	using the module as a jig
7	Terminals, USB, button, fuse clips	6	big pins, big joints
8	Small transistors	3	reading the part number
9	Power parts and heatsinks	4	live metal tabs, heat



splay the legs under the board so the part cannot fall out

- Fit all of one kind, splay the legs so they stay put, turn the board over, solder them all, trim them all. Then tick them off on the map.
- From kind 3 on, the board no longer lies flat on its face. A vise or a "helping hands" stand, or a strip of masking tape over each part, keeps it steady.
- When kind 9 is done, the quick check (step 39) lights the board for the first time. Then section C tests it properly, **before** any module goes in.

25 Resistors: how

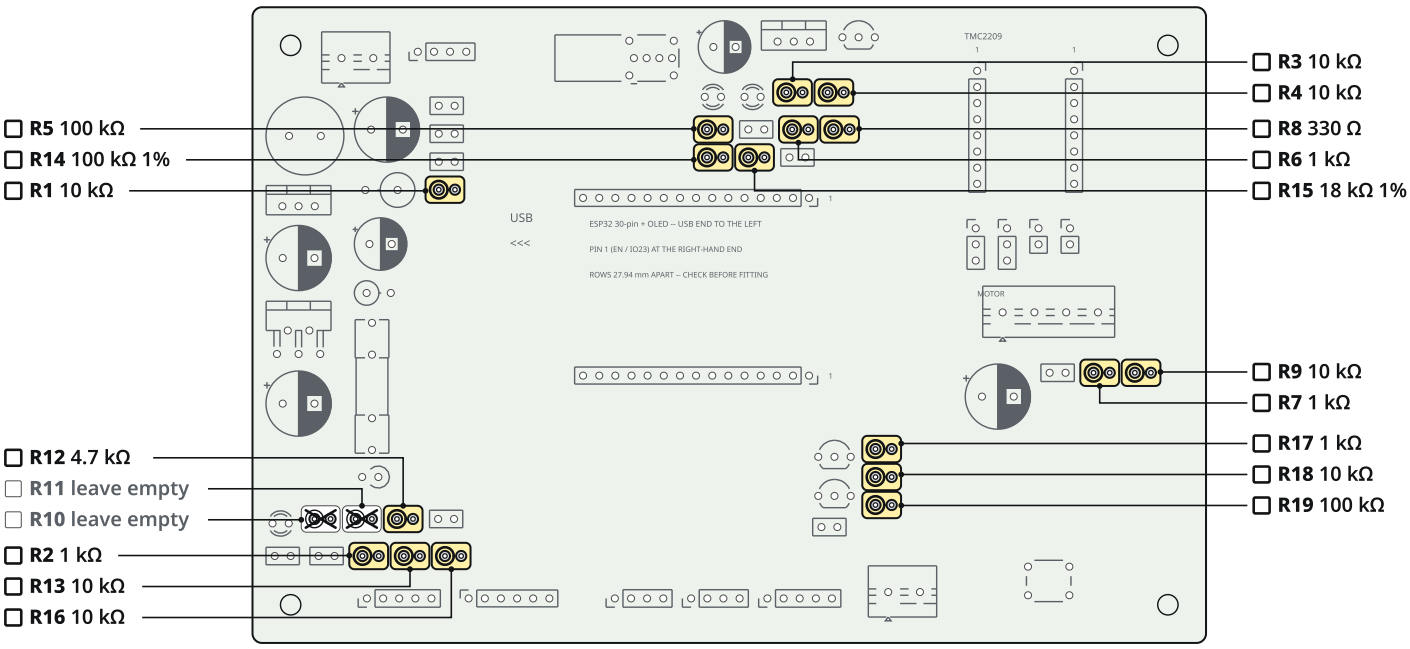


- These resistors **stand up**. Bend one lead tightly over the top of the body and down beside it. The body stands over the **circle**; the bent lead goes into the other hole.
- No polarity: either end can go up.
- **Measure every resistor** with the meter on Ω before it goes in. It is quicker and surer than reading colours, and the colours below confirm it.

	ordinary (4 bands, 5 %)	precision (5 bands, 1 %)
10 kΩ	 brown · black · orange · gold	 brown · black · black · red · brown
1 kΩ	 brown · black · red · gold	 brown · black · black · brown · brown
100 kΩ	 brown · black · yellow · gold	 brown · black · black · orange · brown
4.7 kΩ	 yellow · violet · red · gold	 yellow · violet · black · brown · brown
330 Ω	 orange · orange · brown · gold	 orange · orange · black · black · brown
100 kΩ 1%	—	 brown · black · black · orange · brown
18 kΩ 1%	—	 brown · grey · black · red · brown

⚠ **R14 and R15 are the 1 % pair** that scales the battery voltage for the ESP32. Keep them apart from the ordinary 100 kΩ resistors: a swap makes every battery reading wrong.

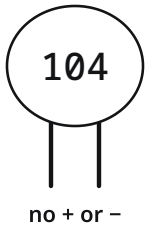
26 Resistors: where



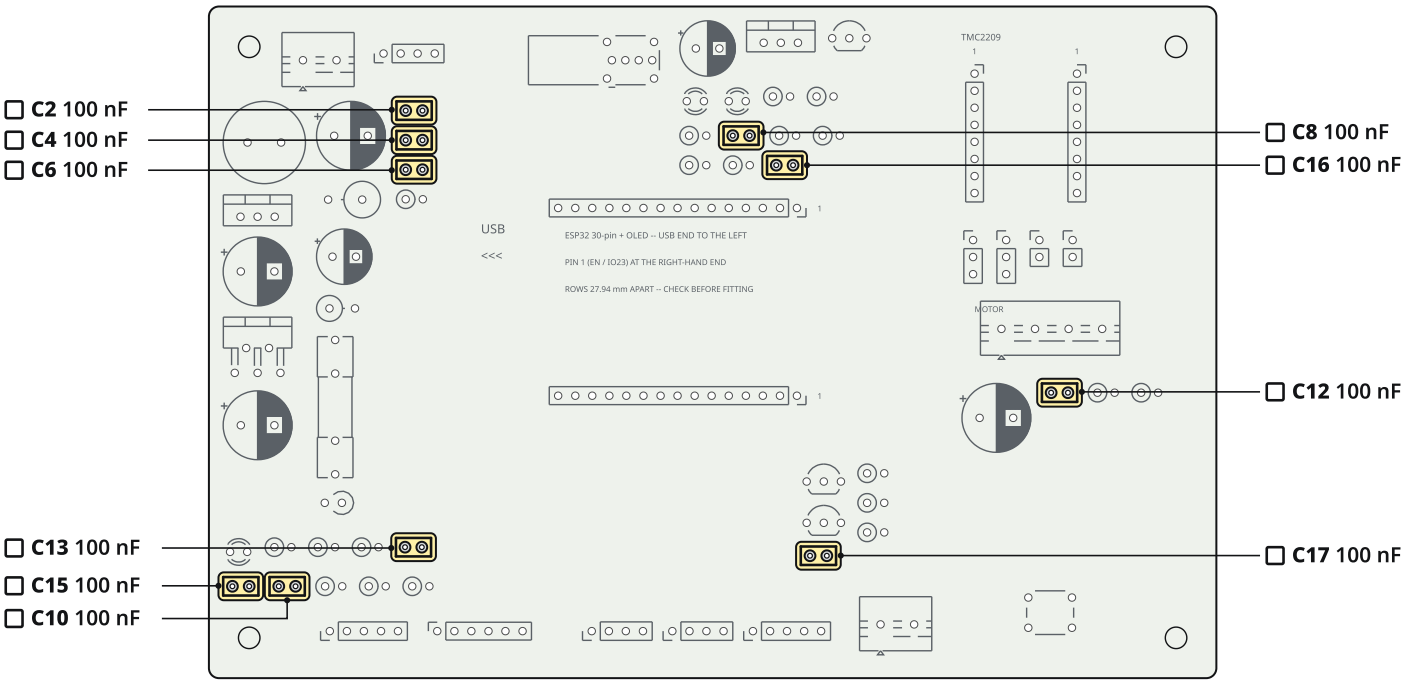
Value	How many	Where
10 kΩ	7	R1 R3 R4 R9 R13 R16 R18
1 kΩ	4	R2 R6 R7 R17
100 kΩ	2	R5 R19
4.7 kΩ	1	R12
330 Ω	1	R8
100 kΩ 1 %, 18 kΩ 1 %	1 + 1	R14, R15
leave empty		R10 R11: the DS3231 module has its own I^2C pull-ups

Work one value at a time: all seven 10 kΩ, then all four 1 kΩ, and so on. The value bag never has to be put down mid-row.

27 Ceramic capacitors: what and where

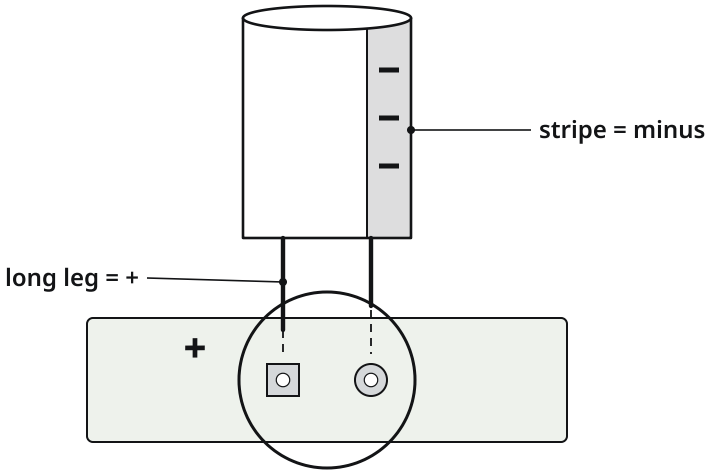


`10 then 4 zeros`	the first two digits, then a count of zeros
`100 000 pF`	picofarads
`= 100 nF`	nanofarads: the value in the parts list
`= 0.1 µF`	microfarads: how some sellers list it



- **Ten, all the same: 100 nF, printed 104.** Small blue or yellow discs. Each one sits beside a chip or module pin and smooths its supply.
- No polarity. Push it down until the kinks in its legs stop it, a millimetre or two above the board.

28 Electrolytic capacitors: how



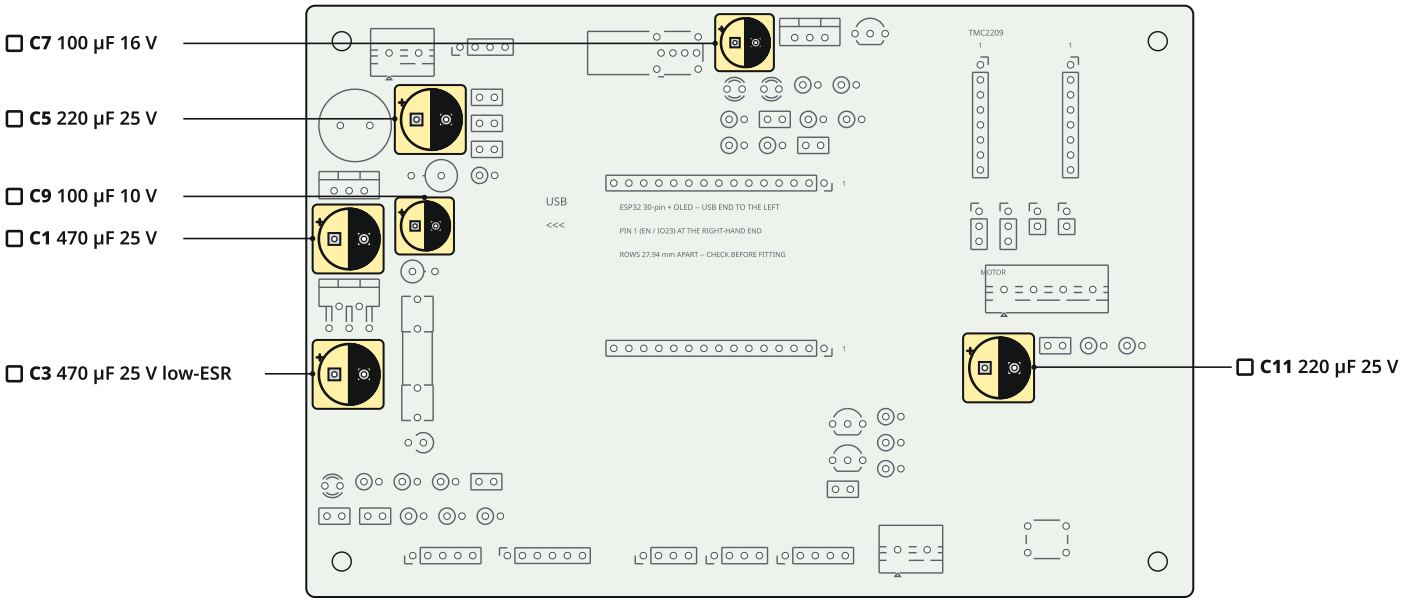
- These have **polarity**, the first parts on the board that do. The **long leg is +**. The **stripe** down the sleeve marks **−**.
- On the board, **+** is the **square pad**, with a small **+** printed beside the circle. Long leg into the square pad, stripe away from it.
- Sit each one flat on the board before soldering.

Where	Value	Size	
C1	470 µF 25 V	10 mm	12 V input
C3	470 µF 25 V **low-ESR**	10 mm	regulator input
C5	220 µF 25 V	10 mm	5 V output
C11	220 µF 25 V	10 mm	motor driver supply
C7	100 µF 16 V	8 mm	camera 5 V
C9	100 µF 10 V	8 mm	3.3 V

⚠ An electrolytic fitted backwards can swell or burst when power goes on. Check every one against its **+** before soldering. A higher voltage rating than listed is always fine; a lower one never is.

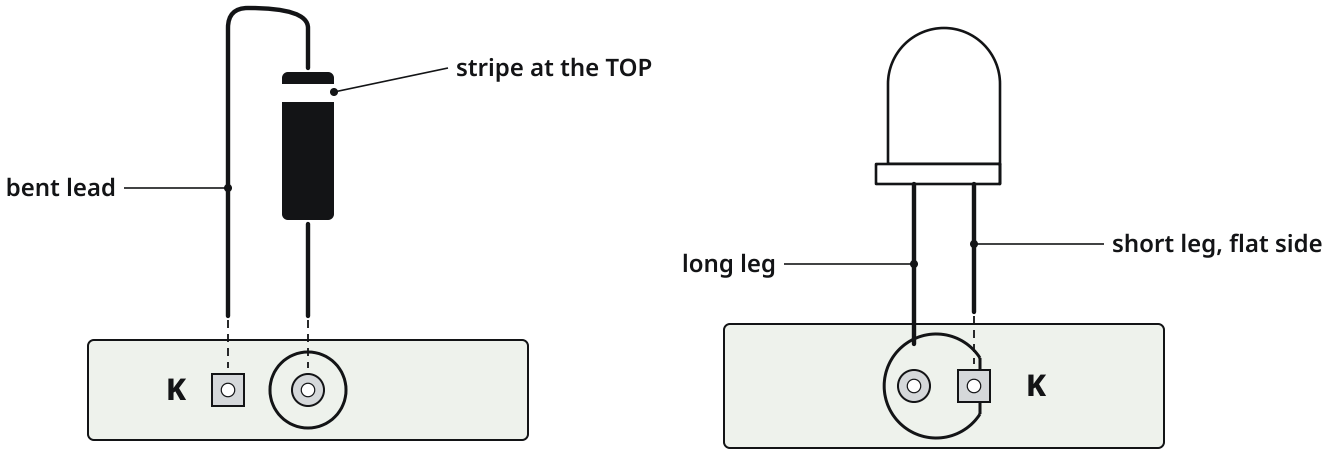
ⓘ C5 and C11 are the 220 µF Amazon parts (RESUME.md). If the driver resets or the motor steps noisily later, swap C11 for a low-ESR part first.

29 Electrolytic capacitors: where



The square pad in each circle is **+**. Look for it on the board before each capacitor goes in.

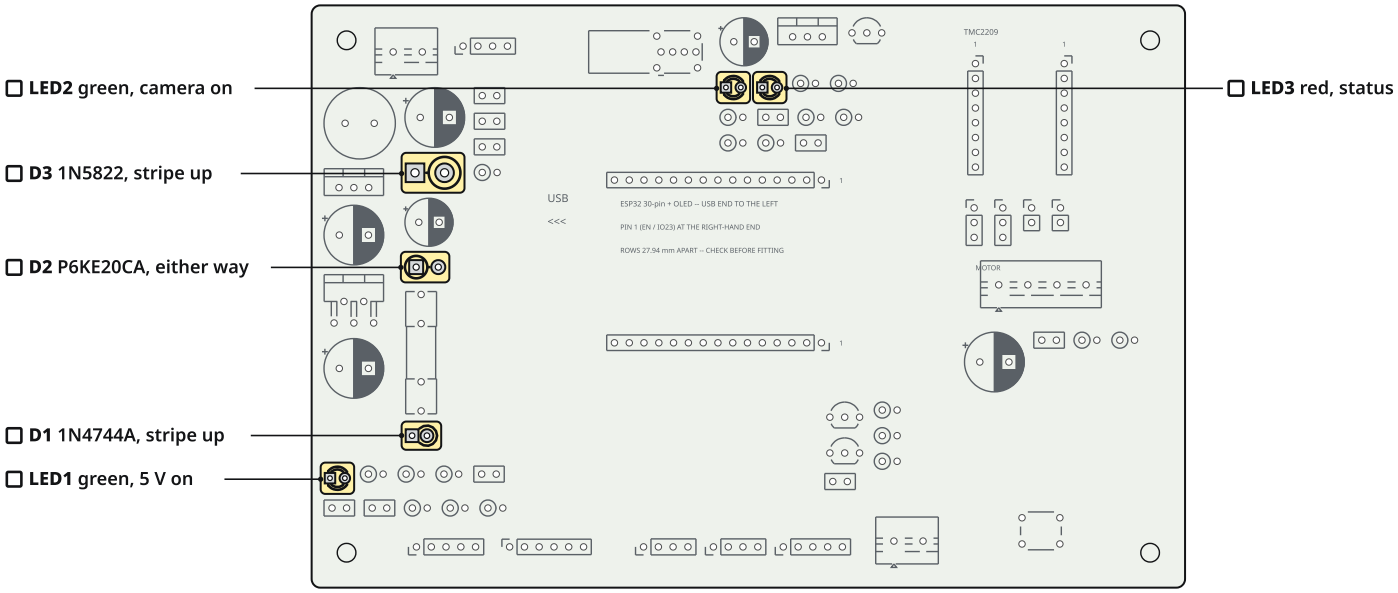
30 Diodes and LEDs: how



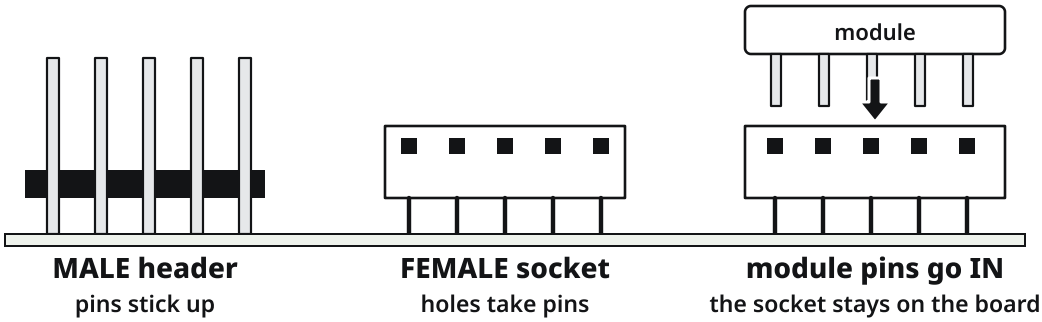
- **Diodes stand like resistors**, with the **stripe at the top**. The bent top lead goes into the **square K pad**.
- **D1** `1N4744A` is small; **D3** `1N5822` is big and black. Both have a stripe.
- **D2** `P6KE20CA` works either way round: it has no stripe to match.
- **LEDs:** the **short leg** and the **flat side** of the rim go to the **square K pad**. Two green, one red.

Not sure which LED leg is shorter after trimming? Set the meter to diode test: the LED glows faintly when the black probe is on K.

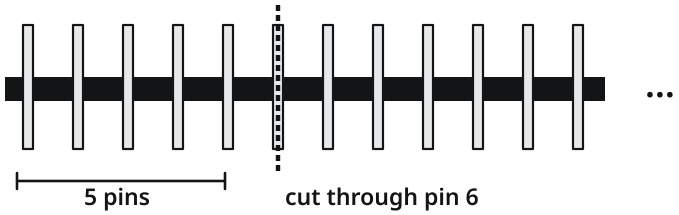
31 Diodes and LEDs: where



32 Male and female



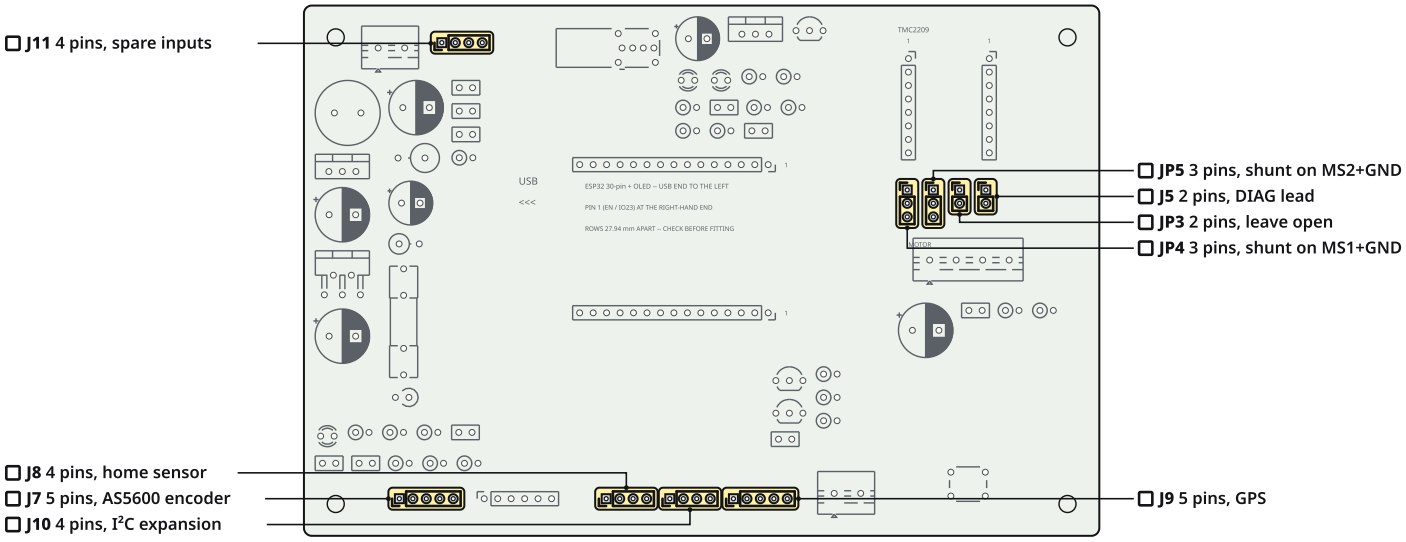
- A **male** header has **pins**. A **female** socket has **holes**.
- On this board, a **female socket** goes wherever a **module plugs in** (ESP32, TMC2209, DS3231). A **male header** goes wherever a **cable plugs on** (encoder, GPS, sensors) or a jumper sits.
- Male headers go in **short ends down**, through the board; the long pins stand up.

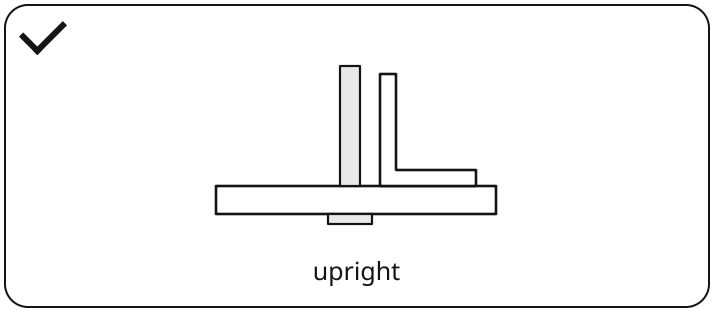


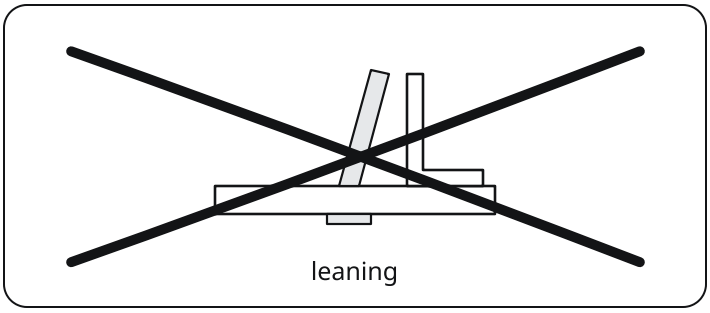
Cut from the 1×40 male strip	Pins
J7 J9	5 each
J8 J10 J11	4 each
JP4 JP5	3 each
J5 JP3	2 each
total	**32 of 40**

Count **pins**, not gaps. Cut through the middle of the next pin with flush cutters, and the edge stays clean.

33 Male headers and jumpers: where

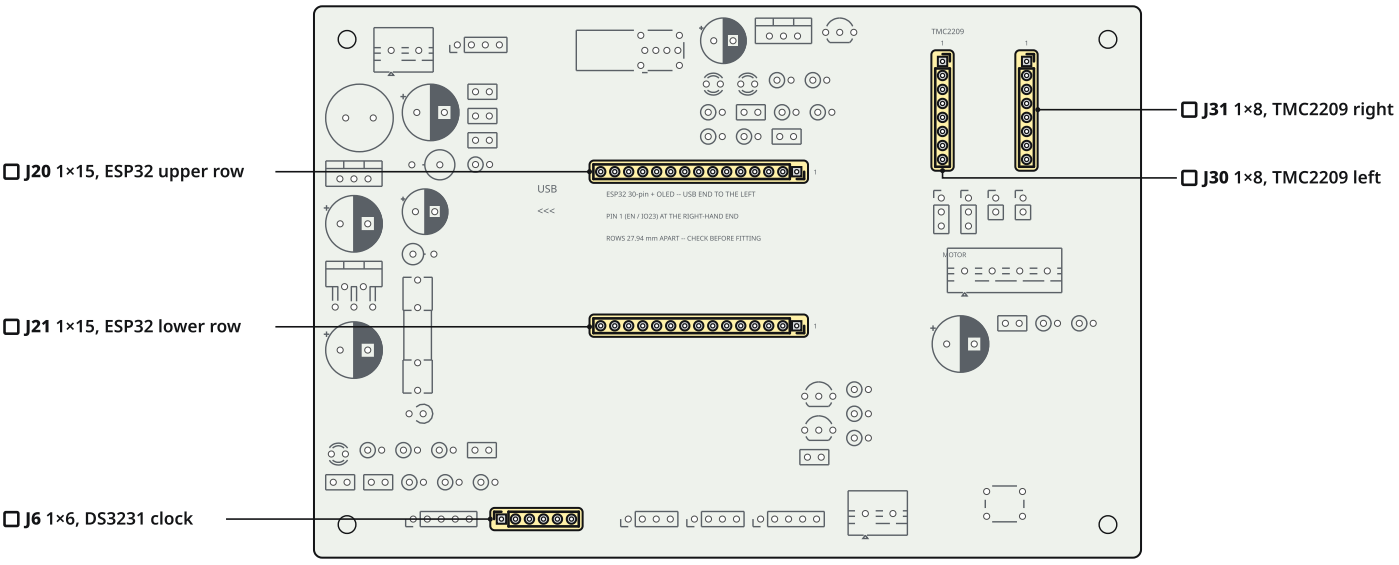


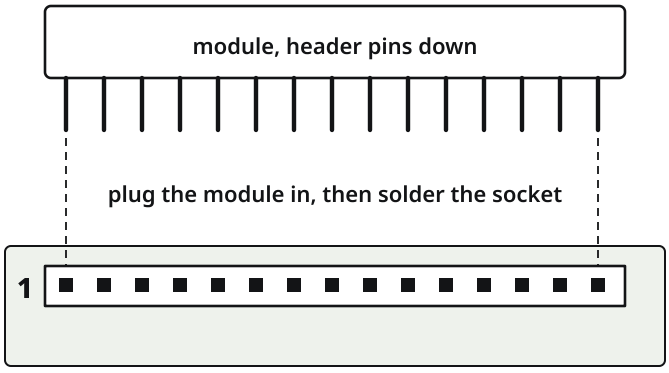




- Solder **one pin**, then look from two sides. Leaning? Melt that one joint and push the header straight with a finger on the plastic, not the pins. Then solder the rest.
- **Jumper shunts** on JP4 and JP5 bridge the **two pins at the GND end** (the lower two, next to the printed GND). JP3 gets **no** shunt.

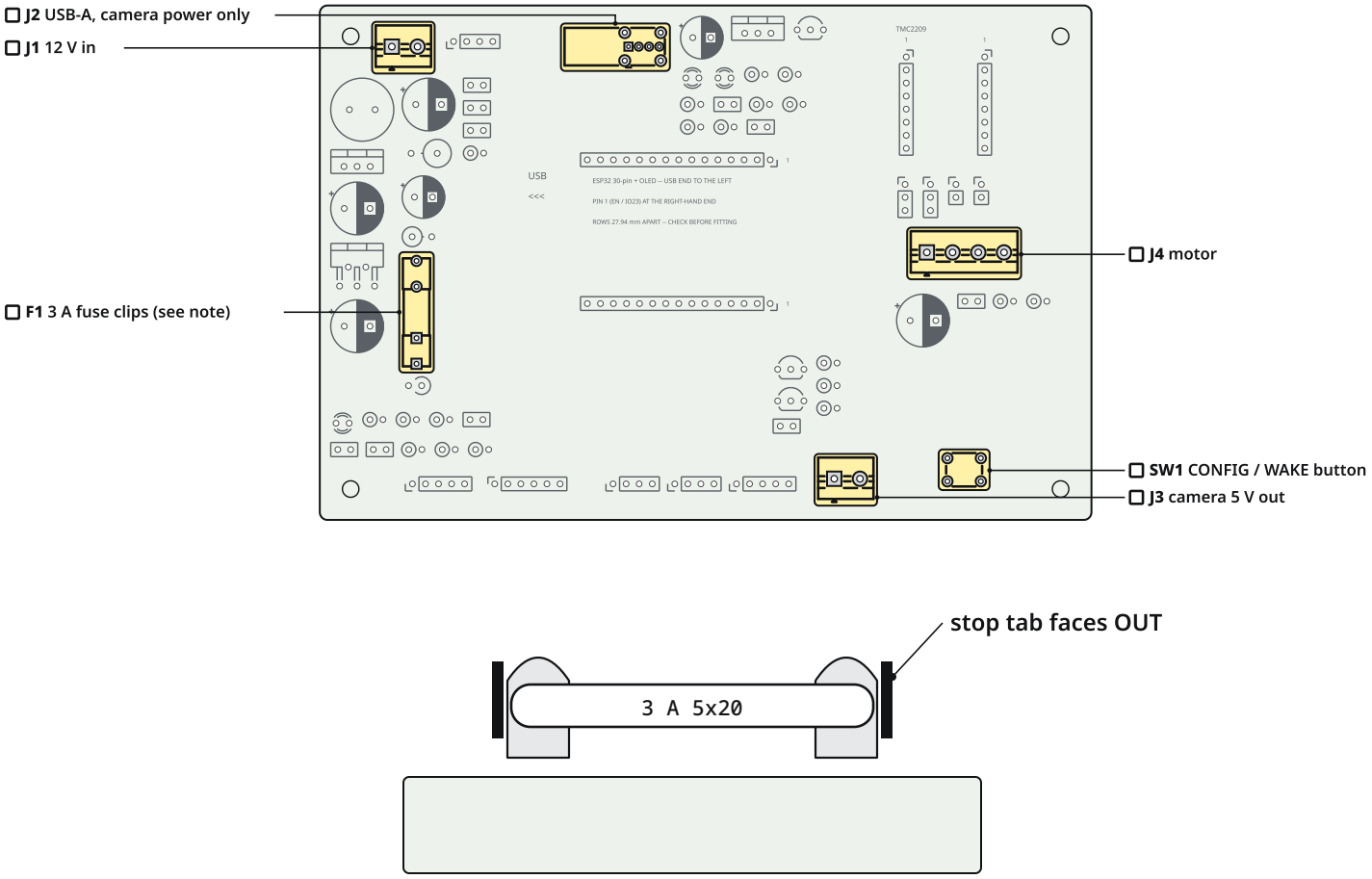
34 Female sockets: how and where





- Push both rows of a pair into the board, **plug the module in** to hold them exactly the right distance apart, and turn it all over.
- Solder the **two end pins** of each row, check everything sits flat and square, then solder the rest. **Unplug the module** afterwards: modules go in during section C, one at a time.
- A socket has no front or back. The module's way round is set later, by the **1** and the arrow printed on the board.

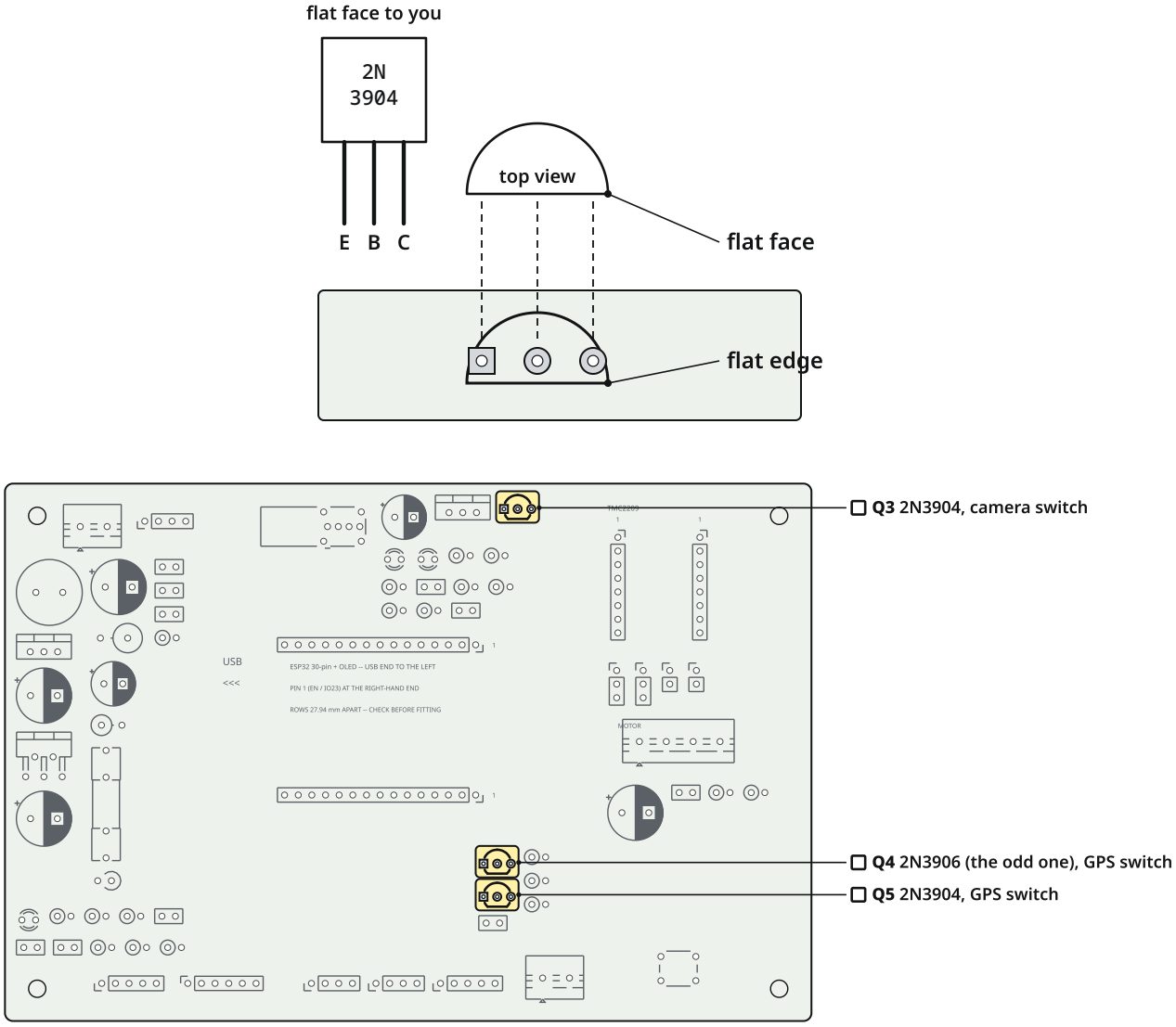
35 Terminals, USB, button, fuse clips



- **Screw terminals** J1, J3, J4: the wire openings face the **bottom edge** of the board as the map shows it, toward the line printed just past the holes. Both ways fit the holes; only that one is right.
- **J2 USB-A**: the two metal shell tabs solder too. They carry the strength when a cable is pulled.
- **SW1** fits two ways and both are correct. Press it flat before soldering.
- These pins are big and soak up heat: hold the iron on a little longer, 4 to 5 seconds, until the solder flows into the hole.

⚠ **Rev C boards:** the F1 clip holes are turned 90° to the clips, so standard clips do not drop in (fixed in Rev D). Bend the clip legs to suit the holes, or wire an in-line 5×20 fuse holder to the two F1 positions. Whichever you choose, the stop tabs face **out** so the fuse cannot slide.

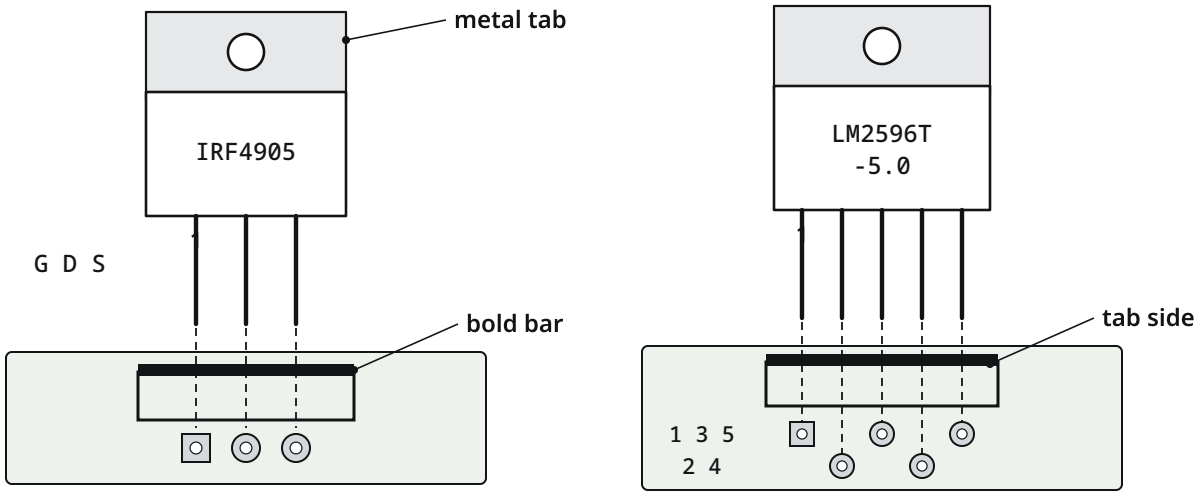
36 Small transistors: how and where



- The **flat face** of the body sits over the **flat edge** of the outline. The legs go straight down, middle leg into the middle hole.
- **Read the print on each one before it goes in.** The 2N3904 and the 2N3906 look identical. Q3 and Q5 are `2N3904`; **Q4 is the 2N3906**.
- Leave 3 to 5 mm of leg above the board. The heat stays in the leg, not the transistor.

⚠ A swapped 2N3904/2N3906 does not smoke. The GPS simply never powers up, and it takes a long time to find. Read the print with a magnifier or a phone camera.

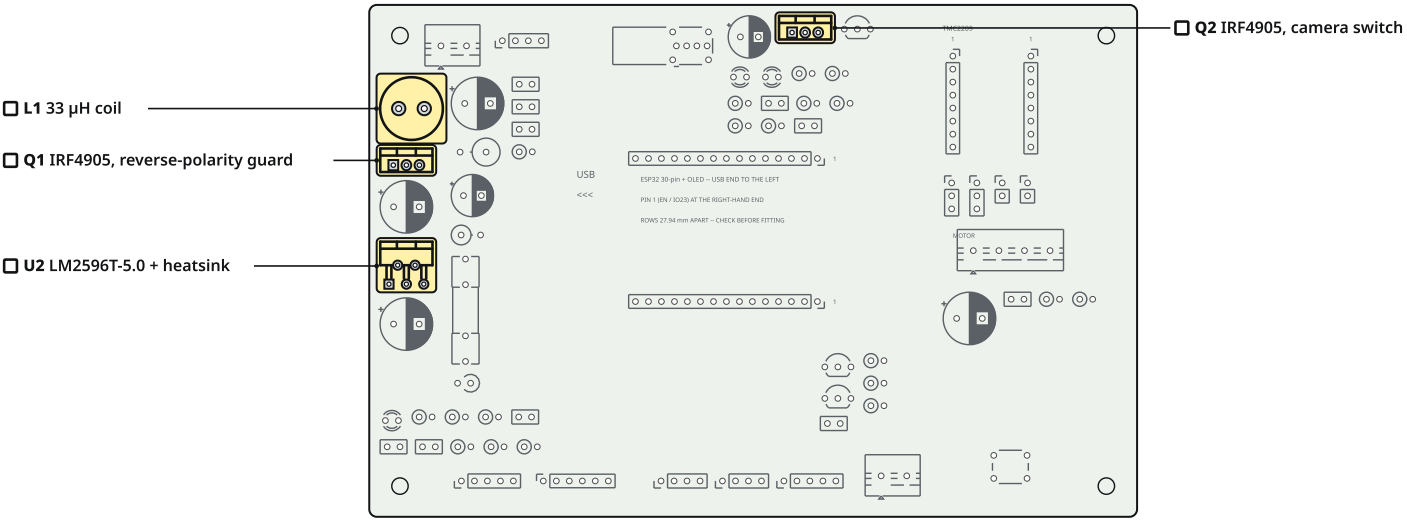
37 Power parts: how



- **Q1, Q2** `IRF4905`: the **metal tab** faces the **bold bar** on the outline, and pin 1 goes in the square pad. Printed face toward you, the legs read G D S.
- **U2** `LM2596T-5.0`: five legs in two rows, 1, 3, 5 in one and 2, 4 in the other. Pin 1 is the square pad. Clip the heatsink on before soldering, so it sits where it will live.
- **L1** 33 µH, the big coil: no polarity. Its legs are 6 mm apart and the holes 5 mm, so squeeze them in gently with pliers.
- Stand each part upright and square, solder one leg, check, then the rest.

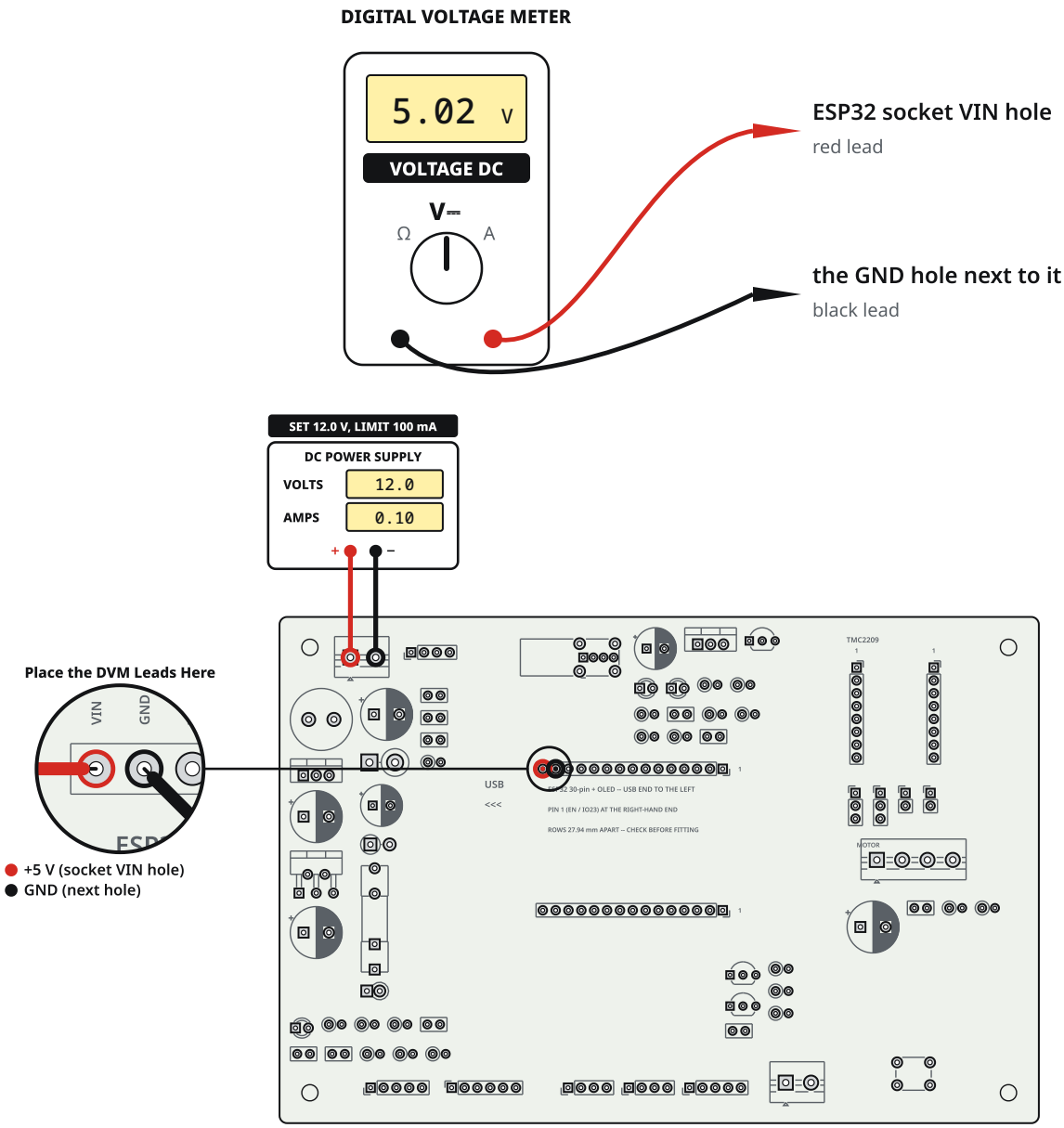
⚠ The metal tabs are **live**. Q1's tab is the 12 V input, Q2's is the camera 5 V, U2's is ground. Heatsinks must not touch each other or anything else.

38 Power parts: where



Q1 is where reverse-polarity protection is won or lost: drain to the supply, source to the load. The footprint fixes that, as long as pin 1 is in the square pad.

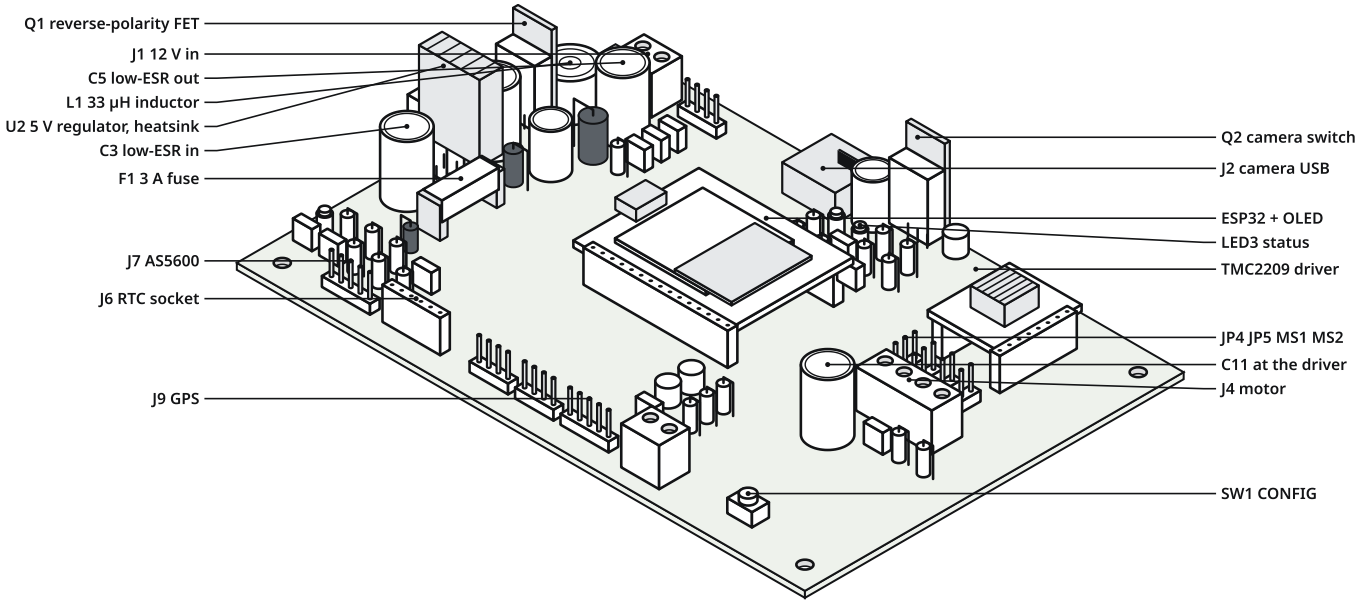
39 Quick check: first light



- **No modules in the sockets.** Meter on Ω across J1: the reading climbs as C1 charges. A steady reading under 10 Ω is a short: find it first.
- 12 V into J1 through a **100 mA current limit**, or a 12 V bulb in series with the lead. **+ to the left screw** (J1 pin 1, the square pad).
- **LED1 lights green.** The meter reads **4.9 to 5.1 V** at the ESP32 socket's VIN hole. LED2 and LED3 stay dark. Nothing gets warm.
- Power off. All good? On to the full tests in section C. Not good? Section C's tests, one at a time, show which part is at fault.

This is the first time the board does something. If LED1 lights, the input protection, the fuse and the 5 V regulator are all in and working.

40 Before you test

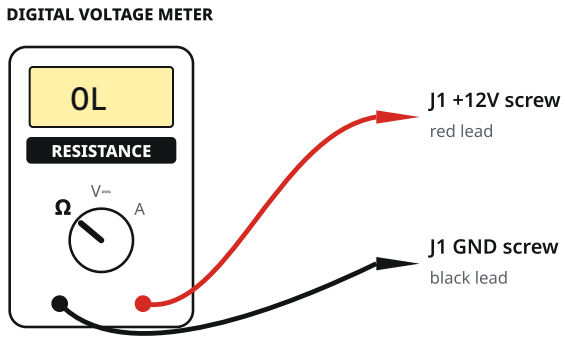


- Every box on every map ticked. The only empty places are **R10 and R11**, and the sockets, which stay empty.
- Turn the board over under a bright light. Every joint a shiny cone, no bridges, every lead trimmed.
- Scrub off the flux with isopropyl alcohol and an old toothbrush, and let it dry.
- Now Test 0. No module goes in until section C says so.

C Smoke test

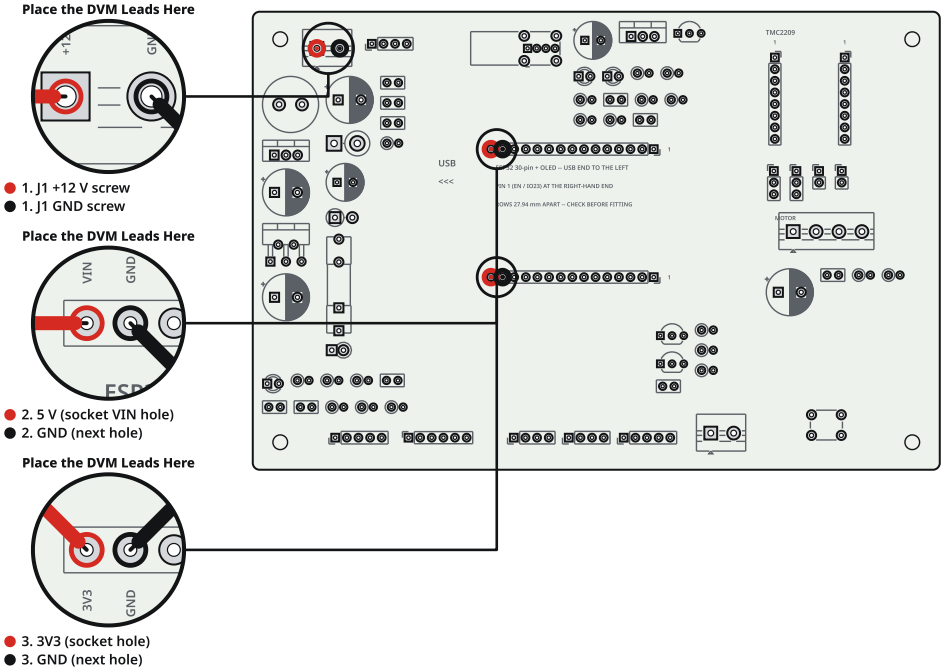
Nothing expensive goes in until every rail reads right. A bench supply with a current limit, or a 12 V bulb in series with the lead, turns a mistake into a reading instead of a smell.

41 Test 0: DON'T HOOK UP POWER YET – Look, then measure resistance



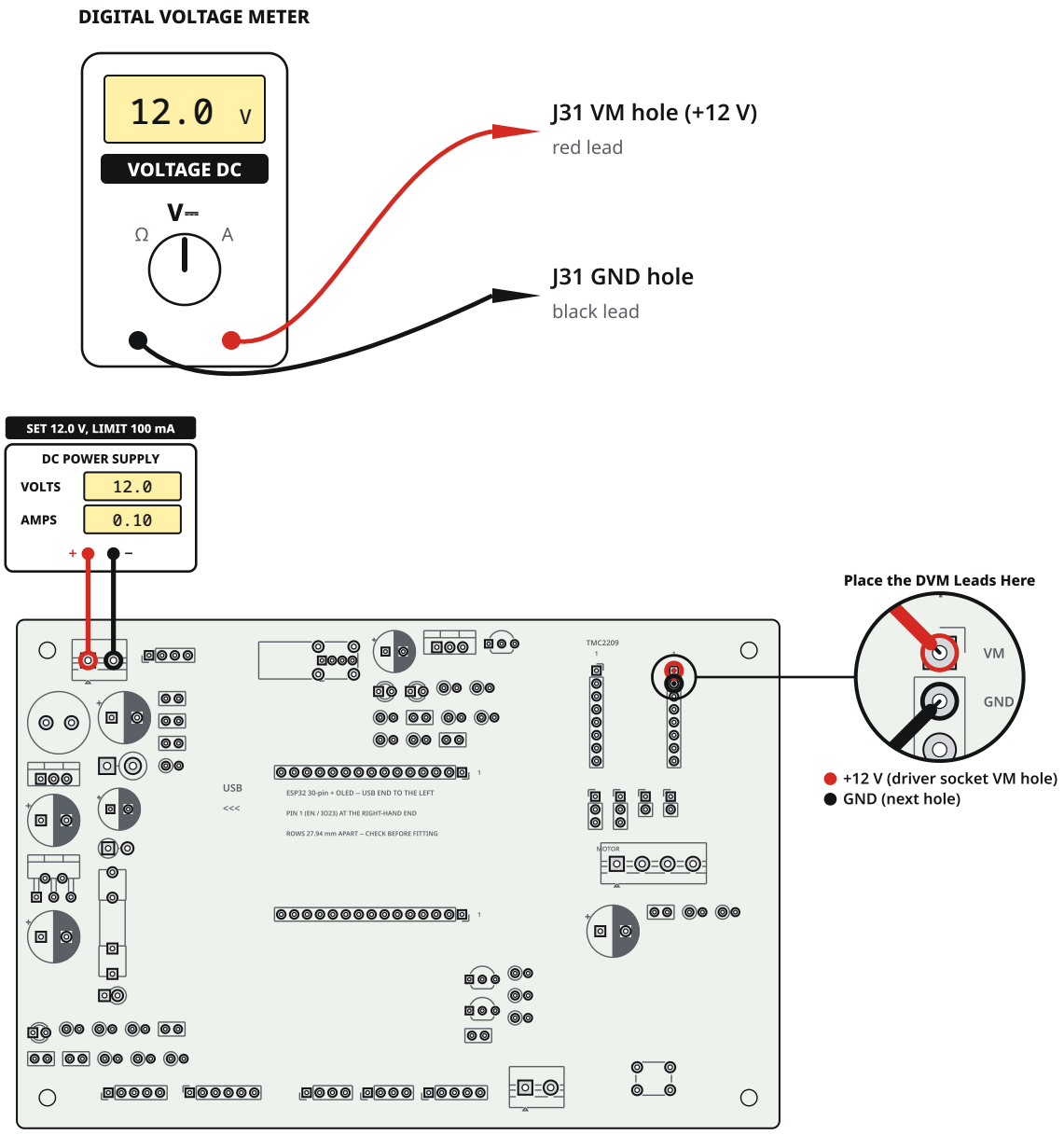
How this works with no power. On Ω the meter sends a small current of its own and reads the voltage it makes. Across J1 that current fills the board's capacitors through the fuse and Q1, so the reading starts low and rises to megohms, then OL. A short never fills and stays low. The test: **does it climb, or stay put?**

Here the leads matter. Unpowered, Q1 passes the meter's current only from J1 + inwards. Red is + on most meters, so red on +12V gives the climb. Swapped, the meter jumps straight to OL: harmless, but no check.



- **1.** Red lead on J1 +12V, black on J1 GND. The reading **starts low and climbs** to megohms or **OL** (see the note). A reading that **stays** low is a short: **find it before any power goes in**.
- **2.** 5 V to GND and **3.** 3V3 to GND, at the socket holes: they climb the same way, or settle at a few kΩ through the resistors on those rails. Neither may stay below 10 Ω.

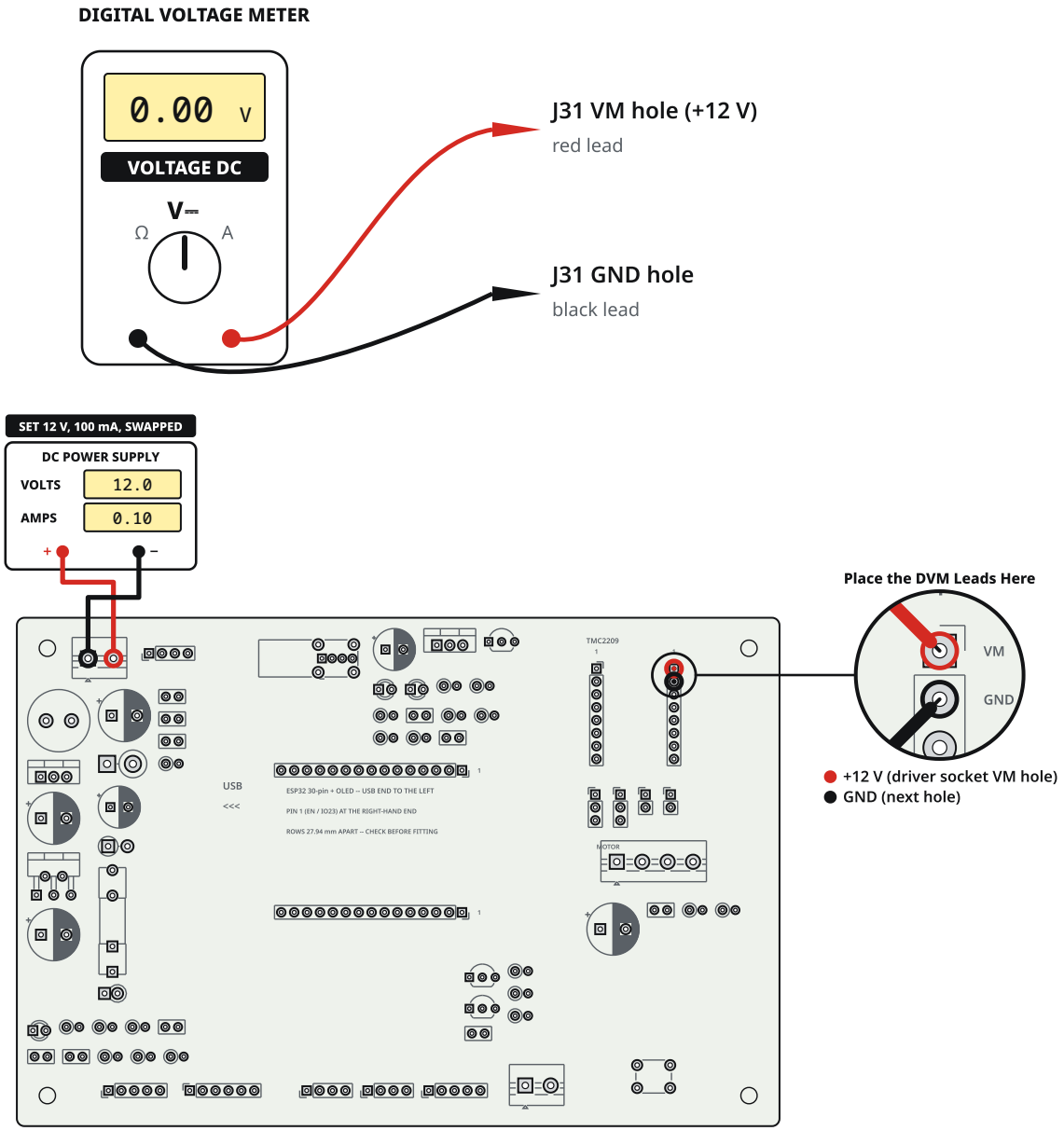
42 Test 1: First power, current-limited



- At this point all the electronic components are installed. **EXCEPT DO NOT YET INSERT THE MODULES:** the ESP32 goes in at step 48, the TMC2209 and DS3231 at step 49, each only after the tests before it pass. Supply at **12 V, 100 mA limit** into J1.
- The driver socket's **VM hole** (the +12 V rail, after Q1) reads the input voltage, within 0.1 V.

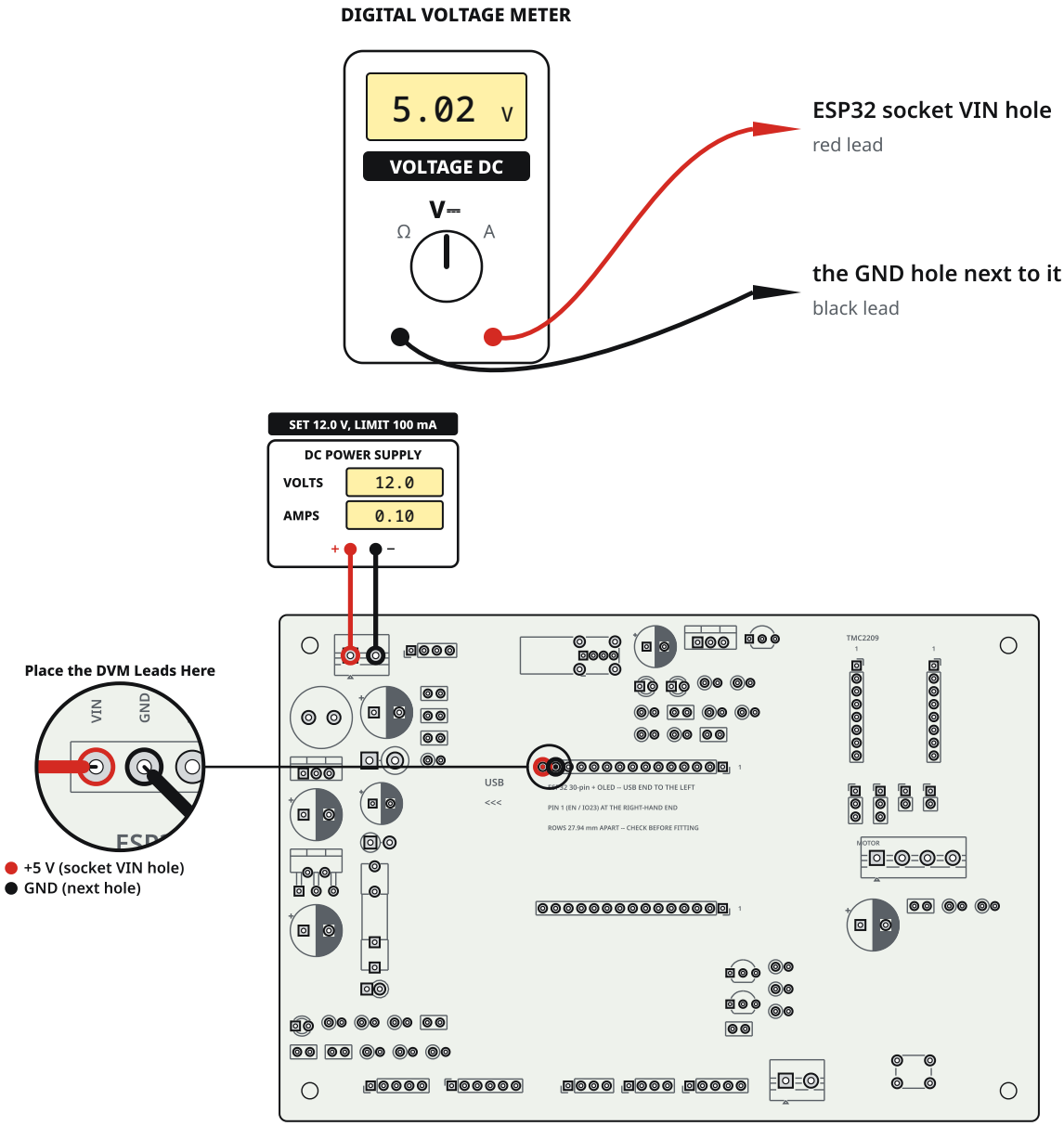
43 Test 2: Verify correct polarity to the chip

Step 42 showed power gets through when the supply is connected the right way. This step connects it the **wrong** way on purpose and shows that Q1 stops it, so a reversed battery can never reach the chips.



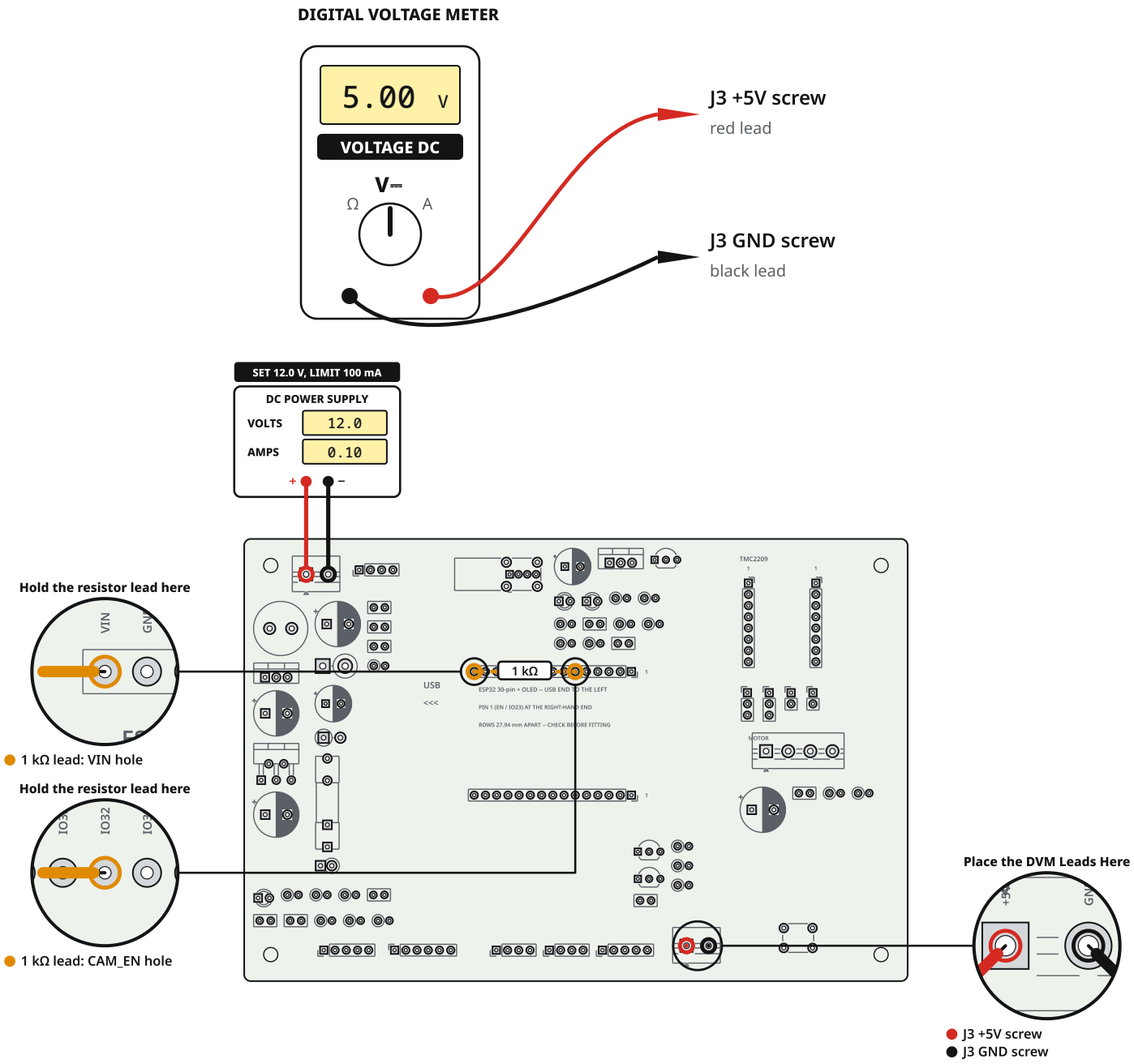
- Swap the leads at J1, still current-limited. The rail reads **0 V** and nothing gets warm.
- If the rail shows −12 V, Q1 is in backwards. Stop here.

44 Test 3: Verify 5 V is correctly converted from the 12 V



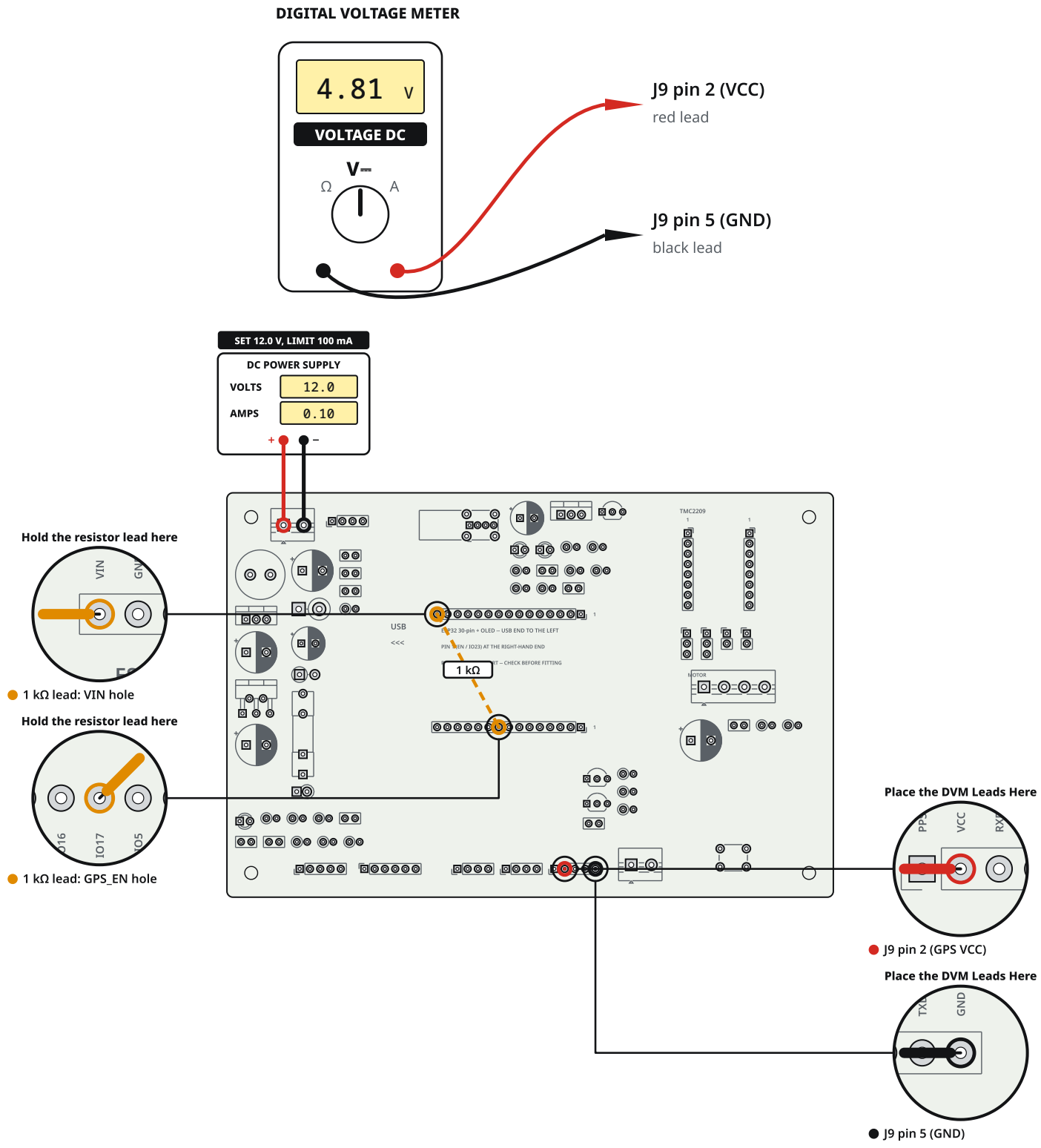
- **4.9 to 5.1 V** at the ESP32 socket's VIN hole. LED1 lights.
- Idle draw at 12 V is around 10 to 20 mA: the regulator's own current plus LED1.

45 Test 4: Camera switch



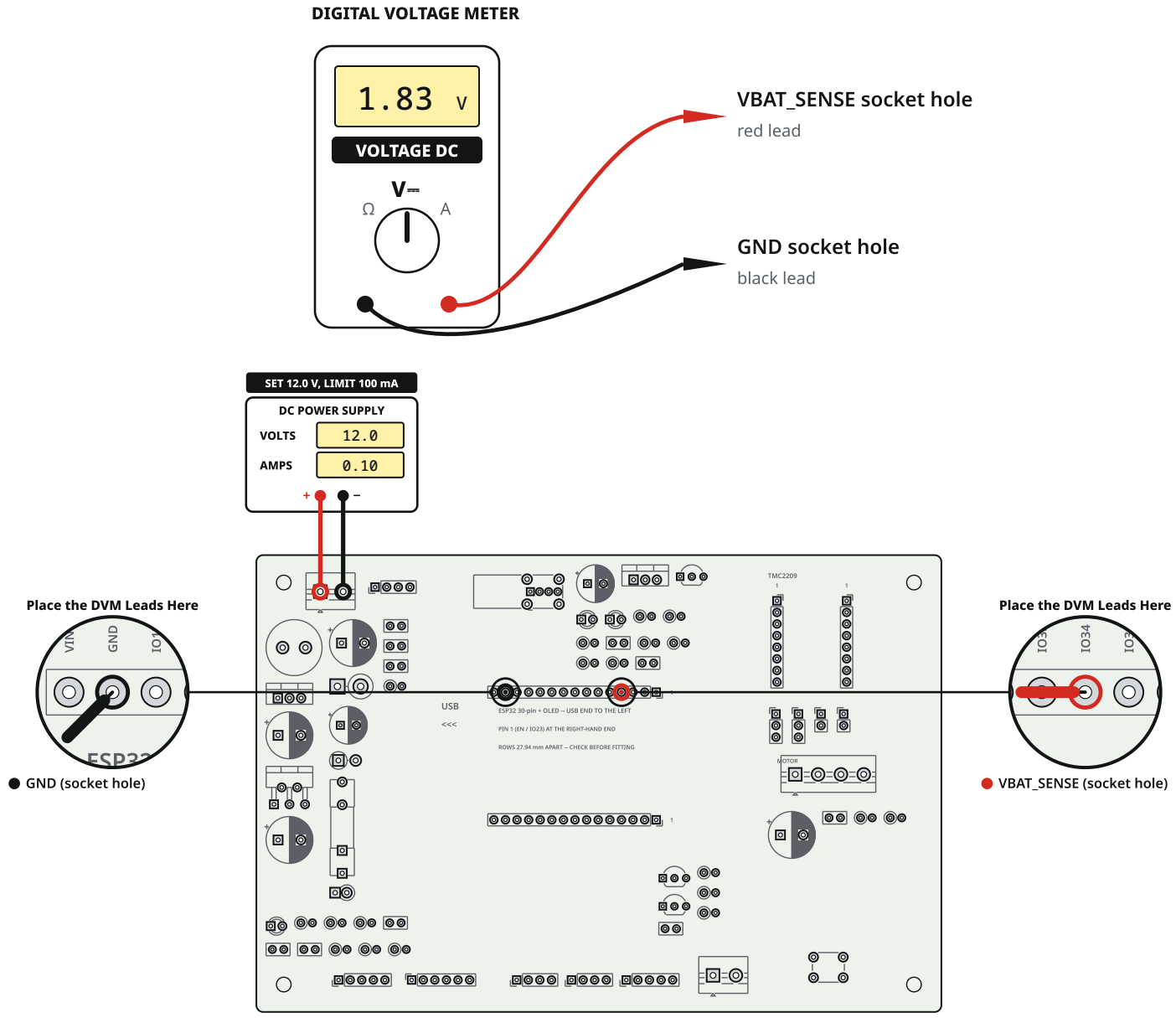
- J3 reads **0 V** at rest: the camera is off by default.
- Hold a **1 kΩ** resistor from the socket's VIN pin to its **CAM_EN** pin. J3 and the USB-A read 5 V, LED2 lights. Take it away and they drop to 0.

46 Test 5: GPS switch



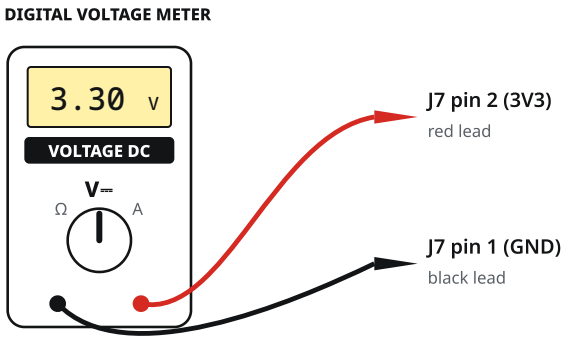
- The same 1 kΩ from VIN to the **GPS_EN** pin: J9 pin 2 reads about **4.8 V**. Without it, 0 V.

47 Test 6: Battery divider

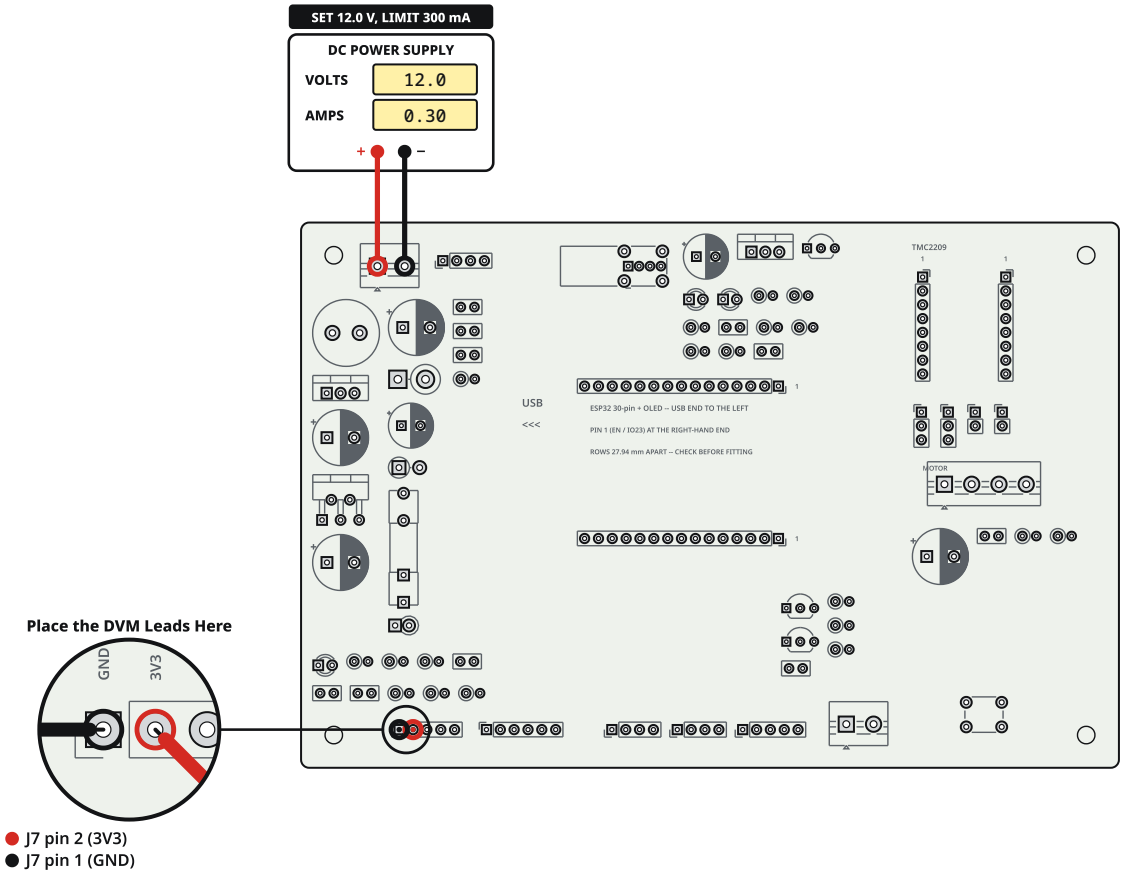


- VBAT_SENSE = input × 18 / 118. **12.0 V in gives 1.83 V**, a full 14.6 V LiFePO4 gives 2.23 V. Both are safely under the ESP32's 3.3 V.

48 Test 7: Warm check, then the ESP32



3.3 V comes out of the ESP32 module's own regulator. With no module fitted, the 3V3 net reads nothing, and that is correct.

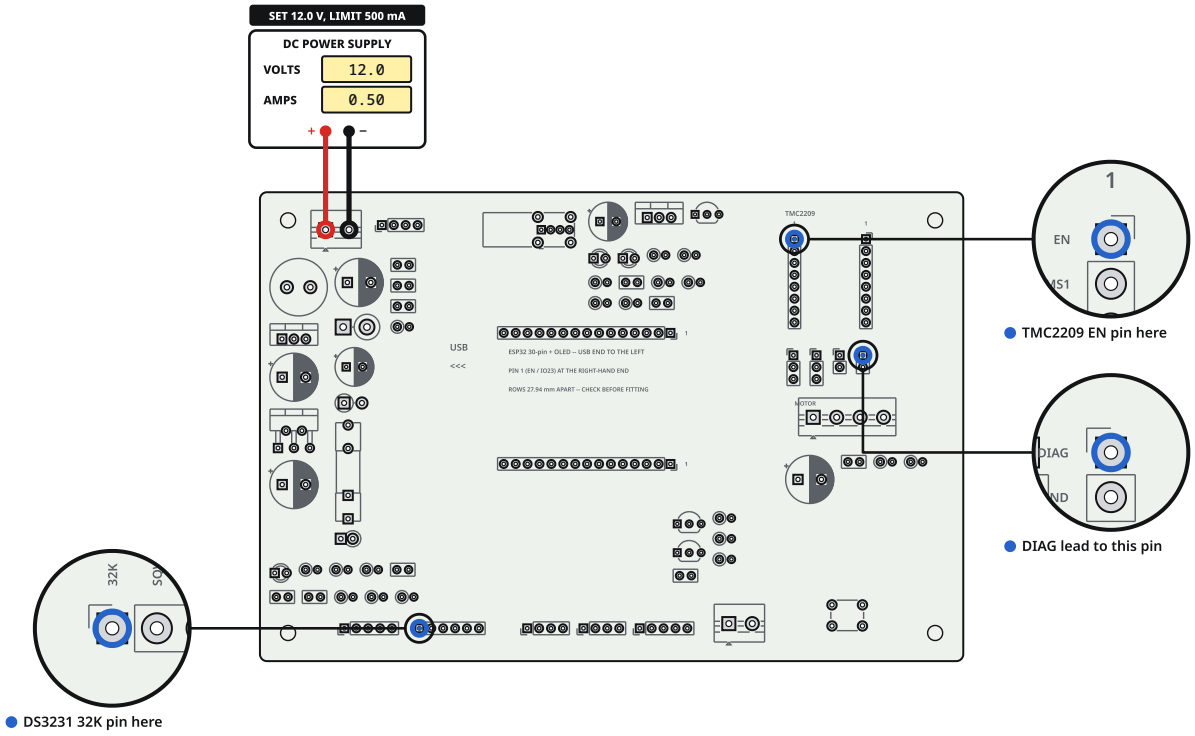


- Five minutes powered with nothing fitted. U2 and Q1 barely warm.
- **Power off.** Fit the ESP32 only: USB **toward the arrow**, its pin 1 at the socket's **1**. If it does not line up, stop.
- Raise the supply's current limit to **300 mA**: the ESP32 draws more than the bare board, in bursts when its radio starts.
- Power on. **3.2 to 3.4 V** at J7 pin 2, J10 pin 2, the DS3231 socket's VCC and the driver socket's VIO. The OLED may light.

49 Fit the driver and the clock

fitting, then a check

Not a measurement: this is where the last two modules go into their sockets. Each one fits two ways round; the blue rings show where its pin 1 goes.



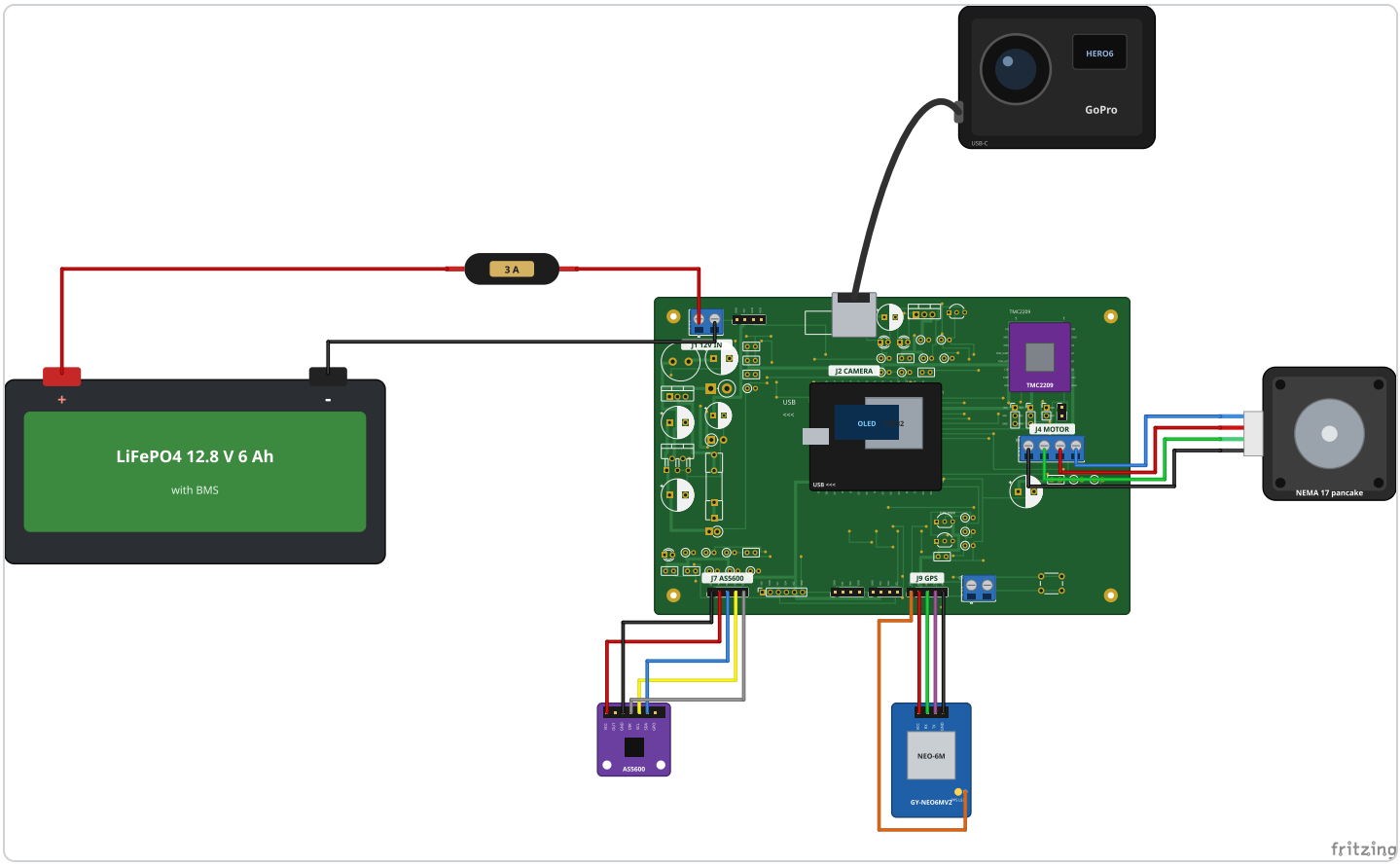
1. **Power off** and unplug the supply.
2. **Wire the motor first:** do step 51, then come back. The driver must never be powered without its motor.
3. Turn the TMC2209's V_{REF} pot to its **low end** (fully anticlockwise) for now.
4. **TMC2209** into J30/J31: the module's **EN** pin into the hole marked **EN**. On BTT modules the chip is on the underside; that is normal.
5. **DIAG:** no module brings it out on its header. Solder a short flying lead from the module's DIAG pad to **J5 pin 1**.
6. **DS3231** (ZS-042) with a CR2032 into J6: its **32K** pin into the first hole. The printed order runs `32K SQW SCL SDA VCC GND`. Its charging circuit is harmless at 3.3 V.
7. **Check:** power on at 12 V with a 500 mA current limit for a minute. Nothing gets warm, the OLED still lights. Power off.

⚠ Never plug or unplug the motor with power on. The driver dies from the inductive kick, and C11 must be fitted at VMOT before the driver sees 12 V.

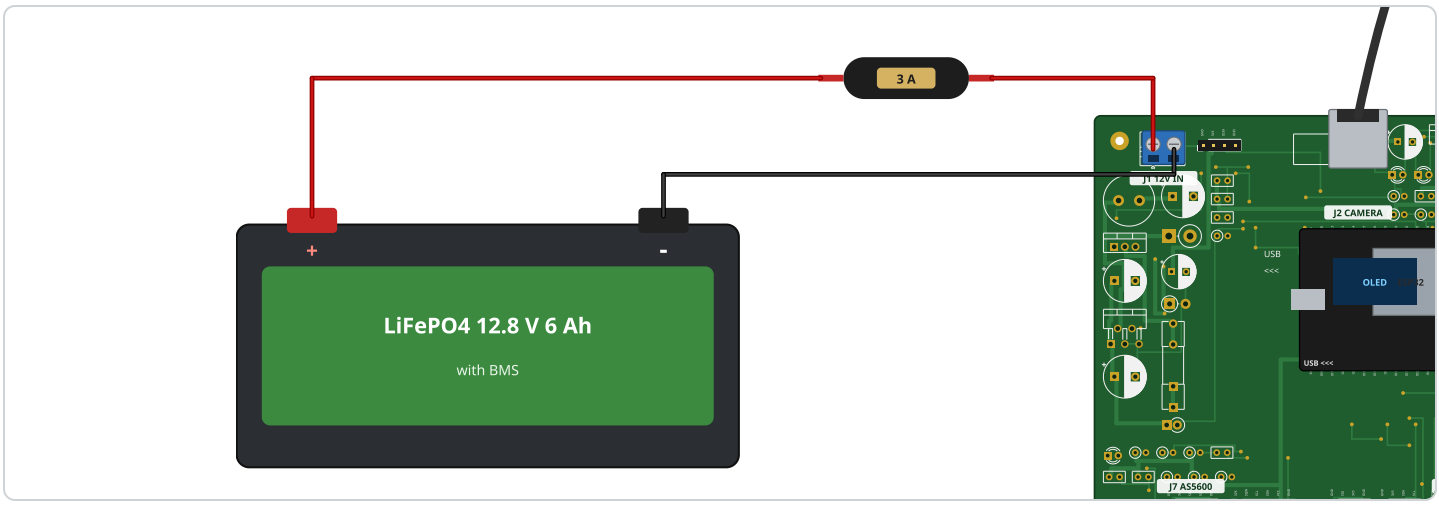
D Wiring

The board, the battery, the motor, the encoder, the GPS and the camera.

50 The whole loom

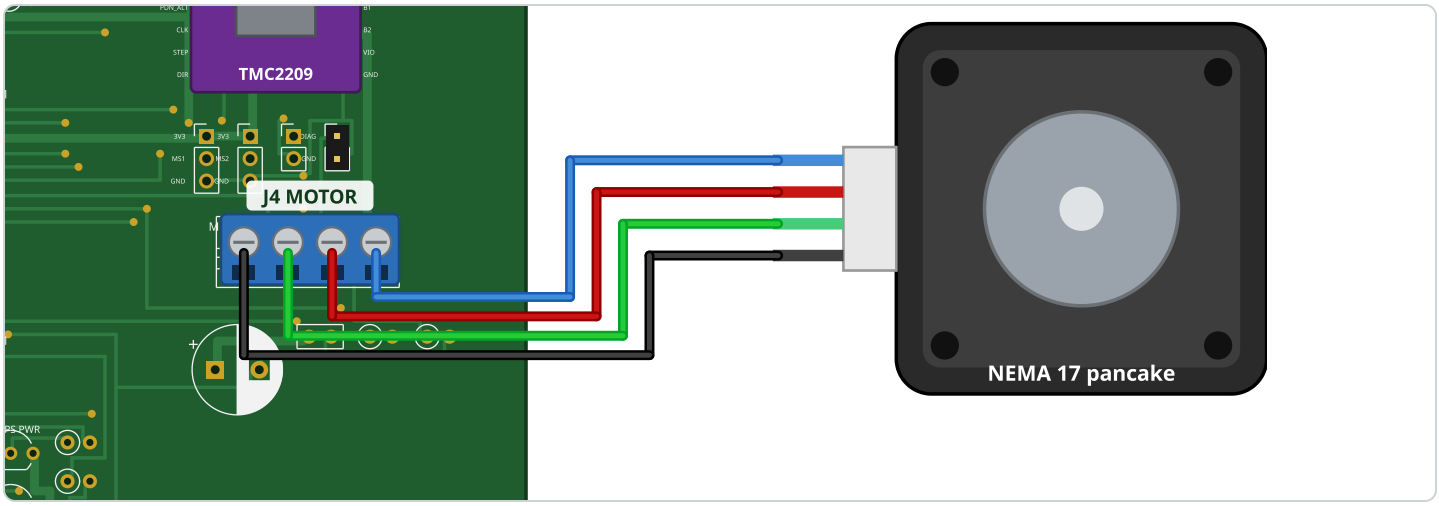


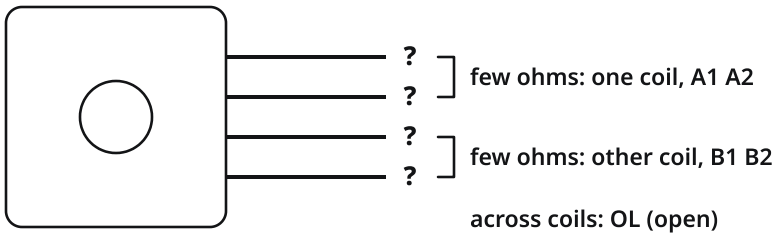
Fritzing, breadboard view. Source: `hardware/fritzing/star_tracker_wiring.fzz`, generated by `gen_fritzing.py` from the KiCad layout.



- **Battery** 12 V LiFePO4 → inline **3 A fuse** → XT60 → **J1**, + to +12V. With solar, feed from the charge controller's load output.
- Board fuse F1 is also 3 A. The inline fuse protects the lead.

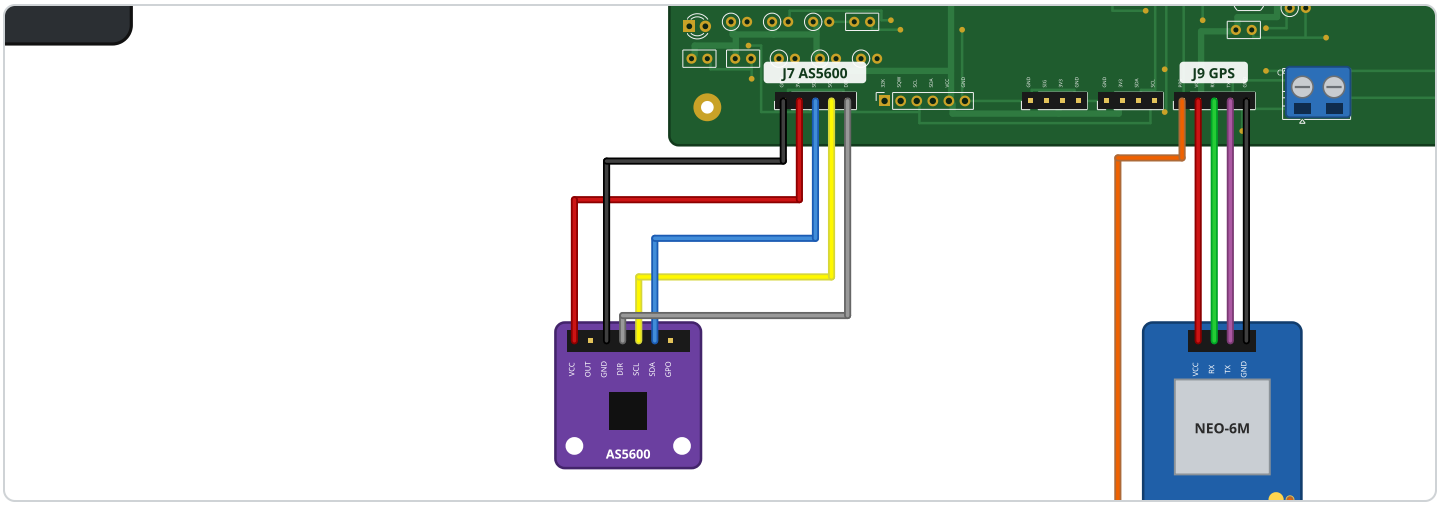
51 Motor coils





- Wire colours vary between pancake motors, so measure. One pair to **A1 A2**, the other to **B1 B2** on J4.
- Wrong direction on first spin? Swap **A1 and A2 only**, with power off.

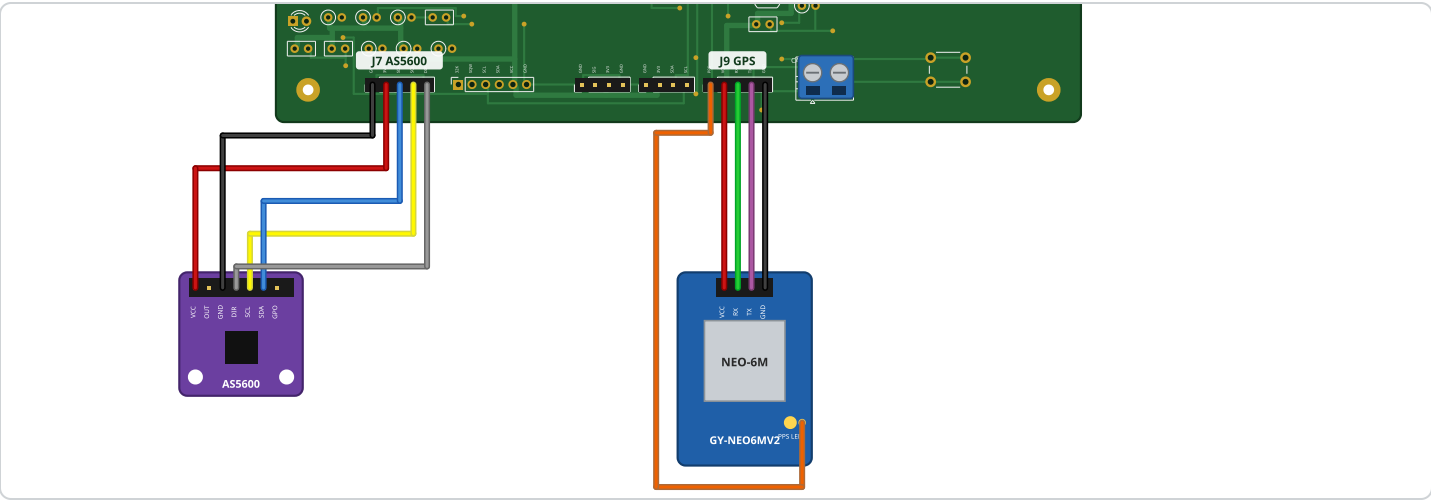
52 Encoder lead



J7	Board	AS5600
1	GND	GND
2	3V3	VCC
3	SDA	SDA
4	SCL	SCL
5	GND	DIR

- Under about 30 cm, SDA and SCL twisted with a ground. The bus runs at 400 kHz.
- The bracket and sensor stay still; only the magnet turns. This lead never has to follow the camera.

53 GPS and the PPS lead



- The GY-NEO6MV2's own 4-wire cable plugs into **J9 pins 2 to 5** (VCC RX TX GND), in the module's own order.
- **PPS** is not on its header. It drives the on-board PPS LED. Solder a flying lead from that LED's driven side to **J9 pin 1**.
- Pick the pad that swings to about 3 V once a second after a fix. If you see only about 2 V, you are on the LED side of its resistor: move to the other end.
- The antenna goes where it sees sky, outside the IP54 box.

PPS measures the crystal and sets the clock to under a second. It does not steer tracking: 20 ppm over a 12-hour night is 3.6 arcsec.

54 Camera cable



- **J2** USB-A → short right-angle USB-C → HERO6. J2's data lines are shorted, which tells the camera it may draw full charge current.
- Route it with the platform at the start angle. A night is **180.5°** of travel, so leave a slack loop that reaches both ends without pulling on the camera.
- Tie the cable to the platform so the loop, not the USB-C plug, takes the bend.

Measure the real arc the loop allows. It becomes `TRAVEL_MIN_DEG` / `TRAVEL_MAX_DEG` in step 61.

E First power-up and calibration

today works with a sketch already in the repo. planned needs the firmware laid out in FIRMWARE.md, which is not written yet.

55 Pin check

today

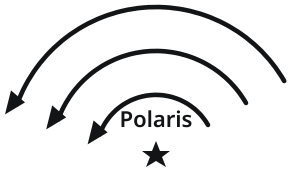
- Flash `firmware/pincheck/pincheck.ino`. Arduino IDE, board "DOIT ESP32 DEVKIT V1", serial monitor at 115200. Nothing else needs to be connected.
- It scans three I²C buses and names what answers, then prints PSRAM.

You see	Means
`0x3C` on GPIO 21/22	OLED shares the sensor bus. WAKE stays on GPIO 4.
`0x3C` on GPIO 4/15	Those pins are taken. WAKE moves to the upper row.
PSRAM: none	WROOM, as expected.
PSRAM: present	WROVER. GPIO 16/17 are taken and the driver UART moves.

Result for this build (2026-09-27): ESP32-D0WD-V3, 4 MB flash, PSRAM none; `0x3C` on GPIO 21/22, nothing on 4/15. Board Rev C is laid out from it. Run it again only if you change ESP32 boards.

56 First spin, and which way

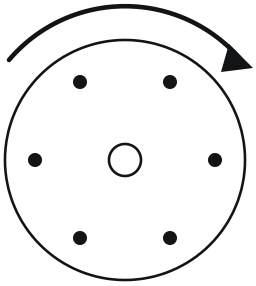
today



facing north, the sky turns

N

counter-clockwise



camera platform, seen from the camera end: clockwise

southern hemisphere: both reversed

- `sidereal_drive.ino` drives STEP 25, DIR 26, EN 27, which the planned pin map keeps.
- V_{REF} pot near its low end. Motor buzzes but does not turn? Raise it a little.
- Wrong way round? Change `digitalWrite(PIN_DIR, …)` or swap A1/A2 with the power off.

⚠ **Without UART setup the TMC2209 runs standalone.** With MS1 and MS2 at GND that is 1/8 microstepping, not the 1/32 the sketch assumes, so the output turns **4 × sidereal**, about 1° a minute. That is fine for seeing the direction, and it is not tracking.

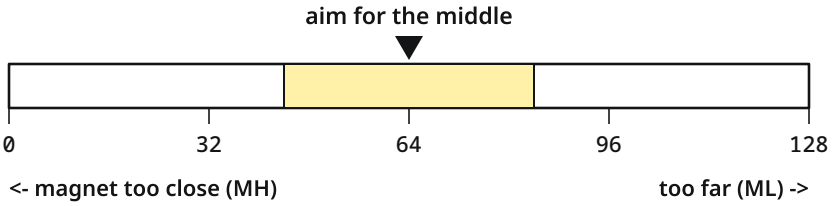
57 Driver current and microsteps

planned

- Over UART: run current **300 to 400 mA**. There is a 250 × torque margin, and low current means less vibration and heat.
- MRES **1/32** for tracking, **1/8** for slews. The position counter always stays in 1/32 units.
- Once UART sets the current, the V_{REF} pot does nothing.

58 Magnet gap

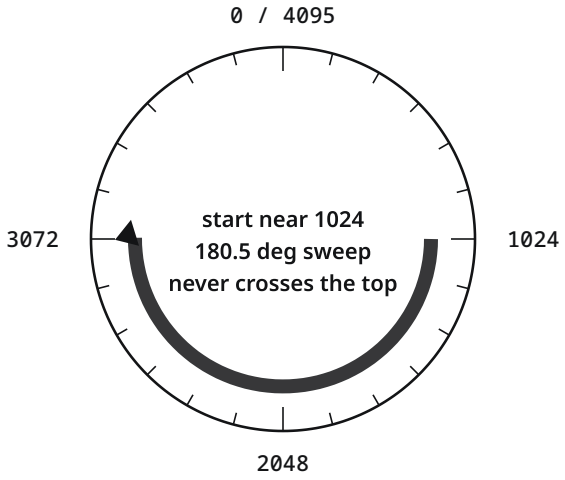
today



- Flash `firmware/as5600_check`. Only J7 needs to be connected. It reads **AGC**, **MAGNITUDE** and the **MD / ML / MH** bits five times a second and shows them on the OLED and in the serial monitor at 115200.
- Want: top line **GAP GOOD**. That means MD = 1, ML = 0, MH = 0 and AGC in the middle of its bar (about 64 of 128 at 3.3 V). The sketch only reads; it never programs the chip.
- **Too strong** (MH = 1, AGC near 0): slide the bracket down a little, away from the magnet.
- **Too weak** (ML = 1, AGC near 128): the bracket is already at its stop, so bring the magnet closer instead. Loosen the hub grub and slide the shaft down 0.5 mm at a time.
- Then lock the bracket with a dab of hot glue.

59 Encoder zero

today



- Same sketch. Park the platform at the start angle. Turn the magnet cap on the shaft until the big number on the OLED reads **ZERO OK**, which means raw is within ±12 counts (1°) of **1024**.
- **SEAM!** in the corner means the night would cross 4095 → 0. Keep turning toward 1024.
- The 180.5° sweep then ends near 3080 and never crosses the 0 / 4095 seam, so the firmware needs no wrap handling.

60 Clock and site

planned

- Set the DS3231 from the phone's clock (`POST /api/time`), or let TIME_SYNC take it from a GPS fix.
- Put your latitude and longitude into the schedule's `site`. The default is 0, 0. With a GPS fix, the tracker overrides it each night.
- Watch `rtc_drift_s` for a few nights before deciding the clock needs correcting at all.

61 Travel limits

planned

- Jog to each end of the arc the camera cable allows (step 54) and note the encoder angle.
- Set `TRAVEL_MIN_DEG` / `TRAVEL_MAX_DEG` in `include/config.h`. They are compile-time on purpose, so a phone cannot widen them.
- A schedule that would run past them is refused at upload, with the reason.

62 Backlash

planned

- Start with `BACKLASH_USTEPS = 10000`. The estimate from 0.35 mm on the final mesh is about 7,600, or 750 arcsec.
- Run `repeat_test`: orient, short track, rewind, read the encoder at the start angle, N times. It reports mean and σ of the return error in arcsec.
- Adjust until the mean return error sits near zero.

ⓘ The start angle is always approached going forward. Stop on a reverse stroke and the first minutes of tracking take up the backlash as drift.

63 Camera profile

planned

- Run `discover`. It joins the HERO6, dumps its `/gp/gpControl` settings tree and serves it at `/api/camera/dump`.
- Fill `camera_profile.json` from your camera's own tree. The IDs in FIRMWARE.md are placeholders.
- Three things to find out on the bench: does wake-on-LAN wake it, what it draws in standby, and whether it stays rejoinable after the ESP32 leaves mid-record.

64 First light

today

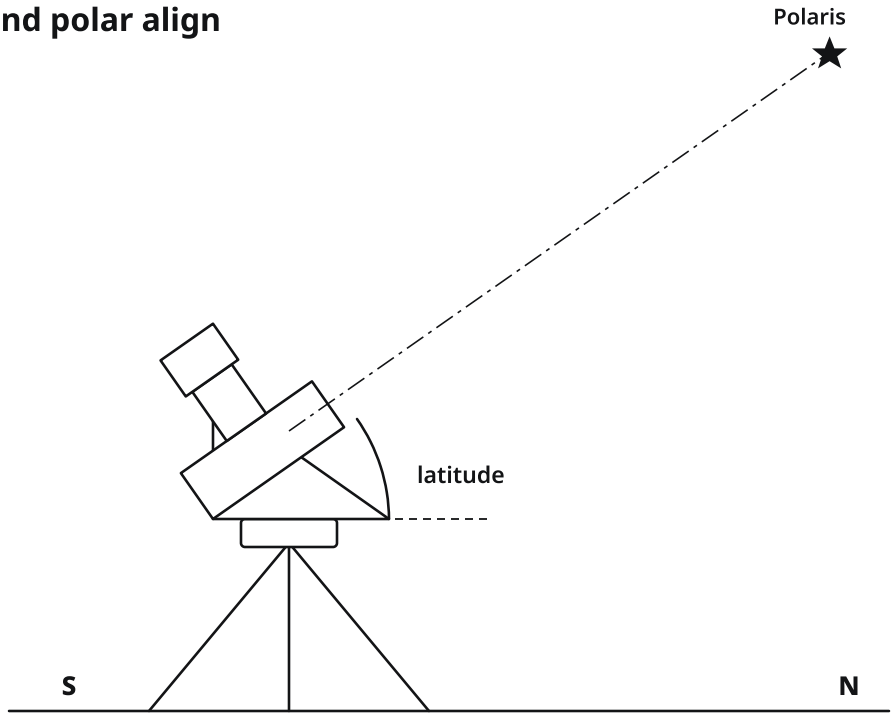
- Open-loop drive, a 50 mm lens, **one 5-minute exposure**. Measure the trailing.
- Only then decide what to fix. Upgrading shaft 3 to ground stock is the first mechanical step if the numbers call for it.

The encoder in this design is for scheduling and stall detection. Using it to correct tracking before you have trailing data solves a problem you have not measured.

F How to use

Set it up at dusk once, then leave it. The phone page and the night report reflect the planned firmware.

65 Set up and polar align

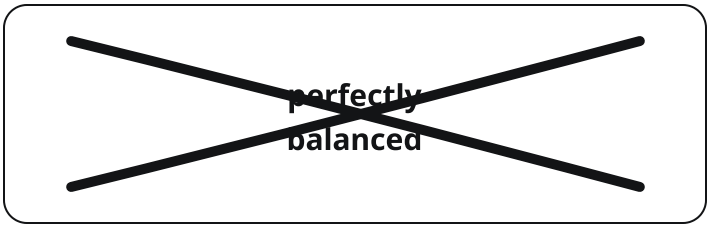


- Level the tripod. Tilt the wedge to your latitude and point the output axis north.
- Sight Polaris down the axis. Aim for 0.25° or better.

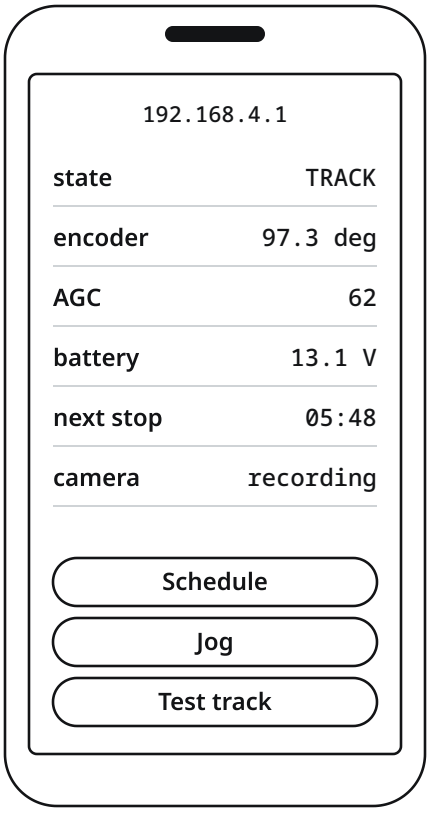
66 Load the camera

- Camera on, cable tied, slack loop free to swing.
- Bias the balance so the load pulls one way through the whole sweep. That keeps the mesh on one flank.
- Card empty, camera battery charged. It still runs on USB, and the battery rides through a hiccup.



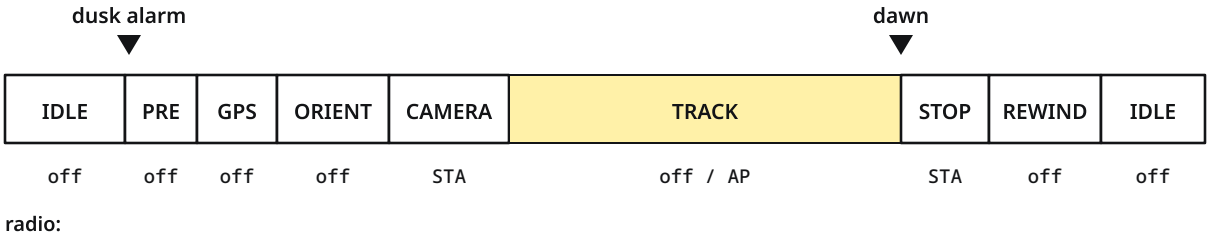


67 Configure from your phone



- Press **CONFIG** (SW1). The tracker raises its own Wi-Fi network for 10 minutes.
- Join it, open the page. Upload a schedule: site, start and stop events (dusk and dawn with offsets), start angle, camera profile.
- A bad schedule comes back with every error listed. One that would pass the travel limits is refused.
- Values on the example screen are illustrative.

68 Walk away

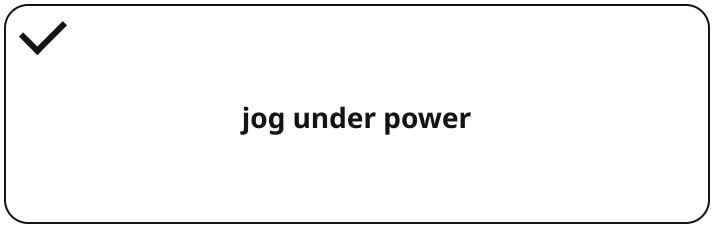


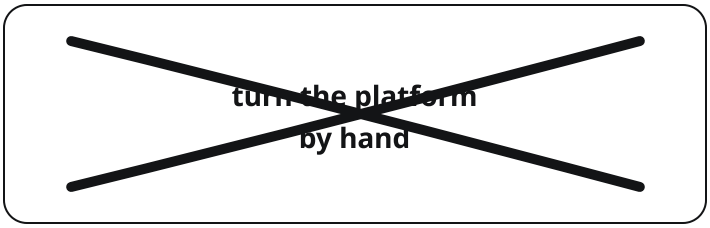
- It sleeps until the dusk alarm, checks itself, takes a GPS fix, slews to the start angle going forward, starts the camera, then tracks.
- At dawn it stops the camera, rewinds with an overshoot and comes forward to the start, then sleeps.
- Camera or GPS trouble is logged and the night tracks anyway. A stall, encoder disagreement over 2° or a flat battery stops the motor first, then the camera.

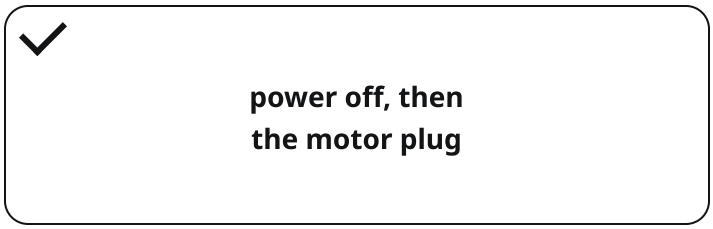
69 Next morning

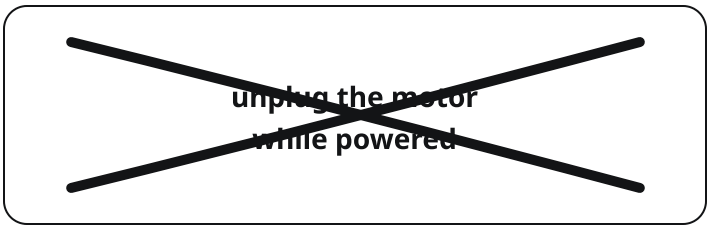
- Read the night report: planned against actual start and stop, the largest encoder disagreement, lowest battery, camera result, every warning.
- Encoder disagreement over **0.5°** is a warning worth a look; the train may be slipping.
- Keep the reports. When something odd happens on Tuesday, you want Tuesday's file.

70 Care









Symptom	First look
Drift from the first minutes	Polar alignment, then the forward approach to start
Once-per-turn trailing wobble	Shaft 3 nail straightness, the 24T pinion
FAULT at boot	Encoder not answering: J7 lead, magnet gap
Night skipped, no fault	Battery low, schedule invalid, or no twilight at your latitude
No frames, tracking fine	Camera profile IDs, wake-on-LAN, camera battery

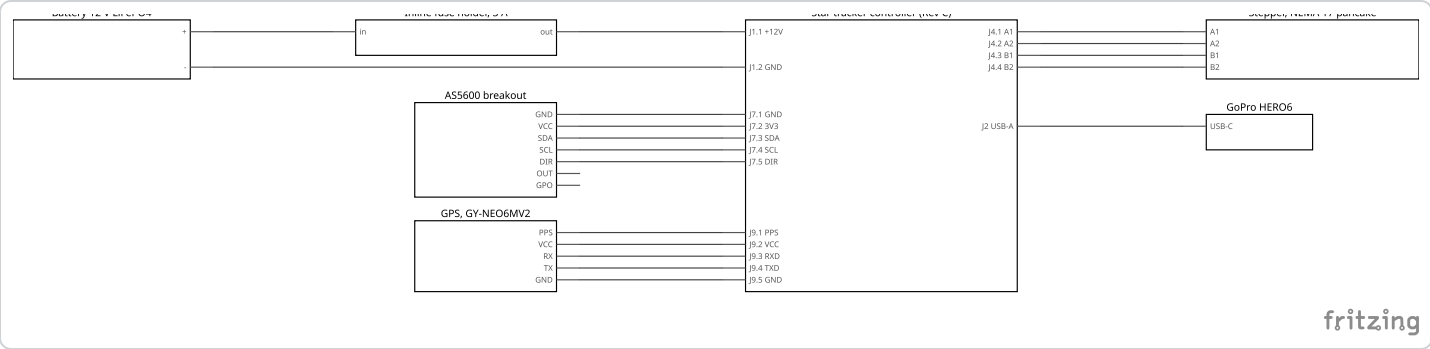
The numbers

Train	15→120, 15→120, 15→120 (module 1), 24→96 (module 2): 8 × 8 × 8 × 4 = **2048 : 1**
Microsteps per sidereal day	13,107,200 at 1/32 (200 × 32 × 2048)
Step interval	6573.80 µs, 152.13 steps/s, motor 1.426 rpm, 0.099 arcsec per microstep
Centre distances	67.5 / 67.5 / 67.5 / 120.0 mm
Shafts (x, y)	S0 (0, 0) · S1 (67.5, 0) · S2 (67.5, 67.5) · S3 (0, 67.5) · S4 (0, −52.5)
Output backlash	≈ 750 arcsec, 0.208°. `BACKLASH_USTEPS` 10,000 to start
One night	12 h, 180.5° of output. About 43 Wh, 69 % of it the camera
Magnet to die	2.0 mm (window 0.5 to 3.0)

G Appendix: schematics

The whole system as one diagram, the controller board as blocks, then the board's full schematic straight from KiCad.

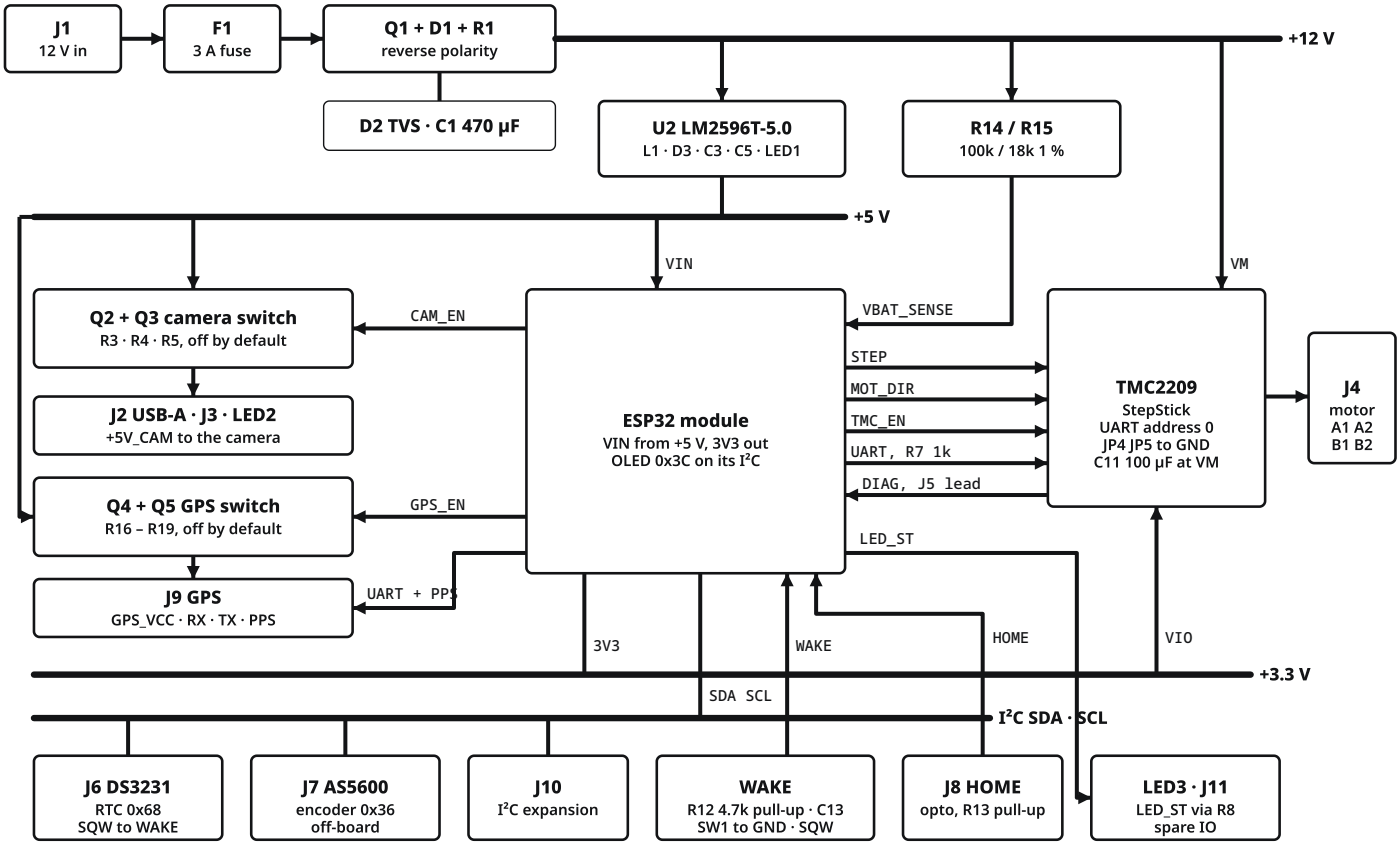
G1 · System wiring, schematic view



Fritzing, schematic view of the same sketch as section D. Each external part sits level with the board pins it meets, so every wire is one straight line.

Harness	Board	Far end	Wires
Power	J1 +12V, GND	battery via 3 A inline fuse	red through the fuse, black direct
Motor	J4 A1 A2 B1 B2	NEMA 17, coil A and coil B	pairs found with a meter (step 51)
Encoder	J7 GND 3V3 SDA SCL DIR	AS5600 GND VCC SDA SCL DIR	DIR is tied to GND on the board
GPS	J9 PPS VCC RXD TXD GND	GY-NEO6MV2, PPS from its LED	module's own 4-wire cable + one lead
Camera	J2 USB-A	HERO6 USB-C	short right-angle cable, slack loop

G2 · Controller board, logical diagram

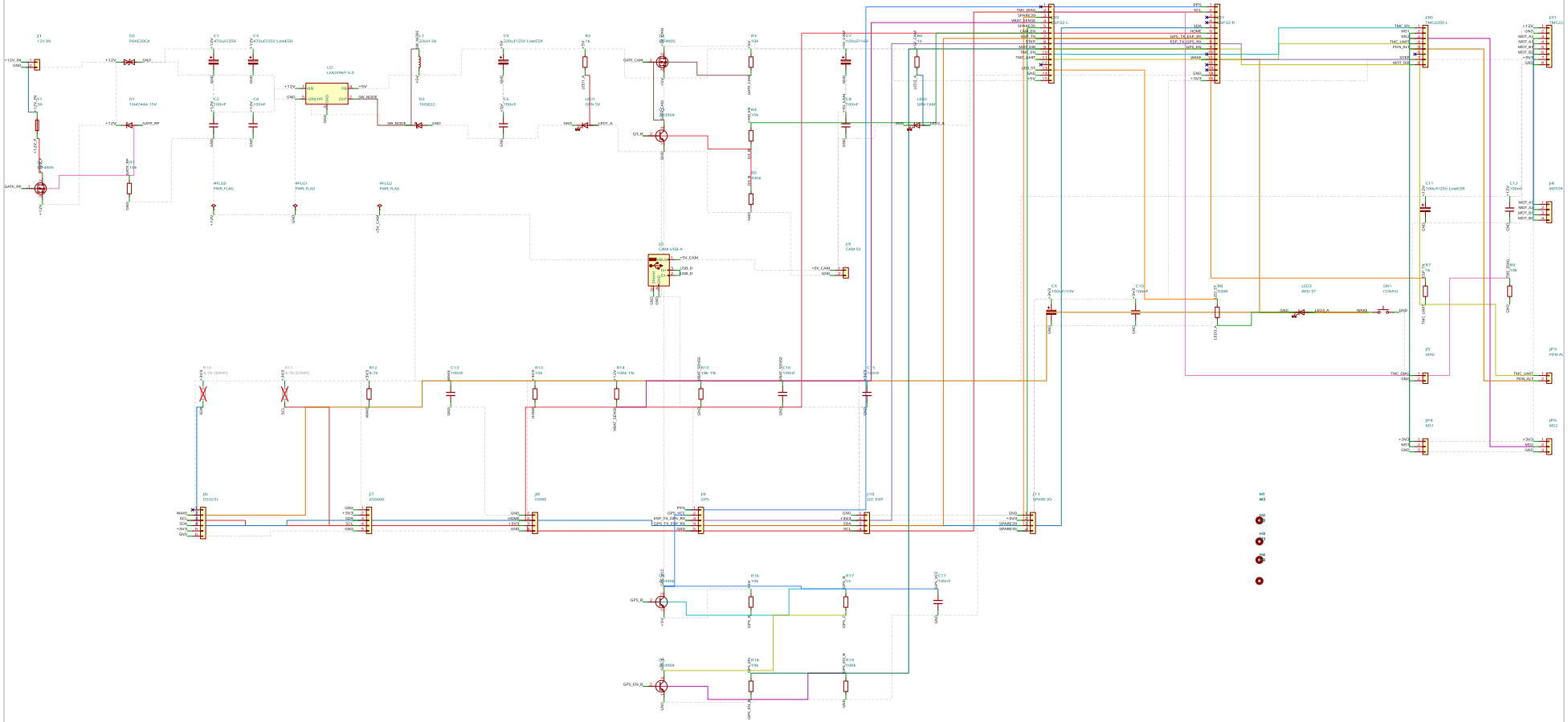


- **Three rails.** 12 V from the battery through F1 and Q1; 5 V from the LM2596 buck; 3.3 V from the ESP32 module's own regulator, about 20 mA of extra load.
- **Two loads are switched and off by default:** the camera's 5 V (CAM_EN) and the GPS's 5 V (GPS_EN). Pull-downs hold both off while the ESP32 boots.
- **One wake line** takes both the RTC alarm and the CONFIG button, wired-OR on a single 4.7k pull-up. Firmware tells them apart from the RTC's alarm flag.
- Signal names, not GPIO numbers: FIRMWARE.md §4 maps each to its GPIO for board Rev C.

GPS UART, fixed. Rev B named the nets GPS_TX / GPS_RX and wired the ESP32's transmit to the module's TXD. `design.py` now names them by direction, `ESP_TX_GPS_RX` and `GPS_TX_ESP_RX`, and sends the ESP32's TX (GPIO 5) to J9 pin 3, the module's RXD. Board Rev C carries the fix.

G3 · Controller board, full schematic

Rev C · 30-pin module



From `hardware/star_tracker_ctrl/star_tracker_ctrl.kicad_sch` via `kicad-cli`. Parts are joined by net label in KiCad; the coloured lines redraw those connections for this page (one colour per signal net; faint dashed grey for GND and the power rails). The drawing is vector: zoom in on it as far as you like, and search its text.

Board Rev C, as generated from `hardware/design.py`. Regenerate the board and rebuild the guide, and this page follows.